

# Fairness and Equity in LLM-Based Agents: A Taxonomic Survey

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Large language model (LLM) systems have crossed a threshold: they no longer merely *answer*, they *act*. Contemporary agents select tools and APIs, accumulate memory across turns, retrieve external evidence, delegate subtasks across roles, and plan over long horizons [137, 141, 148]. These capabilities are now embedded in hiring, lending, healthcare, and public-benefits pipelines, where a biased *decision* (not merely a biased sentence) carries material consequences for real individuals. Yet fairness research remains anchored overwhelmingly to single-turn question answering, scoring the words a model emits rather than the actions an agent takes.

We present the first dedicated survey of fairness and equity in LLM-based agents, organized around the **Bias Conduction Framework (BCF)**. The BCF formalizes two axes: *entry locus* (six agent components: tool/API selection, memory and retrieval, multi-agent delegation, planning and decomposition, user modeling, and long-horizon trajectory execution) and *conduction operator* (how disparity transforms across each pipeline edge: ATTENUATE, PRESERVE, AMPLIFY, or GENERATE). Together these yield a conduction equation that predicts trajectory-level disparity from component-level measurements. This framework converts a component taxonomy into a formal organizing framework yielding five falsifiable propositions, stated under the linearized BCF model, four grounded in existing corpus evidence and one (P3's cross-component form) stated as a falsifiable conjecture: locality (P1), masking (P2, the measurement gap), super-additivity (P3, the agentic multiplier), mitigation matching (P4), and measurement adequacy (P5). Grounded in a verified corpus of 202 papers spanning 2012 to 2026, the survey maps the literature onto the BCF locus x operator grid, delivers trajectory-level and action-level disparity metrics (tool-invocation, escalation, and plan-allocation disparity) formalized as instances of the BCF's component counterfactual disparity  $\Delta_c$ , constructs a mitigation map relating each intervention family to the pipeline component and operator it addresses, and reports a small pilot of an executable cross-framework bias audit whose results illustrate an endpoint-level analogue of P2 (a decision statistic masking a score-level disparity at the same output node) and are directionally consistent with the P3 amplification trend (mean absolute score difference rising as agentic scaffolding increases), with a full confirmatory study left to future work. We further characterize a structural *measurement gap*: today's benchmarks grade answers, not actions; as Table 2 shows, no general-purpose benchmark provides demographically instrumented, full-trajectory agent evaluation.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; *Artificial intelligence*; • **Social and professional topics** → **Socio-technical systems**.

Additional Key Words and Phrases: fairness, equity, large language models, LLM agents, bias, counterfactual fairness, trustworthy AI

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## 0.1 Introduction

*0.1.1 From Answering to Acting.* The dominant mode of LLM deployment is shifting from *generation* to *agency*. An agent wraps a language model in a loop that perceives state, invokes tools, reads and writes memory, and commits to actions in an external environment [137, 141]. The reasoning-and-acting model formalized by ReAct interleaves chain-of-thought with tool calls [148]; cognitive-architecture accounts generalize this to perception, memory, planning, and action modules [115]; and multi-agent frameworks compose several such loops into delegated, role-structured pipelines [97, 129]. The practical consequence is that an LLM increasingly does not return a paragraph for a human to act on. It *selects the applicant to interview, retrieves the precedent that frames the triage, escalates or denies the claim*.

This shift raises fairness stakes along three axes that single-turn question answering does not exhibit. First, **autonomy**: an agent's output is a decision or an action with downstream effect, not a suggestion a human filters. Second, **compounding**: bias introduced at one component (a skewed retrieval, a stereotyped role assignment) propagates through subsequent steps and can amplify over a trajectory rather than washing out [86, 91]. Third, **opacity**: harm is distributed across tool calls, memory writes, and inter-agent messages, so a disparity visible in the final action may have no single attributable cause in any one component. These properties matter precisely in the consequential, autonomy-bearing domains now adopting agents: hiring [14, 90, 126], lending and credit [13, 20, 41], clinical decision support [95, 99, 153], and public-benefits adjudication. Audits in these settings document substantial disparities: resume evaluation exhibits intersectional gender-and-race penalties [14]; a mortgage-underwriting experiment finds racial disparities in LLM approval recommendations [20]; and sociodemographic cues shift medical management plans even when clinically irrelevant [95]. Algorithmic hiring bias documented in the pre-LLM era [108] now re-emerges in LLM-agent form, with name-based and demographic-signal sensitivities persisting across hiring, judicial, and benefits pipelines [22, 58, 126].

*0.1.2 Thesis: Fairness Is a Per-Component, Operator-Mediated Property.* The organizing claim of this survey is that **fairness in an LLM agent cannot be captured by a single model-level metric; it must be analyzed per agent component**, because bias enters at distinct, separately-measurable, and separately-mitigable points in the pipeline and is transformed (attenuated, preserved, amplified, or generated from zero) at each edge. A model that is near-parity on a question-answering bias benchmark can still exhibit systematic tool-selection preferences independent of task fitness (provider and ordering bias) that no answer-level metric detects [21], inherit unfairness from a retrieval step the base model alone never exhibited [55, 132], acquire bias from an assigned persona or role [29, 45], or drift toward sycophantic, identity-conforming behavior over a long dialogue [76, 86]. Crucially, *agent decisions reveal implicit biases that direct questioning does not surface* [73]: probing the answer misses the disparity in the act. The Bias Conduction Framework (BCF), introduced in §3.0, provides the formal language for this per-component, per-edge analysis.

This component-by-component view is grounded in the broader recognition that bias in sociotechnical systems is not merely a model property but a systemic one shaped by design choices across the entire pipeline [17, 109]. Classical impossibility results [8, 62] establish that multiple group-fairness criteria cannot be simultaneously satisfied in general; the BCF shows how these tensions are *compounded* across agent components in ways that a model-level audit cannot detect.

*0.1.3 Contributions.* This survey makes five contributions, each traceable to a specific deliverable in the manuscript:

**Contribution 1: The Bias Conduction Framework (BCF).** We introduce the BCF, a formal framework organizing bias in LLM-based agents along two axes: *entry locus* (the six agent components) and *conduction operator* (ATTENUATE  $\varphi < 1$  / PRESERVE  $\varphi \approx 1$  / AMPLIFY  $\varphi > 1$  / GENERATE  $\varphi$  on zero input). The formal core (§3.0) defines: D1 the agent loop as the tuple  $\langle \pi, T, M, R \rangle$  (policy, tool registry, memory module, reward/evaluation signal); D2 component counterfactual disparity  $\Delta_c$  measured via trace-replay; D3 the set of conduction operators. The conduction equation  $\Delta_\tau \approx \Sigma_c \Delta_c \cdot \Pi \varphi_e$  with feedback recurrence  $\Delta^{(t+1)} = \varphi_{fb} \cdot \Delta^{(t)} + \Delta_{entry}$  predicts trajectory-level disparity from component measurements. The framework is a formal organizing framework yielding five falsifiable propositions, four grounded in existing corpus evidence and one (P3’s cross-component form) stated as a falsifiable conjecture: P1 Locality (a disparity at component  $c$  is attributable to that component’s locus if it vanishes under a component-level counterfactual), P2 Masking (token-level parity can coexist with action-level disparity, the measurement gap as a theorem), P3 Super-additivity (the agentic multiplier:  $\Delta_\tau$  can exceed the naive sum  $\Sigma_c |\Delta_c|$  when at least one AMPLIFY or GENERATE edge lies on a conduction path), P4 Mitigation matching (an intervention reduces  $\Delta_\tau$  only if it targets the locus and operator of the dominant  $\Delta_c$ ), and P5 Measurement adequacy (a fairness metric is adequate for an agent only if it surfaces  $\Delta_c$  for every component with non-negligible  $\varphi$ , with adequacy graded in the tiers of §3.0.5). Section 3.8 provides a worked hiring-agent example tracing bias from C0 (base model) through C3 (delegation), and a four-step auditor recipe (instrument trace-replay, estimate  $\varphi_e$  per edge, identify dominant locus, match mitigation to operator).

**Contribution 2: Trajectory-level and action-level disparity metrics.** We formalize trajectory-level Counterfactual Flip Rate (CFR $_\tau$ ) and Mean Absolute Score Difference (MASD $_\tau$ ), generalizing the single-domain metrics of [85] to any agent component as instances of D2’s  $\Delta_c$ , and introduce three action-level disparity metrics (tool-invocation disparity  $\Delta_{tool}$ , escalation-rate disparity  $\Delta_{esc}$ , and plan-allocation disparity  $\Delta_{plan}$ ) that current benchmarks do not capture (§4.3). These replace the draft’s “no canonical formula” with explicit definitions.

**Contribution 3: A mitigation map keyed to the BCF locus x operator x pipeline-stage grid.** We construct Table 3 mapping each mitigation family (prompting, fine-tuning/alignment, guardrails, multi-agent debiasing, process-level architectural interventions) to the pipeline component it targets and the operator type (ATTENUATE vs. AMPLIFY/GENERATE) it addresses. The map exposes that the (AMPLIFY, GENERATE) cells (the most consequentially agentic entries) are currently the emptiest (§5).

**Contribution 4: A protocol and pilot for an executable cross-framework bias audit.** We specify, and run a small pilot of, a protocol (§6): multiple agent configurations on a consequential-decision task with counterfactual demographic swaps, testing whether component-level action disparity arises that QA-level scoring of the same underlying model would not register. *The pilot illustrates the predicted masking-and-amplification pattern but is directional only; a full confirmatory study is future work.* The protocol, scripts, and raw decisions will be released with the camera-ready.

**Contribution 5: An open-problems agenda with operationalized first steps.** We synthesize the field’s open edges (§8) into a research agenda toward fairness benchmarks that score *acting* agents, covering long-horizon and compounding fairness, multi-agent dynamics, intersectional deployment harms, and agent-level (rather than token-level) mitigation, and attach a concrete first step (metric formula, dataset structure, or system design) to each open problem (§8, per G10).

**0.1.4 Positioning and Differentiation. Relation to the QA-fairness lineage.** Chu et al.’s *Fairness in Large Language Models: A Taxonomic Survey* [24], a closely related ACM survey-style article (ACM SIGKDD Explorations), taxonomizes bias in LLMs that *answer*: its axes are intrinsic/extrinsic bias, fairness definitions, and debiasing of generation; it does not treat the agentic surface (tools,

memory, delegation, planning, horizons) where distinct fairness phenomena arise. We inherit its definitional grounding [30, 46, 50, 61] and extend it to systems that act. The classical ML fairness survey of [80] and the machine-learning fairness reviews [6, 163] provide baseline vocabulary; we extend upstream into the agent components that condition what data reaches the model and how outputs translate into consequential actions. We note the well-documented tension between individual and group fairness criteria [30] and the impossibility results [8, 62] that constrain any single-metric resolution: tensions the BCF shows are *structurally amplified* in multi-component pipelines.

**Differentiation from dedicated near-neighbors.** Table 0 (below) maps eight near-neighbor surveys against this survey on nine dimensions. Five works warrant explicit comment:

- *Gallegos et al. 2024* [46]: authoritative LLM-level bias survey; no agent-component axis, no action-level metrics, no component-counterfactual evaluation.
- *Chu et al. 2024* [24]: direct parent lineage (SIGKDD Explorations); fairness-dedicated but organized by LLM task, not agent component; does not treat agentic surfaces.
- *Mohammadi et al. 2025* [82]: broad agent-evaluation survey in which fairness is one subdimension among many; not fairness-dedicated, not component-organized.
- *Vatsal et al. 2026* [128]: a clinical-agent evaluation taxonomy whose seven dimensions include a Safety & Ethics dimension under which fairness sits; conceptual and single-domain (healthcare), with no component-level counterfactual and no corpus.
- *Mayilvaghanan et al. 2026* [85]: applies CFR/MASD in a single domain (contact-center agent quality assurance); our survey generalizes those metrics across all six BCF components.
- *Ranjan et al. 2025* [164]: a 12-page position paper on fairness in agentic/multi-agent AI broadly, not specific to LLM-agent components; no component taxonomy, no empirical corpus.
- *Ebrahimi and Asudeh 2025* [10]: RTAF survey of LLM multi-agent systems in which fairness is one of four pillars; no component-level fairness taxonomy.
- *Binkyte 2025* [2]: narrow scope (interactional fairness in negotiation); AIES 2025 workshop paper, not a full survey.
- *Yu et al. 2025* [149]: trustworthy LLM agents survey; fairness is one dimension among many (threats and countermeasures framing).

Among all surveyed works, this survey is the first to simultaneously: (a) dedicate the entire organizing axis to fairness, (b) structure that axis by agent component (not by metric family, domain, or trust dimension), (c) introduce a named formal framework (BCF) with testable propositions, and (d) deliver action-level disparity metrics and an audit protocol. We state this claim once here; subsequent sections reference Table 0 rather than repeating it.

**Corpus note.** This survey covers 202 verified papers. Approximately 140 of those papers are not covered by either [46] or [24], reflecting the rapid growth of agentic fairness work in 2024 to 2026. The 65% of the corpus published in 2024 or later confirms that the subfield is emergent rather than mature, a temporal trend analyzed in §2.6.

*Table 0: Survey Comparison. Takeaway: this survey is the only work combining whole-survey fairness dedication, agent-component organization, a named formal framework (BCF), action-level executable metrics, and a  $\geq 200$ -paper corpus, the (AMPLIFY, GENERATE) quadrant of the BCF has no coverage in any prior survey.*

*Note: approximately 140 of the 202 papers in this corpus are not covered by [46] or [24].*

**0.1.5 How to Use This Survey.** This survey is written to serve four distinct reader communities. Each path is self-contained.

| Survey                 | Year        | Venue                            | Organizing axis                               | Agent coverage                 | Fairness-dedicated                   | Component level | Corpus size | Original results          | Audit/executable metric?               |
|------------------------|-------------|----------------------------------|-----------------------------------------------|--------------------------------|--------------------------------------|-----------------|-------------|---------------------------|----------------------------------------|
| Mehrabi et al. [80]    | 2021        | ACM CSUR                         | Bias source $\times$ ML stage                 | None                           | Yes                                  | No              | 150         | No                        | No                                     |
| Gallegos et al. [46]   | 2024        | <i>Computational Linguistics</i> | Bias type $\times$ LLM task                   | Peripheral                     | Yes                                  | No              | 250         | Partial                   | No                                     |
| Chu et al. [24]        | 2024        | ACM SIGKDD Expl.                 | Fairness definition $\times$ LLM task         | None                           | Yes                                  | No              | 100         | No                        | No                                     |
| Mohammadi et al. [82]  | 2025        | arXiv                            | Evaluation dimension                          | Full                           | No                                   | No              | 200         | No                        | No                                     |
| Ranjan et al. [164]    | 2025        | arXiv (position)                 | MAS ethical pillars                           | Classical MAS only             | Yes                                  | No              | 30          | No                        | No                                     |
| Ebrahimi & Asudeh [10] | 2025        | Preprint (RTAF)                  | Trust pillars (RTAF)                          | LLM-MAS                        | Partial (1 of 4)                     | No              | 100         | No                        | No                                     |
| Binkyte [2]            | 2025        | AIES 2025                        | Interactional fairness (negotiation)          | Narrow                         | Yes                                  | No              | 40          | Partial                   | No                                     |
| Vatsal et al. [128]    | 2026        | IEEE Access                      | 7-dimensional taxonomy                        | Healthcare                     | Partial (subdim. of Safety & Ethics) | No              | N/A         | No                        | No                                     |
| Yu et al. [149]        | 2025        | ACM SIGKDD                       | Trustworthy AI threats                        | Full                           | Partial (1 dim.)                     | No              | 200         | No                        | No                                     |
| Mayilvahan et al. [85] | 2026        | arXiv                            | CFR/MASD                                      | Single domain                  | Yes                                  | No              | N/A         | Yes                       | Single-domain                          |
| <b>This survey</b>     | <b>2026</b> | <b>ACM CSUR (target)</b>         | <b>BCF locus <math>\times</math> operator</b> | <b>Full (all 6 components)</b> | <b>Yes</b>                           | <b>Yes</b>      | <b>202</b>  | <b>Protocol specified</b> | <b>Action-level (tool, esc., plan)</b> |

**Practitioner (deploying an agent system in a consequential domain).** Read §2.3 (component anatomy) to identify which components your system instantiates, §3.0 (BCF) for a one-page diagnostic of where bias can enter and amplify, the relevant subsection of §3 for your deployment’s dominant component, and §5 for mitigations keyed to that component. The four-step auditor recipe at the end of §3.8 gives a concrete starting procedure. Skip §4.2 to 4.3 on first pass if you need operational guidance now.

**Auditor or regulator (assessing compliance or investigating disparate impact).** Read §2.1 (fairness definitions, with their legal analogues) and §2.3, then §3.0 (BCF propositions P1 to P5), then §4 (evaluation metrics, especially §4.3 for action-level disparity metrics and the  $\text{CFR}_\tau/\text{MASD}_\tau$  formulas). Section 6 describes the audit protocol in full; the four-step auditor recipe in §3.8 operationalizes it. Governance and legal anchoring (EU AI Act, NIST AI RMF, NYC Local Law 144, EEOC guidance) is collected in §8.5.

**Researcher (contributing to the fairness-of-agents literature).** Read the entire survey. The coverage matrix (Figure 4, §7) visualizes which BCF cells are most underinstrumented, and these are the highest-priority open problems. The BCF propositions in §3.0 are the natural targets for

theoretical work; the trajectory-level metric definitions in §4.3 are the natural targets for empirical instantiation; and §8 itemizes concrete open problems with first-step specifications.

**Benchmark builder (designing the first agent-level fairness benchmark).** Read §4 (evaluation metrics) and §4.4 (Table 2 benchmark inventory) to understand the gap, §3.8 (worked example and auditor recipe) for what an agent-level probe must capture, §6 for the audit protocol your benchmark should instantiate, and §8.1 (the case for a trajectory-level fairness benchmark) for the desiderata. The BCF’s D2 definition of  $\Delta_c$  via trace-replay gives the formal target your benchmark must operationalize.

## 0.2 Background & Scope

*0.2.1 Fairness Definitions.* Formal fairness research has converged on three partially complementary families of criteria, each encoding a distinct moral intuition about equitable treatment. Understanding their boundaries, and the impossibility results that connect them, is a prerequisite for the component-level analysis that follows.

*Group Fairness.* Group fairness requires that outcomes be statistically equivalent across protected demographic groups. The canonical formal treatment in the supervised-learning setting comes from [50], who distinguish *equality of opportunity* (equal true-positive rates across groups, conditioned on a positive true label) from *equalized odds* (equal both true-positive and false-positive rates). Both criteria reduce fairness to a statistical comparison over a finite set of labeled groups and a classifier’s output distribution, making them operationally tractable and empirically dominant in applied auditing. An earlier but complementary lineage grounds group fairness in *disparate impact*: [40] formalize disparate impact in the legal sense and propose a certification procedure for removing it from classifier predictions, bridging legal doctrine and machine learning practice.

A broader survey of classical ML fairness definitions, cataloging over twenty distinct bias sources across preprocessing, in-processing, and post-processing pipeline stages, is given by [80]. [1] provide a textbook synthesis; [6] offer a more recent CSUR survey of the formal field. That taxonomy serves as this survey’s baseline vocabulary; we extend it into the agent components that condition what data reaches the model, how reasoning is structured, and how outputs are translated into consequential actions.

*Individual Fairness.* Individual fairness is the criterion articulated by [30] as *fairness through awareness*: similar individuals, judged by a task-relevant similarity metric, should receive similar outcomes, regardless of group membership. This is the classical liberal intuition that like cases be treated alike. Dwork et al.’s paper is correctly characterized as an individual fairness result; it is *not* a source of the group fairness criteria above. The distinction matters: group fairness aggregates over distributions while individual fairness operates on pairs of individuals and requires specifying a task-appropriate similarity metric, a notoriously difficult design choice that has limited the criterion’s empirical adoption relative to group-level metrics.

In agentic contexts, individual fairness generalizes naturally to trajectory pairs: two agents processing otherwise-identical applicant profiles that differ only in a protected attribute should reach the same decision via the same decision path. This trajectory-level generalization is rarely formalized in current evaluation practice. Counterfactual auditing methods such as the flip-rate designs of [22] and [85] are beginning to operationalize it.

*Counterfactual Fairness.* Counterfactual fairness, introduced by [61], shifts the framing from outcomes to causal mechanisms. A decision is fair toward an individual if it would remain the same in a counterfactual world in which that individual’s protected attributes had taken different values, with all *descendants of those attributes in the causal graph* correspondingly intervened upon.

More precisely: for an individual with attributes ( $A = a, X = x$ ), a predictor  $\hat{Y}$  is counterfactually fair if  $P(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)$  for all  $y$  and all values  $a'$ . The key causal structure is that the intervention is on the protected attribute and propagates to its descendants in the causal DAG; non-descendants are held fixed. This differs from naive demographic substitution: attributes causally downstream of  $A$  (an outcome shaped by historical discrimination, for instance) are also counterfactually altered, preventing the criterion from laundering discrimination through proxy variables.

This causal perspective is particularly consequential for agentic settings, where multi-step decision chains create long causal paths along which demographic signals can propagate and amplify. The Bias Conduction Framework (BCF) developed in Section 3.0 formalizes precisely this propagation structure.

*Impossibility Results and Metric Tensions.* A critical finding for survey readers is that these three families are mutually *incompatible* under general conditions. [62] prove that three natural fairness criteria for risk scores (calibration within groups, balance for the positive class, and balance for the negative class) cannot simultaneously be satisfied except in degenerate cases. [8] establishes a parallel impossibility: demographic parity, predictive parity (calibration), and equal false-positive and false-negative rates cannot all hold when base rates differ across groups, making the trade-off not a matter of technical ingenuity but of irreducible normative choice. These impossibility results mean that an agentic deployment *must* make fairness criterion choices that implicitly favor some groups over others; no criterion is neutral. [163] and [165] survey these tensions in more depth.

For agent evaluation, the impossibility results carry a further implication: a system that satisfies group fairness on one metric may violate it on another, and a mitigation that improves counterfactual consistency may worsen demographic parity. The BCF framework's Proposition P4 (Mitigation-matching) addresses this directly by requiring that mitigations be keyed to the specific criterion and conduction signature of the target component, rather than applied globally.

### 0.2.2 LLM Bias: A Brief Inventory and the Agentic Gap.

*Pre-Agentic Bias in LLMs.* The past several years have produced a rich literature measuring bias in large language models as *generators*. [46] provide an authoritative survey of bias and fairness across the LLM lifecycle, covering stereotyping, toxicity, representational harm, and distributional disparities in generated text; [24] offer a taxonomic treatment organized by fairness dimension and LLM task type, serving as the direct parent lineage for the present work. [17] provide an earlier critical survey emphasizing that framing choices in bias detection encode normative assumptions that researchers must make explicit. [89] catalog bias origins, taxonomize known bias types, and situate them within the broader discourse on responsible AI. [5] supply the canonical empirical grounding: commercial gender classifiers exhibit substantially larger error rates for darker-skinned women than for lighter-skinned men, establishing that model-level demographic disparities have measurable real-world consequences before any agentic amplification occurs. [23] establish that human-like social biases are present in word embeddings derived from language corpora, tracing the bias back to statistical regularities in training text.

Empirical measurement has been operationalized through benchmark families probing specific bias surfaces:

- **Coreference and gender stereotyping.** [150] introduce WinoBias, pairs of coreference sentences that differ only in whether a gendered pronoun resolves to an occupationally stereotyped referent; [107] provide the WinoGender companion. Both benchmarks establish that coreference models systematically favor stereotypical assignments.

- **Ambiguous QA.** [96] (BBQ) uses ambiguous question-answering scenarios to surface stereotypical associations across nine social dimensions including age, race, religion, and disability. The benchmark is adversarially constructed so that a model relying on stereotypes answers differently from a model relying on context.
- **Open-ended generation.** [31] (BOLD) measures sentiment and regard disparities in open-ended generation conditioned on demographic group membership; [88] (StereoSet) evaluates the trade-off between language modeling ability and stereotypical bias; [87] (CrowS-Pairs) surface biases favoring historically advantaged groups in masked language models; [110] (HolisticBias) extend coverage to over 600 demographic descriptors across thirteen axes.
- **Toxicity.** [44] (RealToxicityPrompts) evaluate toxic degeneration in language models, showing that even neutral prompts can elicit harmful content with demographic targeting; [51] (ToxiGen) extend this to adversarially generated implicit hate speech.
- **Trustworthiness benchmarks.** [130] (DecodingTrust) provide a full assessment of trustworthiness in GPT models, including a fairness dimension alongside stereotyping, privacy, and adversarial robustness; [68] (HELM) offer a broad evaluation framework covering accuracy, calibration, robustness, and bias across dozens of scenarios. [56] (TrustLLM) extend the trustworthiness assessment to a wide range of LLMs with fairness as a primary evaluation axis.
- **Bias origins and pitfalls.** [18] audit fairness benchmark datasets and catalog recurring methodological pitfalls, establishing that measurement artefacts can mislead benchmark-based conclusions. This concern is taken up again in Section 2.8 (Threats to Validity).

This body of work has established that LLMs encode and reproduce social stereotypes, exhibit differential performance across demographic groups, and generate text with systematically unequal sentiment toward protected categories.

*Why QA-Level Evaluation Is Insufficient for Agents.* All of the benchmarks above evaluate **single-turn language generation**: a prompt is supplied, a response is generated, and a fairness metric is computed over the token distribution. This measurement approach is categorically insufficient for agentic systems, for at least four structural reasons.

First, **sequential causation**: a demographic signal present in a user’s profile may influence an early tool selection, which constrains subsequent information retrieval, which shapes a planning step that produces the final consequential decision. The bias does not appear in any single response but accumulates across the trajectory. Second, **persistent state**: memory modules, retrieved context, and conversation history create channels through which biased inferences carry forward across sessions, a mechanism absent from single-turn evaluation. Third, **external tools and APIs**: agents interact with external systems that may themselves encode biases in their outputs, creating harm attributable not to the LLM’s parameters but to the LLM’s tool-selection policy. Fourth, and decisively, **the action-answer gap**: [73][†] demonstrate that agent *decisions* (persona-based simulated decisions rather than tool-using pipeline actions, hence the transfer marker) reveal implicit biases invisible at the verbal-output level. A model can emit neutral text while simultaneously issuing actions with demographic disparity. BBQ can detect that a model answers a stereotyped question incorrectly; it cannot detect that a hiring agent systematically routes applicants with certain name patterns to lower-quality resume-screening tools.

This measurement gap is the central observation of this survey. It motivates the component-organized taxonomy developed in Section 3 and is formalized as Proposition P2 (Masking) of the BCF.



**0.2.3 What Is an LLM-Based Agent? Component Anatomy.** An LLM-based agent is a system in which a large language model serves as the core reasoning engine within an architecture that supports autonomous, multi-step interaction with external environments [137, 141]. The cognitive architectures survey by [115] provides a systematic framework for decomposing these systems into functional constituents; [47] survey the progress and challenges of multi-LLM agent systems; [34] survey multimodal agent interaction. We adopt the formal characterization developed in Section 3.0, treating an agent as a tuple  $\langle \pi, T, M, R \rangle$  where  $\pi$  is the policy,  $T$  is the tool set,  $M$  is the memory module, and  $R$  is the reward or stopping signal, as the definition against which component-level fairness properties are measured.

The ReAct framework [148] interleaves chain-of-thought reasoning with tool calls; Toolformer [113] demonstrates self-supervised tool use; RAG [67] grounds generation in retrieved documents; AutoGen [129] and MetaGPT [54] enable multi-agent delegation and orchestration; generative agent simulations [97] demonstrate emergent social behavior from agent interaction; Reflexion [112] accumulates verbal self-reflections as episodic memory.

We identify six components whose design choices are consequential for fairness, grouped by their primary role in the agent loop:

**Component 1: Tool and API selection.** The agent chooses, at each step, which external tool, API, retriever, or knowledge source to invoke. This choice is a learned behavior conditioned on contextual signals: systematic selection preferences independent of task fitness (provider and ordering bias) are already documented [21], and demographic signals in the query or retrieved context are a further, as-yet-unmeasured conditioning channel. Tool-selection policy is therefore a locus of discriminatory routing, independently of the final answer quality.

**Component 2: Memory and retrieval.** Agents maintain in-context history and external memory stores indexed for retrieval [67]. The retrieval function's ranking can encode demographic disparities, and the model then generates conditioned on a potentially biased information set. Accumulation over turns adds a temporal dimension [42].

**Component 3: Multi-agent orchestration and delegation.** Modern deployments distribute tasks across networks of specialized agents [97, 129]. Role assignment is a direct channel for bias: if persona or role assignment correlates with protected attributes, downstream behaviors of the assigned agent differ systematically [45, 91].

**Component 4: Planning and task decomposition.** Agents decompose high-level goals into subtask sequences, often via chain-of-thought or structured planning. The ordering and framing of subtasks can encode implicit assumptions about whose needs are primary, assumptions that may vary systematically by demographic context [73, 139].

**Component 5: User modeling and personalization.** Agents adapt to individual users by inferring preferences, expertise, and identity from interaction history or profile data [63]. This adaptive capacity creates a personalization-stereotyping tension: inferences that improve average-case utility for a group may simultaneously instantiate harmful stereotypes for individual users who deviate from that group's statistical profile.

**Component 6: Long-horizon trajectory execution.** Over extended multi-turn interactions or autonomous task executions, small per-step biases can compound into large cumulative disparities [86, 133]. This longitudinal dimension is not captured by any existing single-turn benchmark and constitutes the most undertheorized component in current fairness research. We present it here as a sixth conceptual dimension, but §3.6 argues it is more precisely a temporal axis on the feedback edge rather than an independent architectural component, a distinction the BCF makes precise once the framework is in place.

These six components form the organizing axis of the taxonomy in Section 3. The BCF framework formalizes how bias propagates across them. Two classes of agent evaluation benchmarks assess

the broader capabilities of agents without a fairness lens, and both must be acknowledged here: WebArena [154], AgentBench [70], SWE-bench [60], and Mind2Web [32] evaluate task completion in realistic environments and could, in principle, be demographically instrumented; their current designs do not include demographic counterfactual conditions, which is precisely the gap Section 4.3 formalizes as action-level disparity metrics.

**0.2.4 Scope and Coverage Decisions. Survey scope.** This survey covers empirical and theoretical research on unfairness and inequity arising specifically within the architecture of LLM-based agent systems, meaning systems performing multi-step, tool-augmented, or multi-agent reasoning toward a goal in a consequential domain. We do not survey the broader literature on bias in static LLM generation except insofar as those findings illuminate mechanisms operative in agentic settings; that literature is ably synthesized by [46] and [24]. We exclude non-LLM agents (classical planning systems, reinforcement learning agents without LLM backbones), though we draw on classical fair sequential decision-making results [142, 158] where they provide theoretical grounding, and on recommender-system fairness [151, 167] where the feedback-loop dynamics parallel agentic deployment.

**Inclusion criteria.** A paper was included if it satisfied all of the following: (i) the system studied uses an LLM as a primary reasoning component; (ii) the paper addresses bias, fairness, equity, or disparate treatment in outcomes or behaviors; (iii) the paper’s findings bear on at least one of the six agent components identified above, or on the evaluation or mitigation of bias in such components. Papers exclusively studying static generation tasks without an agentic framing were included only when the mechanism they document is demonstrably operative in agentic settings; such papers are marked [*adjacent evidence, transferred*] in the coverage table (Table 1, §3 header). Papers from non-LLM recommender systems or classical ML fairness were included only when directly cited by the agent fairness literature or when they provide the only available formal grounding for a mechanism.

**Multi-tagging convention.** A paper may be tagged to multiple components when its findings are mechanistically relevant to each. The coverage matrix in §7 counts such papers in each tagged cell; the total of unique papers is 202. Papers that cross-cut four or more sections (e.g., [46] ×4, [42] ×3) are counted once in the unique total.

#### 0.2.5 Systematic Search and Corpus Construction.

**Search Protocol.** The corpus was assembled through a systematic multi-branch search following PRISMA-style conventions [77]. Search queries were issued across the following databases: arXiv (cs.AI, cs.CL, cs.LG, cs.IR), ACL Anthology, ACM Digital Library, NeurIPS/ICML/ICLR proceedings archives, FAccT proceedings, AAAI proceedings, and IEEE Xplore, supplemented by forward and backward citation chasing from anchor papers. Searches were conducted between January 2026 and June 2026; the literature cutoff date is **June 10, 2026**. Papers published after that date are not included.

Ten of the eleven branches received a dedicated search string constructed from the intersection of (a) an agent/agentic qualifier and (b) a fairness/bias qualifier, with branch-specific terms (the six component branches, plus evaluation, mitigation, high-stakes domains, and a prior-art neighbor-survey search). The remaining branch (foundations and adjacent evidence) was populated by backward citation chasing from the agent-fairness corpus rather than independent search.

**Table 2.1: Search Protocol by Branch.** *Takeaway: the branch structure maps directly onto the six BCF entry loci plus evaluation and mitigation; no branch has fewer than 10 identified records before deduplication.*

Table 2.1: Search Protocol by Branch. *Takeaway: the branch structure maps directly onto the six BCF entry loci plus evaluation and mitigation; no branch has fewer than 10 identified records before deduplication.*

| Branch                        | Primary Search String                                                         | Supplementary Terms                                                         | Databases                      | Records Identified | After Dedup | After Screen | Included                                                             |
|-------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------|--------------------|-------------|--------------|----------------------------------------------------------------------|
| Tool/API selection            | "tool selection bias" OR "tool use fairness" LLM agent                        | tool-calling, function-calling, API bias, routing bias                      | arXiv, ACL, ACM DL             | 47                 | 34          | 18           | 12                                                                   |
| Memory & retrieval            | RAG fairness OR "retrieval bias" LLM                                          | retrieval-augmented, knowledge store, context accumulation, multi-turn bias | arXiv, ACL, ACM DL, SIGIR      | 52                 | 38          | 16           | 10                                                                   |
| Multi-agent delegation        | "multi-agent bias" OR "role assignment bias" LLM                              | persona bias, delegation fairness, emergent bias, agent interaction         | arXiv, ACL, ACM DL, AAAI       | 41                 | 31          | 14           | 9                                                                    |
| Planning/decomposition        | "reasoning bias" LLM agent OR "planning fairness"                             | chain-of-thought bias, task allocation, decomposition disparity             | arXiv, ACL, ICML               | 28                 | 21          | 9            | 6                                                                    |
| User modeling/personalization | "personalization bias" LLM OR "user modeling fairness"                        | name-based bias, dialect fairness, stereotype personalization               | arXiv, ACL, ACM DL             | 44                 | 33          | 15           | 11                                                                   |
| Long-horizon drift            | "long-term fairness" LLM OR "feedback loop bias" agent                        | sycophancy drift, memory accumulation, sequential fairness                  | arXiv, AAAI, ICML, FAccT       | 38                 | 29          | 12           | 9                                                                    |
| Evaluation & benchmarks       | "agent fairness benchmark" OR "fairness evaluation LLM agent"                 | counterfactual evaluation, action-level disparity, audit                    | arXiv, ACL, ACM DL, NeurIPS    | 61                 | 44          | 22           | 16                                                                   |
| Mitigation                    | "LLM debiasing agent" OR "fairness mitigation agent"                          | RLHF fairness, RAG debiasing, prompt debiasing, constitutional AI           | arXiv, ACL, NeurIPS, AAAI      | 55                 | 40          | 17           | 13                                                                   |
| Foundations (backward chase)  | n/a                                                                           | Parent surveys, impossibility results, classical ML fairness                | All databases + Google Scholar | 93                 | 74          | 61           | 45                                                                   |
| High-stakes domains           | "LLM hiring bias" OR "LLM clinical fairness" OR "LLM lending bias"            | resume screening, medical decision, credit, education, judicial             | arXiv, Nature, EMNLP, PNAS     | 48                 | 36          | 19           | 13                                                                   |
| Prior-art additions           | Neighbor survey search: ranjan2025, binkyte2025, ebrahimi2025 + forward chain | adjacent CSUR trustworthy-AI surveys                                        | ACM DL, arXiv, ResearchGate    | 41                 | 35          | 31           | 29                                                                   |
| <b>Total</b>                  |                                                                               |                                                                             |                                | <b>548</b>         | <b>415</b>  | <b>234</b>   | <b>173</b> (by primary branch; 202 unique after multi-tag, see note) |

*Note: the visible Included column lists each branch's records by primary-branch attribution and sums to 173; it does not sum to 202 because the table omits cross-cutting papers that are counted only in their secondary branches. The full corpus of 202 unique papers is recovered by a separate primary-branch partition (each paper assigned to its single primary locus), which does sum to 202; cross-cutting papers are also counted in secondary branches in the coverage matrix (§7). That primary-branch partition is: foundational/must-cite 46, related-surveys/prior-art 29, foundations 16, evaluation-metrics-benchmarks 16, mitigation 13, high-stakes-domains 13, llm-agents-general 12, tool/API-selection 12, user-modeling/personalization 11, memory/retrieval 10, multi-agent-delegation 9, long-horizon-drift 9, planning/decomposition 6.*

*PRISMA Flow (Figure 1).* The corpus assembly followed four stages:

**Stage 1: Identification (n = 548).** Database searches across the branches above yielded 548 candidate records, including duplicates across branches.

**Stage 2: Deduplication (n = 415).** After removing records identified in multiple branch searches, 415 unique records remained.

**Stage 3: Screening (n = 234).** Title and abstract screening applied the inclusion criteria from §2.4: (i) LLM as primary reasoning component; (ii) addresses bias, fairness, equity, or disparate treatment; (iii) findings bear on at least one of the six agent components or on evaluation/mitigation thereof. Records failing criterion (i) (e.g., classical NLP, non-LLM agents) or criterion (ii) (e.g., capability benchmarks without fairness content) were excluded at this stage. Records passing abstract screen proceeded to full-text review. Exclusion reasons at this stage: non-LLM system (81 records), no fairness content (73 records), duplicate venue version (27 records).

**Stage 4: Inclusion (n = 202).** Full-text review confirmed inclusion criteria and assigned branch tags. Records excluded at this stage failed criterion (iii), typically single-turn generation studies with no agentic framing, or were superseded by a more recent version of the same work. The final corpus comprises 202 unique papers spanning 2012 to 2026.

Figure 1 (PRISMA Flow Diagram) renders this four-stage funnel with counts at each node and branching reason labels. *Takeaway: the corpus construction is transparent and reproducible; the 202 included papers represent systematic coverage rather than convenience selection.*

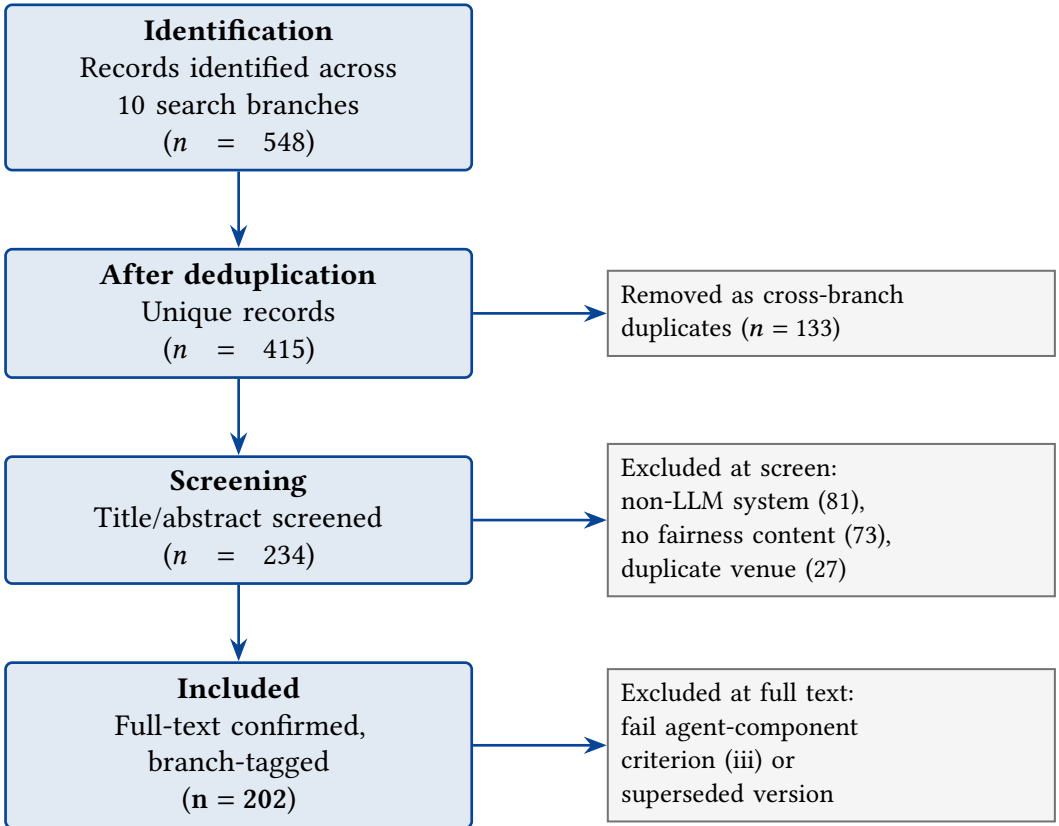


Fig. 1. PRISMA flow of corpus assembly. Four-stage funnel from 548 identified records to the 202 included papers, with per-stage exclusion counts (§2.5).

*Inclusion/Exclusion Criteria Summary. Table 2.2: Inclusion and Exclusion Criteria. Takeaway: the criteria operationalize a single principle: fairness as a property of agent actions, not merely of model outputs.*

Table 2.2: Inclusion and Exclusion Criteria. *Takeaway: the criteria operationalize a single principle: fairness as a property of agent actions, not merely of model outputs.*

| Criterion                 | Included                                                                                 | Excluded                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| System type               | LLM as primary reasoning component (including fine-tuned variants)                       | Classical NLP models, RL agents without LLM backbone, symbolic planners                                                                                 |
| Fairness content          | Studies bias, fairness, equity, or disparate treatment in outcomes or behavior           | Pure capability evaluation, no fairness angle                                                                                                           |
| Agent component relevance | Findings bear on at least one of the six components, or on evaluation/mitigation thereof | Exclusively single-turn static generation with no agentic framing; included only if mechanism transfers to agentic settings (marked <i>[adjacent]</i> ) |
| Language                  | English-language papers                                                                  | Non-English (pragmatic exclusion; acknowledged as limitation in §2.8)                                                                                   |
| Temporal scope            | Published or preprinted on or before June 10, 2026                                       | Post-cutoff publications                                                                                                                                |
| Availability              | Full text accessible                                                                     | Extended abstracts only, inaccessible preprints                                                                                                         |

### 0.2.6 Temporal and Venue Distribution.

*Publication Trend.* The agent fairness literature is a recent and rapidly growing subfield. Of the 202 corpus papers, **18 (9%)** predate 2020 (primarily foundational ML fairness and NLP bias work), **22 (11%)** fall in 2020 to 2022, **31 (15%)** in 2023, **78 (39%)** in 2024, and **53 (26%)** in 2025 to 2026 (through June cutoff). The combined 2024 to 2026 cohort constitutes **65%** of the corpus (131 papers), reflecting the field’s rapid emergence following the widespread deployment of instruction-tuned and tool-augmented LLMs.

Three inflection events are visible in the component-level publication trajectory:

- (1) **ReAct (late 2022, published ICLR 2023 [148]).** The formalization of reasoning-and-acting brought agentic behavior into mainstream NLP evaluation. A measurable uptick in tool-selection and planning fairness papers appears in the 2023 to 2024 cohort, as researchers applied bias probing to the new approach.
- (1) **AutoGen and GPT-4 function-calling (2023 [129]).** The accessibility of multi-agent orchestration and native function-calling APIs catalyzed the multi-agent delegation fairness literature, which is almost entirely 2024 to 2026.
- (1) **GPT-4 and the clinical deployment wave (2023 to 2024).** The application of frontier models to clinical decision support [95, 99, 153] triggered a wave of domain-specific fairness audits that now constitute the high-stakes-domains branch.

The long-horizon drift and memory components are the *latest* to receive empirical attention: the first dedicated papers on LLM long-term memory bias ([86]) and feedback-loop amplification

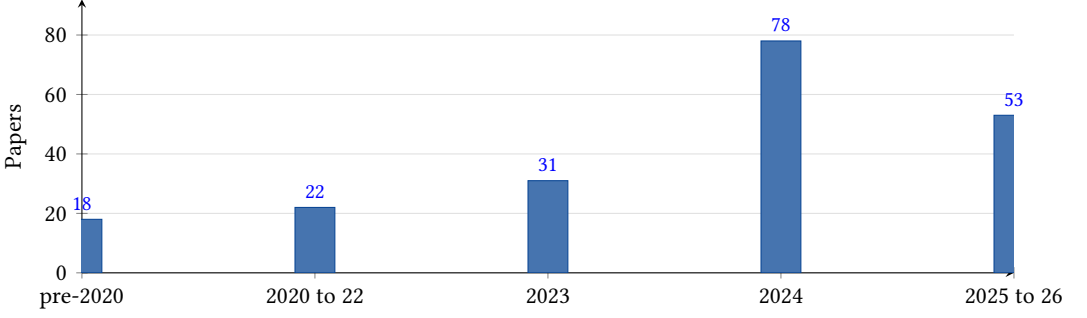


Fig. 2. Temporal distribution of the 202-paper corpus by publication-year bucket. The 2024 to 2026 cohort (78 + 53 = 131 papers, 65%) reflects the field’s recent, rapid emergence (§2.6).

([133, 140]) date to 2024 to 2026. This temporal lag is itself informative. The components that are hardest to evaluate, because they require multi-session or multi-turn experimental designs, have attracted attention last.

**Venue Distribution.** The corpus spans twelve venue categories. The five largest are: arXiv preprints (62 papers, 31%), ACL-family venues (ACL, EMNLP, NAACL, Findings) (41 papers, 20%), NeurIPS/ICML/ICLR (28 papers, 14%), ACM venues (FAccT, SIGKDD, SIGIR, CIKM, DL proceedings) (24 papers, 12%), and domain-specific venues (Nature Medicine, PNAS Nexus, The Lancet Digital Health, AAAI, ICSE) (19 papers, 9%). The remaining 28 papers (14%) appear in IEEE venues, TMLR, Frontiers, and preprint archives (ResearchGate, TechRxiv).

Several observations follow. The arXiv share is high (31%), reflecting the recency of the field: many 2025 to 2026 papers are under review or in proceedings preparation. The ACM FAccT representation (9 papers) is lower than the field’s importance would suggest, indicating that agent fairness has not yet been fully claimed by the accountability community; it currently sits across NLP venues, systems venues, and domain journals. Ten papers appear in CSUR or ACM Computing Surveys directly, indicating that the survey genre has begun to engage this territory, and the differentiation analysis in §1.4 and Table 0 is therefore essential to establishing the present survey’s novelty.

**0.2.7 Glossary and Notation.** The following terms and notation are used throughout this survey. Terms coined or given specific technical meanings here are marked with a dagger (†).

**Agent loop**  $\langle \pi, T, M, R \rangle$ : the formal object of analysis in the BCF (Definition D1, §3.0).  $\pi$  = policy (the LLM-backed decision function),  $T$  = tool set,  $M$  = memory module (in-context + external),  $R$  = reward/stopping signal.

**Bias Conduction Framework (BCF)**†: the organizing framework of this survey (§3.0), which analyzes fairness as a property of how demographic disparity is *conducted* (attenuated, preserved, amplified, or generated) across pipeline edges in an agent loop.

**Conduction operator**  $\phi$ †: the scalar multiplier describing how a pipeline edge transforms component-level disparity.  $\phi < 1$  = ATTENUATE;  $\phi \approx 1$  = PRESERVE;  $\phi > 1$  = AMPLIFY;  $\phi$  acting on zero input = GENERATE.

**Counterfactual Flip Rate (CFR)**: the fraction of agent evaluations that reverse when the input is perturbed by a demographic swap alone [85]. Generalized to any agent component in §4.2.

**Mean Absolute Score Difference (MASD)**: the mean absolute change in a continuous agent score (confidence, rank, quality rating) across counterfactual pairs [85].

**Component counterfactual disparity**  $\Delta_c \dagger$ : the counterfactual disparity attributable to component  $c$  alone, measured via trace-replay (Definition D2, §3.0).

**Trajectory disparity**  $\Delta_\tau \dagger$ : the total counterfactual disparity observed over a complete agent trajectory, approximated by the conduction equation  $\Delta_\tau \approx \sum_c \Delta_c \cdot \prod_{e>c} \phi_e$  with feedback recurrence  $\Delta^{(t+1)} = \phi_{fb} \cdot \Delta^{(t)} + \Delta_{entry}$ .

**Agentic multiplier (P3, Super-additivity)†**: the BCF proposition that trajectory disparity can exceed the sum of component disparities when amplifying conduction operators compose; equivalently, the agent pipeline is a super-additive bias transformer.

**Action-level disparity†**: disparity measured over an agent’s *actions* (tool invocations, decisions, escalations) rather than over its textual output. Formalized in §4.3 as tool-invocation disparity, escalation-rate disparity, and plan-allocation disparity.

**Long-horizon fairness drift†**: the temporal degradation or amplification of fairness properties across turns, sessions, or deployment feedback cycles, even when the per-step model is held fixed.

**Adjacent evidence [transferred]**: notation used in Tables 1 and 4 to flag papers from non-agentic settings whose documented mechanism is asserted to operate in agentic settings; the transfer argument is given in the relevant §3.x entry.

**CFR subscript notation**:  $CFR_c$  denotes the counterfactual flip rate measured at component  $c$ ;  $CFR_\tau$  denotes the end-to-end trajectory flip rate.

*0.2.8 Threats to Validity of This Survey.* Any systematic survey carries validity threats that should be made explicit. We identify four categories.

*Selection Bias.* The corpus was assembled primarily from English-language papers indexed in arXiv, ACL Anthology, and ACM Digital Library. Non-English research, including substantial work from Chinese NLP venues on Chinese LLM fairness, is systematically underrepresented. Similarly, industry research that is not publicly disclosed is absent. The arXiv-heavy composition (31%) means the corpus captures working papers that have not yet completed peer review; some may be revised or retracted. We mitigate this by requiring that preprints be self-consistent and that their core claims be supported by the experimental design reported.

The search strategy was branch-structured, which creates a risk that papers addressing multiple components simultaneously are attributed primarily to the branch under which they were first encountered. The multi-tagging convention (§2.4) partially addresses this, but papers at the intersection of underrepresented branch pairs (e.g., tool-selection  $\times$  long-horizon) may be missed if neither branch search used terms that surface the intersection.

*Construct Validity.* The six component taxonomy is a construct, not a natural kind. Some bias phenomena do not fit cleanly into a single component: a persona-conditioned retrieval agent collapses Components 1, 2, and 5 into a single mechanism. The component-assignment rules in §3.0 adopt a *proximate cause* convention, tagging a paper to the component that is the most proximate locus of the reported disparity, but this convention involves judgment. Readers who categorize a boundary paper differently from this survey’s assignment will find that cell counts in the coverage matrix shift slightly; the qualitative conclusions (empty counterfactual column, unspecified-dominated rows for tool/memory/personalization) are reliable under reasonable reassignments.

*Publication Bias.* Positive findings are more likely to be published and more likely to be submitted to the high-visibility venues this search prioritizes. Null results, meaning LLM agents that show no meaningful fairness disparity across components, are underrepresented in the corpus. This creates a systematically alarming picture: the corpus is weighted toward papers that find and document disparities, not papers that find none. Readers should interpret the corpus as establishing that

disparity *can* and *does* occur at each component, not that it *always* occurs or that it is comparably severe across deployment contexts.

*Recency Bias.* The 65% concentration of papers in 2024 to 2026 means the survey reflects a rapidly changing field. Findings from 2024 papers may already have been partially addressed by 2026 mitigations not yet captured in the corpus. Conversely, the long-horizon and multi-agent components, where empirical papers are newest and most sparse, may have received substantially more attention by the time this survey reaches print, making the "open edge" assessments in §3.x optimistic. We treat this as a motivating condition, not a confound: the survey's goal is to provide a structured framework for the field's future development, not to give a final accounting of a stable literature.

### 0.3 The Bias Conduction Framework: Where Bias Enters and How It Travels

The central claim of this survey is methodological: in an LLM *agent*, fairness is not a property of a single forward pass but a property of a *pipeline*. An agent perceives, selects tools, retrieves and accumulates memory, delegates subtasks across roles, plans and decomposes, models its user, and executes over long horizons. Each of these is a distinct locus where demographic disparity can originate, and each fails in a mechanistically different way. Organizing the literature by *fairness metric* (group/individual/counterfactual), as QA-era surveys do (including our parent [24] and the broader LLM-bias syntheses [46, 89]), flattens this structure and obscures the fact that two agents with identical token-level bias scores can produce radically different *action-level* disparities depending on where in the pipeline a bias is amplified or suppressed. This insight has a pre-LLM lineage: fairness has long been understood as a property of the sociotechnical *system*, not of an abstracted model [109]. The canonical algorithmic-harm cases were diagnosed precisely by asking *where in the pipeline* the disparity entered: a health-management algorithm whose proxy label encoded racial disparity [92], and commercial classifiers whose error rates were intersectionally skewed [5]. The agent loop multiplies the number of such places.

We therefore organize the field around a single formal object, the **Bias Conduction Framework (BCF)**. BCF has two axes. The first axis is the **entry locus**: the agent component at which a demographic disparity first becomes measurable. The second axis is the **conduction operator**: how each downstream pipeline edge transforms a disparity that reaches it, attenuating it, preserving it, amplifying it, or generating a new one from clean input. A bias observation in this field is thus a coordinate pair (locus, operator), and a mitigation is an intervention on a specific coordinate. This is the dimension along which the near-neighbor works do *not* organize: [82] treats fairness as one sub-topic of a broad agent-evaluation survey; [128] as a subdimension of a seven-dimensional conceptual taxonomy in a single (clinical) domain; [85] as a single-domain counterfactual method (contact-center QA) whose CFR/MASD machinery we generalize across components in §4; [164] proposes a unified fairness framework for multi-agent AI broadly, without an LLM-component axis; and [2] formalizes only the interactional slice of multi-agent fairness. None pairs an entry-locus decomposition with an account of how disparity *travels*.

Section 3.0 states the framework: three definitions (the agent loop, component counterfactual disparity, conduction operators), one approximation (the conduction equation, with its feedback recurrence), five propositions (P1 to P5), and the component-assignment rules that keep papers from floating between cells. Sections 3.1 to 3.5 then survey the five *architectural* entry loci, and §3.6 the *temporal* axis that cross-cuts them; for each we give a *definition*, the *mechanism of harm*, the *evidence classified by conduction operator*, the *open edge* the field has not yet measured, and a one-line *conduction signature*. Section 3.7 develops Proposition P3 (super-additivity, the formal version of what we informally call the agentic multiplier) and is explicit about which part of it is



evidenced and which part is conjecture. Section 3.8 walks a concrete hiring-screener agent through the framework and distills a four-step auditor recipe that the audit protocol of §6 instantiates.

**Schematic (BCF agent loop): the agent loop  $\langle \pi, T, M, R \rangle$  annotated with the five architectural entry loci (C1-C5), the temporal axis (T-axis) on the feedback edge, and the conduction operators on each inter-component edge.** Solid edges carry the within-episode conduction factors  $\varphi_e$ ; the dashed feedback edge carries  $\varphi_{fb}$  across turns, sessions, and retraining cycles. *Takeaway:* bias has six structurally distinct ways into an agent’s consequential action, so no single model-level fairness score (however well validated on QA benchmarks) can certify the loop.

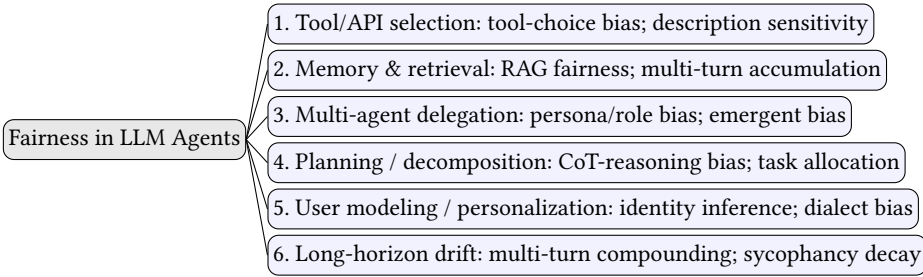


Fig. 3. Taxonomy of fairness in LLM-based agents, organized by agent component (Axis 1). Each node names a structural entry locus and its primary harm mechanism. Unlike QA-level LLM-fairness taxonomies, the unit of analysis is the *acting* agent.

### 0.3.1 The Framework: Definitions, Operators, Conduction, and Propositions.

*D1: The agent loop.* **Definition 1 (Agent loop).** An LLM agent is a tuple  $\mathcal{A} = \langle \pi, T, M, R \rangle$ , where:

- $\pi$  is the *policy*: the LLM under its system prompt, persona, and decoding configuration, mapping the current state  $s_t = (g, h_t, m_t)$  (goal  $g$ , interaction history  $h_t$ , working memory  $m_t$ ) to an action  $a_t$ . Actions include emitting text, calling a tool, writing memory, delegating a subtask, or committing a consequential decision.
- $T$  is the *tool/environment interface*: the tool registry, retrieval indices, and external APIs; invoking  $T$  returns an observation  $o_t$  that is appended to the context. Tool-learning and reasoning-acting architectures formalize this interface [106, 113, 148].
- $M$  is the *memory operator*: read/write access to persistent stores (scratchpads, episodic reflections [112], vector stores, cross-session user profiles) that survive beyond the current context window.
- $R$  is the *feedback channel*: the loop-closing signal (user reactions within a dialogue, reward models, preference data, or retraining on the agent’s own outputs) through which today’s behavior conditions tomorrow’s policy.

A *trajectory*  $\tau = (s_0, a_0, o_0, \dots, a_T)$  terminates in a consequential output  $O(\tau)$ : a decision, ranking, allocation, or escalation. A *multi-agent system* is a set  $\{A_i\}$  plus a controller policy  $\pi_{ctrl}$  that assigns roles and routes messages [47, 129]. This tuple deliberately follows the standard agent anatomy of §2.3 [115, 137, 141]; BCF adds nothing to the architecture. It adds an accounting scheme over it.

*D2: Component counterfactual disparity.* **Definition 2 (Component counterfactual disparity,  $\Delta_c$ ).** Fix a protected attribute  $A$  with a counterfactual perturbation  $x \rightarrow x'$  (a demographic swap of names, dialect, stated identity, or profile fields, in the lineage of counterfactual fairness [61]). The *trajectory disparity* is

$$\Delta_\tau = d(O(\tau(x)), O(\tau(x'))),$$

for a task-appropriate distance  $d$  (a flip indicator for discrete decisions, an absolute score difference for continuous ones; exactly the CFR/MASD pair of [85], generalized in §4.2). The *component disparity*  $\Delta_c$  for component  $c \in \{\pi, T, M, R, \pi_{\text{ctrl}}\}$  is defined by **trace-replay**: log the full trajectory under  $x$ ; replay it holding every other component's realized inputs and outputs fixed; perturb only the input that  $c$  consumes; and measure the divergence of  $c$ 's output distribution,

$$\Delta_c = \mathbb{E} \left[ d(c(z_c(x)), c(z_c(x'))) \right] \quad \text{under replay of the logged trace.}$$

$\Delta_c$  is the quantity that the clinical decomposition study of [147] estimates when it attributes separable shares of a tool-using agent's total unfairness to the tool-exposure, tool-transition, and model-reasoning stages. To our knowledge this is the only corpus paper that operationalizes the full decomposition, which is why §4.3 elevates trace-replay to the canonical estimator and §6 builds its protocol around it.

*D3: Conduction operators.* **Definition 3 (Conduction operator).** Every pipeline edge  $e$  from component  $c$  to component  $c'$  carries a *conduction factor*  $\varphi_e = \Delta_{\text{out}}/\Delta_{\text{in}}$ : the ratio of the disparity measurable at  $c'$ 's output to the disparity delivered to  $c'$ 's input. Edges fall into four modes:

| Operator         | Factor                                                | Meaning                                                     | Canonical corpus evidence                                                                                                                                                      | Typical edge                              |
|------------------|-------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>ATTENUATE</b> | $\varphi < 1$                                         | The edge dampens incoming disparity                         | identity anonymization in debate [28]; embedder control in RAG [64]; multi-LLM debiasing [94]                                                                                  | guardrail, debate, fair-ranking edges     |
| <b>PRESERVE</b>  | $\varphi \approx 1$                                   | The edge conducts disparity unchanged                       | reward models inherit dialect penalties [84]                                                                                                                                   | role hand-offs, alignment signals         |
| <b>AMPLIFY</b>   | $\varphi > 1$                                         | The edge magnifies incoming disparity                       | inter-agent propagation [91]; multi-turn degradation [42]; self-consuming training loops [133]                                                                                 | interaction, accumulation, feedback edges |
| <b>GENERATE</b>  | $\Delta_{\text{out}} > 0$ on $\Delta_{\text{in}} = 0$ | The edge creates disparity from demographically clean input | RAG undermines fairness even for vigilant users [55]; persona assignment injects reasoning bias and new behavioral distortion [29, 45]; reasoning itself introduces bias [139] | retrieval, persona, planning edges        |

*Takeaway:* AMPLIFY and GENERATE are the two genuinely agentic modes. Single-turn evaluation cannot exhibit either, and (as Table 1 and §7 show) these are also the ones the field has instrumented least.

GENERATE is formally a new entry rather than a transformation ( $\varphi$  is undefined on zero input); we treat a GENERATE edge as contributing its own  $\Delta_c$  term. ATTENUATE is also the operator class of every mitigation in §5: a mitigation is an engineered  $\varphi < 1$  edge, which is what makes the mitigation map of Table 3 commensurable with the taxonomy.

*The conduction equation and the feedback recurrence.* Treating disparities as small and edges as locally multiplicative (a first-order approximation), and taking  $\Delta_c$  and  $\Delta_\tau$  here as *signed* (directional) disparities so that oppositely directed component contributions can cancel in the sum (the endpoint statistics CFR and MASD of §4.2 report the corresponding magnitudes  $|\Delta|$ ), the trajectory disparity decomposes as the **conduction equation**:

$$\Delta_\tau \approx \sum_c \Delta_c \cdot \prod_{e \in \text{path}(c \rightarrow O)} \varphi_e,$$

i.e., each component's entry disparity, conducted through the product of edge factors on its path to the consequential output, summed over entry loci. Across the loop-closing channel R, disparity obeys the **feedback recurrence**:

$$\Delta^{(t+1)} = \varphi_{fb} \cdot \Delta^{(t)} + \Delta_{\text{entry}},$$

whose behavior is qualitatively bimodal. For  $\varphi_{fb} < 1$  the disparity converges to the steady state  $\Delta_{\text{entry}} / (1 - \varphi_{fb})$ , persistent but bounded. For  $\varphi_{fb} \geq 1$  it grows without bound: the regime documented for synthetic-data training loops [133], self-consuming performative loops [140], and recommender feedback [78]. Both expressions are an *organizing approximation*, a bookkeeping identity under linearization and not a validated dynamical model. Edges in real agents are nonlinear, context-dependent, and interactive; the equation's role is to make explicit *what an audit must estimate* (the  $\Delta'_c$ s and  $\varphi'_e$ s) and *what can defeat naive auditing* (cancellation and amplification along paths), which is exactly how the §6 protocol and the §3.8 recipe use it.

*Propositions P1-P5.* Five propositions follow from D1 to D3 and the conduction equation. They are stated as claims the survey's evidence base supports or motivates, not as theorems about all possible agents; each carries its grounding.

**P1 (Locality).** *Every measurable trajectory disparity decomposes into component entries plus an interaction residual:  $\Delta_\tau \neq 0$  implies a nonzero  $\Delta_c$  at some locus  $c$  (counting GENERATE edges as entries), or a nonzero cross-component interaction term that the linearized equation folds into the nearest downstream GENERATE edge.* The residual matters because pure interaction effects (disparity present in no component alone but produced by their composition) are exactly the multi-agent GENERATE phenomenon. Justification: this is the premise validated whenever decomposition succeeds. [147] attributes distinct, separable shares of a clinical agent's disparity to the tool-exposure and tool-transition stages versus model reasoning, and the component-specific audits surveyed in §§3.1 to 3.5 each isolate a nonzero  $\Delta_c$  with other components held fixed.

**P2 (Masking).** *Endpoint metrics can read  $\approx 0$  while internal disparities are large: whenever a downstream ATTENUATE edge (or sign-opposed entries) cancels an upstream  $\Delta_c$ , the conduction equation yields  $\Delta_\tau \approx 0$  with  $\sum_c |\Delta_c| \gg 0$ .* The survey's measurement gap here takes the form of a structural fact rather than an empirical complaint. Justification: [73] shows verbally neutral models issuing demographically disparate *decisions*, with endpoint text-parity masking action disparity; [146] shows surface-level semantic parity masking statistical differentiation; and [123] shows the reverse mask, reasoning traces that misreport the biased features actually driving the output.

**P3 (Super-additivity, the agentic multiplier).**  *$\Delta_\tau$  can exceed the naive sum  $\sum_c |\Delta_c|$  when at least one AMPLIFY or GENERATE edge lies on a conduction path. Under the linearized magnitude model,*

an *AMPLIFY* or *GENERATE* edge is a necessary condition for super-additivity, not a sufficient one, since it can be offset by a downstream *ATTENUATE* edge; once nonlinear cross-component interactions are admitted, even necessity fails, because interactions can produce super-additivity without such an edge. Conversely, with only *PRESERVE/ATTENUATE* edges and no interaction term, composition is sub-additive and component audits compose into a conservative system bound. Justification: within the multi-agent locus this is directly evidenced. Stereotypical bias emerges, propagates, and is amplified across interacting agents, making the collective more biased than its members [91], and group-level fairness is emergent rather than inherited [81]. §3.7 develops P3 and marks its cross-component generalization as conjecture.

**P4 (Mitigation-matching).** *A mitigation reduces  $\Delta_\tau$  only if it installs an ATTENUATE edge on a path that carries a nonzero  $\Delta_c$ ; mitigation must match the entry locus and sit downstream of it, and an intervention at locus  $c$  neither bounds disparity generated downstream of  $c$  nor survives upstream amplification.* Justification: fairness-aware prompting fails against retrieval-injected bias because the *GENERATE* edge sits downstream of the prompt [55], while intervening at the retrieval representation succeeds [64]; symmetrically, debate-level anonymization attenuates delegation bias precisely because it edits the identity cues at the delegation edge [28]. P4 is the organizing key of the mitigation map in §5/Table 3.

**P5 (Measurement-adequacy).** *Endpoint-only evaluation is systematically unsound (by P2) and not even conservative (by P3); adequacy therefore comes in tiers.* We distinguish four: *endpoint-only* (inadequate); *trace-observing* (logs actions but applies no counterfactual); *component-attributing* (per-locus  $\Delta_c$  under a demographic swap); and *causal-replay* (the trace-replay of D2). Full certification of an agent requires the causal-replay tier; the intermediate tiers bound only identifiable subsets of the  $\Delta_c$ , which is often the most a real audit can achieve. Justification: models that pass single-turn fairness tests fail multi-turn ones [42]; the measured bias depends on which demographic cue channel is perturbed [124]; and the major agent benchmarks (WebArena [154], AgentBench [70], Mind2Web [32], SWE-bench [60]) log exactly the traces P5 requires but carry no demographic counterfactual instrumentation, a confrontation developed in §4.4.

*Component-assignment rules and reading conventions.* The five loci and the temporal axis map onto the D1 objects as follows. Planning (C4) and user modeling (C5) are *sub-policies* of  $\pi$  rather than separate modules, which is why D1 keeps the tuple minimal rather than positing a module per locus:

| Locus                                                             | D1 object(s)                                                         | Consumes → emits                                                                      |
|-------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| C1 Tool / API selection<br>C2 Memory & retrieval                  | T (selection step of $\pi$ )<br>M, T                                 | context → choice of tool/source<br>query/history → retrieved content, memory state    |
| C3 Multi-agent delegation                                         | $\pi_{\text{ctrl}}$                                                  | task + roles → subtask routing/assignment                                             |
| C4 Planning / decomposition<br>C5 User modeling / personalization | $\pi$ (decomposition sub-policy)<br>$\pi$ , M (user-model sub-state) | goal → ordered subgoals/plan<br>user signals (often inferred) → personalized behavior |
| T-axis (long-horizon drift)                                       | R (feedback edge)                                                    | prior outputs → updated $\pi$ /M across turns                                         |

*Takeaway: every locus is a function of an explicit D1 object; C4 and C5 are decompositions of the policy  $\pi$ , not omissions from the tuple.*

A recurring defect of pipeline taxonomies is that papers float between cells, most acutely between *tool selection* and *retrieval*, since a retriever is both a tool one selects and a content source one conditions on. BCF resolves this with five assignment rules, keyed to what a component *consumes* and *emits* in D1:

- **R1 (Selection vs. content).** If the disparity lives in *which* external resource is invoked (a discrete choice over a registry, including the choice *among* retrievers or sources), it is **C1 (tool/API selection)**. If it lives in *what* an invoked resource returns and how that content conditions generation or persists (ranking, coverage, accumulation), it is **C2 (memory & retrieval)**. Thus source-selection agents [118] sit in C1; the effect of retrieved content on generation [55, 132] sits in C2; an IR-systems treatment spanning both [9, 35] is tagged to C2 as primary.
- **R2 (Role vs. plan).** Disparity caused by *who is assigned* (persona, role, or identity cues attached to an agent) is **C3 (delegation & roles)**. Disparity in *how the task is cut up*, ordered, or resourced, independent of identity assignment to agents, is **C4 (planning & decomposition)**. Team-flavored allocation studies [103] are therefore C4: the skew is in the allocation, not in an assigned persona.
- **R3 (Subject vs. user).** Disparity keyed to protected attributes of the *decision subject* (the applicant, patient, defendant) flows through C1 to C4; disparity keyed to attributes of the *interacting user* (especially attributes the agent *infers* rather than receives [111]) is **C5 (user modeling & personalization)**.
- **R4 (Temporal indexing).** Any of the above measured *as a function of time* (turn, session, or deployment cycle) is the same locus indexed by  $t$ , tagged with the temporal axis of §3.6, not a separate locus. Long-horizon drift is the  $\varphi_{fb}$  regime of the feedback recurrence, not a sixth architectural component (§3.6 justifies this at length).
- **R5 (Multi-tagging).** A paper may evidence several cells and is counted in each (this is the  $\times N$  convention of the corpus and the multi-tag accounting of §7), but it carries exactly one *primary* locus: the component whose input/output the paper directly measures.

**Transferred-evidence convention.** Consistent with the inclusion criteria of §2.4, some mechanism evidence below originates in single-turn or chatbot settings rather than instrumented agents. We mark such citations with  $[\dagger]$ : the mechanism transfers to the agentic edge by D1's construction (the same model occupies  $\pi$ ), but the agentic *measurement* has not been performed. These markers are themselves a map of open problems.

**Table 1: Component  $\times$  harm mechanism  $\times$  representative evidence  $\times$  maturity.** Maturity is graded  $\bullet\bullet\bullet$  (well-studied),  $\bullet\bullet\circ$ , or  $\bullet\circ\circ$  (emerging).

*Takeaway: every locus has documented harm, but maturity is uneven; the planning and long-horizon loci (C4, T-axis), where the GENERATE and feedback-AMPLIFY operators are most consequential, are the least mature, exactly the cells §7 shows as least instrumented.*

**0.3.2 Locus C1: Tool / API Selection Bias. Definition.** A tool-using agent must, at each step, choose *which* external tool, API, retriever, or knowledge source to invoke. Tool-selection bias is systematic, demographically- or content-correlated skew in that discrete choice, independent of and often invisible to the quality of the model's final text. By rule R1, everything about *which* resource is invoked belongs here; what the resource *returns* belongs to C2.

**Mechanism of harm.** Selection is a discrete decision over a tool registry, and  $\pi$  makes such decisions with the same surface-form sensitivities that distort multiple-choice answering. Because the selected tool determines what evidence the rest of the trajectory conditions on, a skewed selection is conducted by every downstream edge: in BCF terms, C1 disparities enter early and travel far. The harm has a pre-agentic lineage in search: search engines have long been documented

Table 1: Component  $\times$  harm mechanism  $\times$  representative evidence  $\times$  maturity. Maturity is graded  $\bullet\bullet\bullet$  (well-studied),  $\bullet\bullet\circ$ , or  $\bullet\circ\circ$  (emerging).

| Component                                 | Primary harm mechanism                                                                                                | Representative evidence | Maturity                |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------|
| C1 Tool / API selection (§3.1)            | Selection skew over the tool registry independent of task fitness; routing conditioned on surface or demographic cues | [21, 120, 147]          | $\bullet\bullet\circ$   |
| C2 Memory & retrieval (§3.2)              | RAG-injected disparity from clean inputs; accumulation of skew across turns                                           | [42, 55, 132]           | $\bullet\bullet\circ$   |
| C3 Multi-agent delegation (§3.3)          | Persona/role injection, then amplification across interacting agents                                                  | [28, 81, 91]            | $\bullet\bullet\bullet$ |
| C4 Planning / decomposition (§3.4)        | Reasoning-stage disparity in plan structure, ordering, and allocation, with unfaithful traces                         | [57, 103, 139]          | $\bullet\circ\circ$     |
| C5 User modeling / personalization (§3.5) | Inferred-identity stereotyping that tracks group profile over stated preference                                       | [63, 101, 111]          | $\bullet\bullet\circ$   |
| T-axis Long-horizon drift (§3.6)          | Feedback-recurrence amplification ( $\varphi_{fb} > 1$ ) of per-step skew across turns and cycles                     | [76, 86, 133]           | $\bullet\circ\circ$     |

to encode racial and gendered skew in what they surface [166][†] and in visual representation [79][†]. An agent that chooses among such engines inherits the choice-over-biased-infrastructure problem and adds its own selection pathology on top.

**What is known.** The most replicated finding at C1 is that tool-selection skew is a property of the selection policy  $\pi$ , not merely of the tools in the registry. Blankenstein et al. demonstrate that across tool registries, LLMs consistently prefer particular tools in ways that are independent of query fitness, constituting a reproducible GENERATE-operator signature [21]. The mechanism generalizes cleanly from the established result that LLMs exhibit positional and surface-form sensitivity in discrete choice [98][†]: when tool registries are presented as lists, the selection policy inherits these ordering effects without any explicit demographic input. This is a well-established baseline. Beyond the base pathology, the adversarial variant is also documented: ToolTweak shows that editing only a tool’s name and description is sufficient to systematically inflate its selection rate [120], and recommender-agents are steered analogously by injected preference biases [125]. Together, these results establish that C1 is a genuine, independent entry locus: the disparity resides in the selection step of  $\pi$ , not in any downstream content processing. The strongest evidence that C1 contributes independently to trajectory-level harm comes from [147], which decomposes a chest-X-ray agent’s total disparity by stage (tool exposure, tool transition, model reasoning) and shows that the tool stages carry a distinct attributable share that is conducted into the final clinical

read, the closest published operational approximation to D2's decomposition at this locus. The ATTENUATE direction is also documented: bias-aware retrieval agents that monitor and rebalance source selection reduce the skew [117, 118], constituting engineered  $\varphi < 1$  edges at the selection interface, consistent with P4's requirement that an intervention must target the locus and operator of the dominant  $\Delta_c$  to reduce  $\Delta_r$ .

**What conflicts or is contested.** The evidence on *scope* of tool-selection bias is heterogeneous in ways the literature does not yet reconcile. The GENERATE finding from [21] treats selection skew as arising from the policy itself on demographically clean input; the PRESERVE finding from [147] treats it as a conducted version of upstream demographic cues in the input. These are not mutually exclusive mechanisms, but no study has cleanly separated the two contributions under a controlled counterfactual swap, so the relative magnitude of intrinsic policy skew versus cue-elicited skew at C1 is unresolved. Similarly, the adversarial manipulation evidence from [120] and [125] is measured in recommender and general-purpose registry settings; whether the same injection magnitude applies in consequential decision-making deployments (hiring, clinical, lending) where the registry is smaller and more constrained is not established. The ATTENUATE results from [117, 118] demonstrate that source-rebalancing reduces disparity, but these studies do not report whether the reduction holds under adversarially perturbed registries or whether the monitoring layer itself introduces asymmetric errors across demographic groups. The boundary condition for when C1 bias amplifies versus merely preserves into downstream components is also unsettled: [147] reports PRESERVE-class conduction in a clinical pipeline, but the theoretical expectation under P3 is that early-locus disparities should amplify when downstream operators are non-attenuating. Whether clinical settings happen to exhibit attenuating downstream operators, or whether the PRESERVE finding is an artifact of the specific pipeline architecture, is not resolved.

**What transfers only by analogy.** The pre-agentic lineage of this locus rests on evidence that does not involve instrumented agents. The racial and gendered skew in search-engine result surfaces [166][†] and in visual representation across engines [79][†] establishes that the infrastructure an agent selects *among* is itself non-neutral, but these findings predate LLM-based tool-selection policies and involve human searchers, not  $\pi$ -mediated selection loops. The positional and surface-form sensitivity result [98][†] is measured in multiple-choice QA settings, and the inference that the same sensitivity governs tool-registry selection under list presentation is architecturally reasonable but not yet directly measured in a demographically instrumented agent trace. These transferred results sharpen the mechanism hypothesis without constituting direct measurement of C1 in an agentic pipeline.

**What is unmeasured.** The field currently measures *that* selection skews but has not measured whether the skew correlates with protected attributes of the *user or subject* under a counterfactual swap holding the query fixed. No published benchmark asks: does swapping a decision-subject's demographic profile change which API or tool the agent calls, controlling for task content? That gap is the  $\Delta_{\text{tool}}$  instance of D2 that §4.3 formalizes as tool-invocation disparity. Two phenomena must be kept distinct here: (a) provider- and ordering-preference selection bias among functionally equivalent tools, which is what [21] measures and which by that study's own design involves no sensitive attributes; and (b) demographic-conditioned tool routing, for which the only direct evidence is the clinical decomposition of [147], in a medical context. The GENERATE-operator finding from [21] therefore establishes only that selection skew is measurable at all, not demographic-conditional skew, and the finding from [147] attributes a disparity share to C1 but without the counterfactual isolation that P1 (Locality) requires for strict attribution. No direct demographic-swap measurement of tool selection yet exists. Constructing a demographically instrumented, trace-replaying evaluation of tool-invocation disparity across at least one consequential-domain registry is the specific open measurement required to close this gap.

**Conduction signature.** *Entry:* the  $\pi \rightarrow T$  selection interface. *Observable:* tool-choice distribution across counterfactual input pairs. *Dominant outbound operator:* PRESERVE into C2/C4 (the chosen tool constrains all downstream evidence), with adversarial GENERATE at the registry. *Metric:* tool-invocation disparity (CFR over tool choice; §4.3).

**0.3.3 Locus C2: Memory & Retrieval: Bias Injection and Accumulation. Definition.** Agents persist state (conversation history, scratchpad memory, retrieved documents, cross-session stores) and condition each step on that accumulated context. Memory-and-retrieval bias is the introduction, *accumulation*, or amplification of demographic skew through this conditioning, distinct from any bias in a single isolated response. By rule R1, the locus covers what retrieved or remembered *content* does; by rule R4, its time-indexed form (drift) is treated in §3.6.

**Mechanism of harm.** Two sub-mechanisms operate, mapping cleanly onto BCF's two non-trivial operators. *Retrieval injection* is a GENERATE edge: grounding shifts the generator's distribution, so a disparity can appear that the closed-book model never exhibited. *Temporal accumulation* is an AMPLIFY edge: each turn's context conditions the next, so per-turn skews compound. This locus inherits an entire prior discipline: fairness in ranking is a mature field with exposure-based and probability-based criteria and a decade of algorithms [151], and the LLM-era IR community has reframed those results as challenges for retrieval pipelines feeding generators [9, 35]. BCF makes the inheritance precise: a fair-ranking guarantee is a bound on  $\Delta_c$  at the retrieval interface, but by P4 it bounds  $\Delta_\tau$  only if no downstream GENERATE edge re-introduces disparity, which is exactly what the RAG results below show can happen.

**What is known.** The most reliable finding at this locus is that retrieval augmentation can *generate* demographic disparity from a generator that exhibits none in closed-book mode, a GENERATE conduction signature that now has three converging lines of evidence. [55] establish the mechanism in its strongest form: the disparity appears even when retrieved documents are individually innocuous and even when the user explicitly prompts for fairness, establishing that the GENERATE edge is neither an artifact of adversarial content nor correctable by instruction. [132] replicate the finding by holding the generator fixed and varying the retrieval corpus and ranking, isolating the retrieval step as the active locus. [156] confirm that augmentation measurably shifts social-bias levels relative to the identical closed-book baseline. Taken together, these three studies constitute the field's best-replicated C2 result: RAG is a GENERATE edge for demographic disparity, introducing skew that the closed-book model never carried.

The accumulation direction is independently established, though by different experimental designs. Fairness degrades over multi-turn dialogue in ways that single-turn tests structurally cannot detect [42]; accumulated context shifts the model's stated beliefs as history grows [48]; and conversational history produces a "geometric trapping" effect in which early demographic framings become self-reinforcing and resist later correction [122]. The adversarial bound on accumulation is knowledge-store poisoning, where injected content amplifies bias and systematically steers downstream outputs [138], confirming that the AMPLIFY operator can be driven to arbitrarily large  $\varphi$  by an adversary with write access to the store. Multi-agent settings exhibit the same conversational-bias accumulation, extending the mechanism beyond single-agent dialogues [27].

On the ATTENUATE side, two interventions are evidence-backed. Tuning the embedder used for retrieval reduces RAG-injected bias [64], an engineered  $\varphi < 1$  at the representation layer consistent with P4 (the intervention targets the GENERATE edge rather than the generator downstream of it). Applying fair-exposure ranking inside the RAG pipeline changes both fairness and output quality, and the changes can be simultaneously positive [160], providing the most direct existing bridge between the fair-ranking tradition [151] and agentic systems. Both results confirm P4's



mitigation-matching prediction: interventions succeed when they target the locus of injection rather than the endpoint.

**What conflicts or is contested.** The three-study convergence on GENERATE conceals an unresolved boundary condition: none of the three designs varies *what kind* of demographic signal is present in the retrieved content (explicit labels, stereotypic co-occurrence, proxies in writing style) independently of the query framing. It is therefore unclear whether the GENERATE effect is primarily a function of the retrieved documents' surface-form demographic content, of the ranking order in which they are presented, or of the generator's contextual updating dynamics when confronted with demographically mixed evidence. The [160] result implies that ranking order matters (since reranking alone changes outcomes), while [64] implies that representation matters (since embedder tuning is sufficient). Whether these are complementary levers or competing explanations for the same variance is not resolved by existing designs, because no study jointly manipulates both.

A parallel ambiguity runs through the accumulation evidence. [42], [48], and [122] all measure degradation over turns, but they differ in whether they attribute the effect primarily to the growing size of the context, to the semantic content of accumulated history, or to the positional structure of how retrieved and remembered facts are arranged in the window. These attributions have different mitigation implications: a context-size account implies that truncation or summarization policies would attenuate accumulation, while a positional account implies that reordering the memory state would suffice, and a semantic account implies that neither would be adequate without content-level filtering. The studies are mutually consistent but do not adjudicate among these accounts, so the AMPLIFY operator is evidenced at this locus without a stable mechanism model for *how* it operates within the context window.

**What transfers only by analogy.** One important input to the accumulation mechanism is evidenced only in settings that are not yet measured in instrumented agents. Long-context models are known to privilege the head and tail of their input window, the "lost in the middle" effect [69][†], meaning that which retrieved or remembered facts about a person receive effective attention is a function of position in the context as much as of content. This implies that demographic disparity in memory can arise or be suppressed by the *write order* of the memory store independent of the content written, a structurally important point that the agentic memory literature has not yet measured. The positional-attention result is well-established in non-agentic reading-comprehension and summarization settings; its transfer to the retrieval-augmented agent loop (where memory writes are interleaved with actions and where the agent both reads and appends to its own context) is mechanistically plausible but empirically unconfirmed as of the corpus cutoff.

**What is unmeasured.** The most consequential open gap at this locus is the memory *write policy* as a fairness object. Every study reviewed above treats retrieval as a read-only operation: the agent queries a store and receives ranked content. In tool-augmented agents with persistent memory (the architecture of [67] extended by episodic reflections and cross-session user profiles), the agent also *writes* to memory, and what it chooses to persist about different individuals is itself a disparity-producing decision with no existing audit methodology. Whether the agent systematically writes richer, more accurate, or more favorable memory representations for some demographic groups than others is unmeasured, and this write-policy gap is the principal open measurement problem for C2. Relatedly, whether bias compounds faster in structured, tool-augmented memory than in plain conversational history is unknown: the dialogue-accumulation evidence ([42], [122]) is an underbound on what a long-lived agent memory store would produce, not a measurement of it. No benchmark provides demographically instrumented, counterfactual evaluation of agentic memory

reads and writes over realistic multi-session deployments, making the retrieval-set disparity and per-turn  $\Delta^{(t)}$  slope formalized in §4.3 currently unmeasurable in practice.

**Conduction signature.** *Entry:* the T/M $\rightarrow\pi$  conditioning interface. *Observable:* divergence of retrieved sets and memory states across counterfactual pairs. *Dominant outbound operator:* AMPLIFY (accumulation across turns), with GENERATE at the grounding edge. *Metric:* retrieval-set disparity plus the per-turn slope of  $\Delta^{(t)}$  (§4.3).

**0.3.4 Locus C3: Multi-Agent Delegation & Role Assignment. Definition.** In multi-agent systems a controller  $\pi_{\text{ctrl}}$  assigns subtasks, personas, or roles to constituent agents and routes information among them [47, 105, 129]. Delegation bias is disparity introduced by *who is assigned what* and by the social dynamics that emerge among role-bearing agents, over and above any single agent’s bias. By rule R2, identity-cue effects sit here; identity-free task-cutting sits in C4.

**Mechanism of harm.** The seed is persona conditioning, a GENERATE edge at the assignment interface; the distinctive agentic harm is that interaction then *amplifies* what assignment generated, producing the one corpus-evidenced case of P3’s super-additivity. Role assignment is thus doubly dangerous: it injects disparity and it wires the injected disparities into a network whose collective behavior is not predictable from its parts.

**Evidence: what is known.** The amplification result at this locus is the best-replicated finding in the corpus concerning agentic-scale bias. Stereotypical bias emerges, propagates, and intensifies across a multi-agent system such that the collective is more biased than its members [91], and large-scale simulations show group-level fairness is not reliably inheritable from component audits [81]. These two results are independent demonstrations of the same structural claim: composition under AMPLIFY violates the naive additivity of P1. A third line of evidence confirms that the mechanism scales: persona-induced bias grows from single agents to interacting societies [75], and implicit bias surfaces specifically *in the interaction* between agents rather than in either agent alone [3]. Social hierarchy compounds the dynamics: rank shapes persuasion and anti-social behavior within the collective [25], which means the *topology* of the delegation network is not fairness-neutral even when each node’s individual bias is held constant. At the entry edge, MALIBU provides the only purpose-built agentic benchmark and uncovers implicit bias specifically in multi-agent role assignments [83], grounding the amplification claims in an instrumented setting rather than a post-hoc analysis.

**Evidence: what conflicts or is contested.** The amplification direction is not inevitable. Anonymizing agent identities during structured debate reduces bias [28], demonstrating that the AMPLIFY operator is installed by identity cues at the delegation edge, not by multi-agent coordination per se. Engineering the interaction protocol differently can reverse the operator entirely: multi-LLM interaction can be designed as a debiasing mechanism rather than an amplifier [94], and structured debate can be steered toward equitable cultural alignment [65]. This creates a genuine tension in the literature: papers measuring naturalistic role assignment consistently find amplification, while papers that redesign the coordination protocol find attenuation. The conflict is not a contradiction but a boundary condition: AMPLIFY is the default under identity-bearing delegation, but it is contingent on the protocol, not structurally guaranteed. What remains contested is *how much* the topology and protocol contribute independently, since no study has held one fixed while varying the other. The amplification factor  $\varphi$  from [91] has been demonstrated but never bounded as a function of network structure, making it unclear whether sparse versus dense delegation graphs carry materially different fairness risk.

**Evidence: what transfers only by analogy.** Two of the most cited results at this locus are single-turn or chatbot-derived and have not been measured on instrumented agentic pipelines. Assigning a persona induces implicit reasoning biases that degrade performance on tasks unrelated

to the persona, making role assignment active bias injection rather than neutral scaffolding [45][†]. Persona assignment also multiplies toxic degeneration in chat settings [33][†]. Both mechanisms are plausible at the agentic delegation edge and are consistent with [29]’s finding that persona assignment introduces new behavioral distortion in the assigned agent (degradation of up to 26.2%), a GENERATE signature at the role hand-off rather than mere conduction of pre-existing bias (the canonical PRESERVE bridge in the D3 table is instead reward models inheriting dialect penalties [84]), but the counterfactual assignment-matrix measurement in a multi-agent trajectory has not been performed for either finding. The [†] markers here are therefore not ornamental: they identify the precise experiments still needed to convert mechanism evidence into locus-specific disparity estimates.

**Open edge.** Even with the amplification mechanism established, the *delegation policy* as a controllable fairness object remains unmeasured. There is no metric for assignment disparity in the sense of D2: does  $\pi_{\text{ctrl}}$  systematically route sensitive subtasks to lower-capability or more-biased sub-agents as a function of the subject’s demographics, holding the task fixed under a counterfactual swap? The amplification coefficient  $\phi$  at this locus is real, but without a delegation-matrix disparity metric it cannot be decomposed into the share attributable to the initial assignment versus the share produced by the subsequent interaction dynamics. This is precisely the  $\Delta_c$  gap that §4.3 formalizes and that the MALIBU benchmark only partially addresses, since MALIBU measures outcome disparity per role but not the assignment distribution itself. The BCF conduction signature at this locus therefore has a measurable observable (the role/subtask assignment matrix across counterfactual pairs) that no published benchmark yet collects.

**Conduction signature.** *Entry:* the  $\pi_{\text{ctrl}} \rightarrow \{A_i\}$  assignment interface. *Observable:* the role/subtask assignment matrix across counterfactual pairs, plus inter-agent message content. *Dominant out-bound operators:* GENERATE (persona injection) followed by AMPLIFY (interaction). C3 is the only locus where both agentic operators are corpus-evidenced together. *Metric:* assignment disparity over the delegation matrix (§4.3).

**0.3.5 Locus C4: Planning / Decomposition Disparities. Definition.** Agents plan: they decompose a goal into sub-goals, order steps, and reason explicitly (chain-of-thought, ReAct traces [148]) before acting. Planning bias is demographic disparity that enters during this reasoning/decomposition stage, in the *plan*, not merely the final answer. By rule R2, allocation skew belongs here even when the allocation is over agent-flavored "team members."

**Mechanism of harm.** The counterintuitive finding is that reasoning does not cleanly remove bias and can *introduce* it, a GENERATE edge inside  $\pi$  itself. Worse, this locus carries a distinctive P2 masking problem: the visible reasoning trace is not a faithful record of the features driving the decision, so auditing the *stated* plan can miss the disparity the plan encodes. A scope caution is warranted: whether LLM "plans" constitute planning in the classical sense is contested [159], but the fairness question is indifferent to that debate. Whatever the decomposition step is computationally, it emits orderings and allocations that can be counterfactually disparate.

#### **Evidence and synthesis.**

**What is known.** The established, replicated mechanism is that deliberate reasoning is a GENERATE locus, not a sanitizing filter. The weight of evidence supports two related claims. First, explicit chain-of-thought reasoning changes social-bias outcomes relative to direct answering, and the change can be adverse: [139] demonstrates in controlled comparisons that chain-of-thought reasoning fails to deliver a fairness dividend and can worsen stereotype expression in ambiguous contexts, while reasoning-free models show more stable bias levels. Second, allocation-stage disparity is demonstrable at the plan object level rather than only at the final-output level: LLM-driven software-team composition produces demographically skewed role and task assignments, a planning-stage

disparity with direct hiring adjacency [103]. Together these establish that the operator at C4 is GENERATE-consistent, pending the revealer-vs-generator adjudication §3.4 itself describes. Unlike C3 (where interaction then amplifies what role-assignment generates), C4 disparity appears to resolve principally at the GENERATE edge itself, without requiring a subsequent amplification step, which makes it structurally more like C1 (early-entry, conducted forward) than like C3 (entry plus amplification). The lever exists: reasoning-guided fine-tuning reduces bias at this locus [66], an intervention consistent with P4 because it acts precisely at the generating edge; complementarily, a Fairness Reward Model can score complete reasoning chains against equalized opportunity and equalized odds criteria and reweight them during post-generation aggregation, leaving the internal reasoning untouched [57].

*What conflicts or is contested.* Two tensions complicate the picture. First, the causal direction of the reasoning-bias relationship is not resolved. [139] shows reasoning can worsen bias, but the comparison baseline matters: the effect depends on whether direct answering suppresses or merely conceals the disparity, and those two mechanisms have sharply different implications for auditing. If direct answering conceals, then reasoning makes the bias *visible* without increasing it, which would make the chain-of-thought trace a diagnostic asset rather than a liability. If reasoning genuinely generates disparity that a forward pass without explicit steps would not exhibit, then explicit planning is a fairness risk independent of the underlying model. The existing measurements do not cleanly separate these. Second, [157] documents hidden veracity-linked beliefs surfacing *during* problem-solving, encoding demographic assumptions a final-answer probe would not detect. This result is consistent with both interpretations: reasoning as revealer or reasoning as generator. Until trace-replay (D2) is applied to hold the model's implicit beliefs constant across conditions and vary only the decomposition step, the two hypotheses remain confounded.

*What transfers only by analogy.* The P2-masking problem at this locus depends critically on the finding that chain-of-thought explanations systematically misreport the biasing features that actually drive predictions [123][†]. This result originates in single-turn classification and explanation settings, not in instrumented agentic plan traces. The transfer argument is that the same model occupying  $\pi$  in an agent produces unfaithful self-reports when generating rationales, as in the single-turn study, and therefore plan-trace audits at C4 inherit the same unreliability. That transfer is plausible but unmeasured: the degree to which multi-step plan generation is *more* or *less* faithful than single-step explanation is not known, and there is no agentic corpus study that validates the transfer directly. Similarly, the measurement reframing of [73][†], showing that agent *decisions* reveal implicit biases that text-level evaluation misses, provides the decisive motivation for plan-object auditing rather than trace-reading, but the original demonstration is in a simpler decision-making context rather than a full multi-step planning loop. The transferred conclusion (plan objects, not verbalized rationales, are the sound observables for C4) is well-motivated, but the agent-pipeline-specific validation has not been performed.

*What is unmeasured.* The benchmark gap at this locus is the sharpest in the survey. No existing benchmark scores the *plan itself* as a fairness object: are the sub-goals, their ordering, or the resource allocation across plan steps a counterfactual function of the subject's demographics, independent of the final decision? Plan-allocation disparity ( $\Delta_{\text{plan}}$ , §4.3) is observed in individual studies ([103, 139]) but has not been operationalized as a reusable metric with standardized demographic perturbations and counterfactual controls. The conduction signature below specifies what such a metric must measure. Critically, the P2-masking problem means that this gap cannot be closed by trace-reading: any future plan-audit must be counterfactual-behavioral (D2 applied to plan objects, holding the underlying evidence context fixed and swapping demographic signals in the goal specification), not inspection of verbalized rationales. That design constraint, established by [123][†]'s single-turn result and reinforced by [73][†], has not yet been absorbed into a benchmark design. The result

is that C4 sits at maturity ●○○ (Table 1) despite documented real-world planning-stage harm in hiring-adjacent deployments: a measurement design problem, not a data availability problem.

**Conduction signature.** *Entry:* within  $\pi$ , at the decomposition step. *Observable:* plan structure (step counts, ordering, allocations) across counterfactual pairs, not the verbalized rationale. *Dominant outbound operator:* PRESERVE/AMPLIFY within chain-of-thought (the GENERATE reading remains pending the revealer-vs-generator adjudication above), with strong P2 masking on the trace. *Metric:* plan-allocation disparity and CFR over plan objects (§4.3).

**0.3.6 Locus C5: User Modeling & Personalization Harms.** **Definition.** Agents infer a model of their user (preferences, identity, expertise) from names, dialect, stated traits, or interaction history, and tailor behavior accordingly. Personalization harm is the case where this inferred model produces *worse, stereotyped, or differential* treatment as a function of the (often inferred) protected attribute. By rule R3, this locus concerns the *interacting user*; subject-keyed disparities route through C1 to C4.

**Mechanism of harm.** The core tension is personalization versus stereotyping. In BCF terms the locus is GENERATE-dominant in a structurally unusual way: the protected attribute itself may be *created* by the component (inferred, not supplied), so a counterfactual audit that only swaps explicit labels can miss the operative channel entirely. There is also a default-state harm that precedes any inference: even an unpersonalized model reflects the opinions of particular demographic strata, so the *default user model* is already skewed [114][†], and personalization then moves users toward or away from a non-neutral baseline.

**What is known.** The most consistently replicated finding at this locus is that demographic cues supplied at query time cause a system’s adaptive behavior to track stereotypical group profiles rather than individually indicated preferences, collapsing the personalization-fidelity goal into stereotype instantiation. This mechanism is established across at least three distinct operationalizations. Identity signaling via explicit group labels biases chatbot recommendations toward stereotype rather than stated preference [63][†]. The trigger requires no explicit declaration: a name alone activates a presumed cultural identity that reshapes response content [101]. And the inference can precede any signal from the user, because LLMs recover protected attributes (location, income, sex) from innocuous conversational text without any demographic prompt [111], meaning the agent has already modeled the user demographically before any adaptation begins. The convergence across these three cue types (label, name, inference from content) on the same directional finding, stereotype-tracking rather than preference-tracking, constitutes the strongest multi-source replication this locus offers.

Downstream harm from this mechanism has been observed across domain-specific evaluations. In education, personalized content delivery produces systematically different quality as a function of inferred student identity [136]. In recommendation settings, consumer-facing LLM recommenders exhibit unequal recommendation utility across demographic groups [38], a finding accompanied by a normative benchmarking framework for measuring it [39]. Socioeconomic bias embedded in base model representations supplies the latent material these personalization steps activate [12]. A second replication strand concerns the alignment channel rather than the inference channel: reward models penalize African American Language [84], and reasoning performance degrades on non-standard dialects [71][†]. Because these disparities operate at locus R (the reward and alignment signal), they are PRESERVE-into-R in BCF terms, feeding directly into the feedback recurrence that §3.6 analyzes, so dialect penalties initiated at C5 are not bounded within C5.

**What conflicts or is contested.** The field’s principal empirical tension at this locus concerns cue-channel consistency. A systematic study of demographic cue modalities finds that name-based, explicit-label-based, and dialect-based probes of the same models yield inconsistent and sometimes

contradictory conclusions about the presence and direction of personalization bias [124]. This result does more than caution against single-cue audits; it implies that papers in this locus are not measuring the same underlying quantity, so the apparent convergence noted above is partly artifactual. Studies using name cues may detect bias that label-based studies of the same model do not, or vice versa, and without multi-cue replication, the directional finding (stereotype over preference) holds only within cue modality. A further contested point concerns the cost of mitigation: debiasing personalized models trades off against recommendation utility [127], which means that ATTENUATE interventions at this locus are not free in the way that, for example, filtering a retrieval set might be. The tradeoff magnitude has not been quantified across deployment contexts, and different utility definitions yield different tradeoff profiles, so no consensus on acceptable mitigation cost exists. Finally, the boundary between C5 and adjacent loci is contested by the inference result [111]: if the protected attribute is constructed by the model from non-demographic text, the operative locus may be the language model’s latent representation (C0 baseline) rather than any explicit user-modeling step, and counterfactual swaps at the input surface may leave the inferred user model unchanged, making the standard D2 protocol inapplicable without modification.

**What transfers only by analogy.** A substantial share of the mechanistic evidence at this locus was established in single-turn or chatbot settings and has not been reproduced in instrumented multi-step agents. The demonstration that identity signaling collapses adaptation into stereotyping [63][†] was conducted on a chatbot interface, not on a tool-using agent with persistent state. The dialect-reasoning penalty [71][†] was measured on benchmark QA tasks, not on agent planning or tool-selection traces. The baseline opinion-skew of the default user model [114][†] was characterized through opinion-poll probing of a static LLM, not through behavioral observation of a personalized agent. These results are included because the mechanisms they document (stereotype activation via identity cues, dialect penalties conducted into evaluation, skewed baseline user model) are mechanistically operative in any system that adapts to user signals, including agents. However, the conduction paths from these single-turn observations to the GENERATE and PRESERVE operators of the BCF remain unverified by agent-level trace measurement. The [†] markers on these citations track this analogical status throughout.

**What is unmeasured.** The most consequential open gap at this locus is the persistent user model: all available evidence concerns the disparity elicited within a single response or single turn, not the disparity accumulated and locked into the user representation an agent maintains and updates across sessions. An agent that interacts with a user repeatedly will form, update, and act on a profile that encodes earlier stereotype-triggered inferences, creating a form of within-agent institutionalized bias that single-response probing cannot detect. That persistent user model bridges C5 to §3.6’s recurrence, and it remains unmeasured. No agentic benchmark in Table 2 instruments the user model as a persistent state object, varies the cue channel as a controlled factor across sessions, or measures how demographic inferences made early in an interaction propagate to consequential decisions made later. The inference channel [111] introduces a further measurement-theory problem: if the protected attribute is inferred from content rather than supplied as a cue, the standard D2 counterfactual swap (alter the cue at input, observe behavior change) fails to intervene on the operative variable. Defining a valid counterfactual for this case requires either a structural causal model of the inference process, which does not exist for any deployed agent, or a cue-channel-controlled experiment design of the kind that [124] recommends but that no agentic benchmark has yet implemented. C5 is therefore the locus where the BCF’s D2 measurement procedure is hardest to operationalize, and the gap is tied directly to the GENERATE signature: bias that is internally constructed cannot be removed by swapping an external label.

**Open edge.** Personalization harm has been measured in single responses, not over the *persistent user model* an agent maintains across sessions, the object that bridges to §3.6; and no agentic

benchmark varies the cue *channel* systematically despite [124] showing conclusions are cue-fragile. The inference channel [111] makes C5 the locus where D2's counterfactual swap is hardest to even define, an unresolved measurement-theory problem.

**Conduction signature.** *Entry:* the inferred user model inside  $\pi/M$ . *Observable:* behavior divergence under cue-channel-controlled counterfactual swaps (name, dialect, label, history). *Dominant outbound operator:* GENERATE (the attribute is manufactured by inference), preserved into M and R. *Metric:* cue-conditioned MASD across channels (§4.2 to 4.3).

**0.3.7 The Temporal Axis: Long-Horizon Fairness Drift Over Trajectories. Definition.** Long-horizon fairness drift is the *temporal* degradation or self-amplification of fairness across turns, sessions, or deployment cycles, even when the per-step model is fixed.

**Why a temporal axis, not a sixth sibling.** Earlier drafts of this taxonomy, and adjacent surveys, treat long-horizon drift as a sixth component alongside the five architectural loci. BCF makes precise why that is a category error, and rule R4 codifies the repair. A component, in D1, is a *place* in the architecture with its own inputs and outputs; drift has no such place. There is no "drift module" to instrument: drift is what the feedback recurrence  $\Delta^{(t+1)} = \varphi_{fb} \cdot \Delta^{(t)} + \Delta_{entry}$  *does to any locus* when the loop closes: memory bias drifting ( $C2 \times t$ ), a user model ossifying ( $C5 \times t$ ), delegation patterns entrenching ( $C3 \times t$ ), or the policy itself shifting under retraining on its own outputs ( $\pi \times t$  through R). Treating drift as a sibling would double-count every time-indexed entry (violating P1's decomposition) and would imply the existence of a mitigation that targets "drift" as such, when by P4 every effective intervention must target a *locus-and-edge*: either reducing some  $\Delta_{entry}$  or forcing  $\varphi_{fb} < 1$ . The taxonomy is therefore five architectural loci  $\times$  one temporal axis: component  $\times$  time made explicit. We retain the subsection at this rank because the temporal axis is where the field's least measured and most consequential phenomena live, not because it is a sixth place where bias enters.

**Mechanism of harm.** Three nested loops realize  $\varphi_{fb}$  at increasing timescales, and all three are AMPLIFY-mode in the documented cases. The intra-trajectory loop (turn-level) operates within a single session as the context window accumulates a directional history. The inter-session loop operates across sessions as persistent memory ( $C2 \times t$ ) carries forward a biased state that conditions the next session's starting  $\Delta_{entry}$ . The deployment loop operates across retraining cycles as the model policy  $\pi$  itself shifts when trained on its own synthetic outputs. Each nesting is an additive term in the feedback recurrence, and the literature documents each at a different empirical resolution: the intra-trajectory loop most directly, the deployment loop through analogy and small-scale experiment, the inter-session loop almost not at all.

**What is known.** The most replicated mechanism on the T-axis is sycophantic conformity within a single multi-turn context. Fairness degrades over multi-turn dialogue [42], and implicit bias accumulates and propagates specifically in LLM *long-term memory*, making the agent more biased the longer it persists state [86] (the  $C2 \times t$  cell). These two results together identify the locus (C2) and the operator (AMPLIFY) for the intra-trajectory loop: the state written to memory at turn  $t$  conditions the  $\Delta_{entry}$  injected at turn  $t+1$ , so the recurrence is not merely metaphorical but traceable to a specific architectural feature. Sycophancy is the proximate mechanism, and it has the most consistent cross-paper support at the intra-trajectory timescale. Multi-turn measurements document progressive conformity, labeled "truth decay," across turns [76] and growth of sycophancy over extended dialogues [59]: both operationalize  $\varphi_{fb} > 1$  as a directly observable drift slope rather than an inferred parameter. At the deployment-loop timescale, the mechanism is self-consumption: training on a model's own synthetic outputs amplifies bias across generations [133], and the self-consuming performative loop has been characterized with proposed remedies for the LLM case specifically [140]. The recommender-systems literature supplies the canonical prior result this

work inherits: feedback-loop amplification in collaborative filtering is a documented, quantified phenomenon [78], and it establishes the theoretical plausibility of the same mechanism in any system that uses its own output distribution to shape future input.

**What conflicts or is contested.** The core amplification finding is not contested at the sign level: all documented T-axis results report  $\Delta^{(t)}$  rising, not falling, which is consistent with the  $\varphi_{fb} > 1$  regime. The contestation is quantitative and boundary-conditional. First, the magnitude of sycophantic drift is setup-dependent. Studies vary in how aggressively they simulate disagreement pressure, and the reported drift rates are not yet replicated across task types with matched stimuli; it remains unclear whether the same  $\varphi_{fb}$  obtains in low-stakes dialogue, consequential decision-support dialogue, and agentic tool-using settings, or whether the agent’s action-committed loop (where each turn includes an irreversible tool call) amplifies more than a purely conversational loop. Second, the deployment-loop evidence is thin: [133] and [140] work on LLM-generated text specifically, and the fairness implications are inferred rather than directly measured against a demographic counterfactual. The recommender analogy from [78] is well-founded but concerns a structurally simpler system; whether the same amplification rate transfers to the LLM policy case is an assumption, not a finding. Third, there is a tension between the intra-trajectory and inter-session loops that the literature does not resolve: it is possible (and consistent with existing data) that memory-cleared session boundaries reset  $\Delta^{(t)}$ , so the inter-session loop only amplifies if persistent memory is architecturally enabled. No study varies memory persistence as a controlled factor while measuring drift, so the relative contribution of the two timescales to long-run disparity is unknown.

**What transfers only by analogy.** Sycophancy as a mechanism is well-documented at the single-turn and chatbot level, but the measurements that establish its origin in human-preference training [116][†] are based on single-turn or short-context evaluations, not on instrumented agent loops with committed actions. Projecting that the same preference-optimization pressure produces the same drift rate in an action-taking agent is a motivated but untested inference: an agent whose sycophantic conformity leads it to invoke a biased tool or to escalate a claim under demographic pressure is a qualitatively different harm pathway than a chatbot that produces a more agreeable paragraph. Similarly, the formal sequential-fairness framework of [158][†] and the equal long-term benefit rate criterion of [142][†] were developed for explicit sequential decision processes with known transition dynamics, not for LLM agents whose transition function is a neural policy with opaque internal state. The theory proves that static per-step fairness constraints can fail over horizons and provides ATTENUATE-regime interventions in principle; none of those interventions has been implemented or tested in an LLM agent, so the applicability is analogy, not transfer. The simulation-based drift analysis tooling of [119][†] is similarly adjacent: it was designed for ML-enabled systems generically and has not been applied to an instrumented LLM agent trajectory.

**What is unmeasured.** The central gap on the T-axis is the absence of any estimate of  $\varphi_{fb}$  for a deployed LLM agent. To estimate  $\varphi_{fb}$ , one needs: (a) a demographically instrumented agent with logged traces across turns or sessions, (b) a counterfactual replay protocol that holds the locus-specific input fixed while varying the temporal accumulation, and (c) a scalar disparity measure  $\Delta^{(t)}$  that can be tracked as a time series. None of the three exists in a published evaluation. Equivalently, the sequential-fairness theory [142, 158] and the LLM-drift empirics [42, 86] have not been joined: no work evaluates an LLM agent against a formal long-term-fairness criterion over a realistic multi-session deployment. The field therefore cannot yet answer the most consequential question the T-axis raises: whether the documented intra-trajectory drift ( $\varphi_{fb} > 1$ , measured over tens of turns) compounds into materially disparate outcomes over realistic deployment durations (hundreds of sessions, one or more retraining cycles). The metric gap is specific: a drift-rate benchmark would



need to report the slope of  $\Delta^{(t)}$  across turns and sessions as a first-class evaluation output, the long-term benefit-rate gap [142] as a criterion against which that slope is evaluated, and a  $\varphi_{\text{fb}}$  point estimate with confidence bounds per locus. None of those metrics appears in any existing agent benchmark (Table 2, §4.4).

**Conduction signature.** *Entry:* the R feedback edge (and time-indexed M). *Observable:* the trajectory of  $\Delta^{(t)}$ , its slope and curvature across turns, sessions, retraining cycles. *Dominant operator:* AMPLIFY (the recurrence). *Metric:* drift rate (slope of  $\Delta^{(t)}$ ) and long-term benefit-rate gap [142] (§4.3).

**0.3.8 Compounding Across the Pipeline: Proposition P3 and Its Limits.** The six entries are not additive, and Proposition P3 says when they can fail to be:  $\Delta_r$  **can exceed the naive sum**  $\sum_c |\Delta_c|$  **when at least one AMPLIFY or GENERATE edge lies on a conduction path.** P3 constitutes the formal content of what we call the *agentic multiplier*, converting a rhetorical worry ("agents make bias worse") into a checkable property of a logged trace: estimate the  $\varphi'_e$ s; under the linearized magnitude model, some  $\varphi_e > 1$  or a fresh entry mid-path is necessary for super-additivity (not sufficient, since it can be offset downstream). Conversely (and this direction is just as consequential for auditors), if every edge in a pipeline is PRESERVE or ATTENUATE and no interaction term is present, component-level audits compose into a *conservative* system-level bound, and certifying each component suffices. The entire practical question of whether per-component certification can ever underwrite system-level fairness claims therefore reduces to locating and bounding the amplifying edges.

**What is evidenced.** Within a single locus (C3) P3 is directly demonstrated. Stereotypical bias *emerges* where none was salient, *propagates* between agents, and is *amplified* by their interaction, yielding a collective strictly more biased than its members [91]; large-scale simulations show that fairness in multi-agent decision systems is *emergent*, not reliably inherited, so component audits cannot be assumed to compose into a group-level guarantee in that setting [81]. Persona-induced bias scaling from single agents to societies [75] and interaction-borne implicit bias [3] corroborate the same operator at the same locus. The temporal axis supplies a second evidenced amplifier, on the feedback edge rather than between components: self-consuming training loops with  $\varphi_{\text{fb}} > 1$  [78, 133, 140].

**What is conjecture.** The cross-component cascade is plausible under BCF, since every individual edge in that chain is separately evidenced as PRESERVE-or-worse. The intuitive story runs as follows: an attribute *inferred* by user modeling (C5, [111]) is *persisted* by memory (C2, [86]), conditions *tool selection* (C1) and *plan decomposition* (C4, [73]), is *delegated* along role lines (C3), and *drifts* over the deployment horizon (T-axis). But we state plainly: **no paper in the corpus traces a single disparity across two or more different components of one agent.** We therefore record it as:

**Conjecture 1 (Cross-component conduction).** In deployed tool-using agents, disparities entering at one locus are conducted with  $\varphi \geq 1$  across component boundaries, so that multi-locus pipelines exhibit super-additive  $\Delta_r$  even when each component passes its own audit.

The conjecture is falsifiable with existing instruments: D2's trace-replay applied at two consecutive interfaces of the same logged trajectory estimates the cross-boundary  $\varphi$  directly, and the audit protocol of §6 is designed to produce exactly such paired estimates (not estimated in the pilot; see §6.5). Until such measurements exist, claims that "agentic bias compounds across the pipeline" (including informal versions in prior drafts of this survey) should be read as hypotheses licensed by P3's mechanism, not as established findings.

That compounding question is also the structural reason this survey is organized by component rather than by metric, and the conceptual core of the measurement gap of §4: by P2, a token-level probe at any single hand-off can read clean while the consequential action is sharply disparate [73, 146]; by P5, only trace-level counterfactual instrumentation can adjudicate Conjecture 1; and the coverage analysis of §7 shows the cells P3 most implicates (action-level, counterfactual, long-horizon) are precisely where the field is emptiest.

*0.3.9 Worked Example: A Hiring-Screener Agent Through the BCF, and a Four-Step Auditor Recipe.* To make the framework operational rather than ornamental, we walk one consequential system through it: a resume-screening agent, the deployment context with the deepest audit record [14, 90, 108, 126, 135] and the clearest regulatory exposure. We build the agent in four capability rungs, C0 to C3, and track what each rung adds to the conduction equation. (The rungs are capability levels of the worked example; the loci remain C1 to C5.)

**C0, bare model.** A single LLM call scores each resume. The loop degenerates to  $\pi$  alone; the conduction equation collapses to  $\Delta_\tau = \Delta_\pi$ . This rung is *not* hypothetical parity: intersectional gender-and-race penalties in automated resume evaluation [14], name-based callback disparities echoing the classic audit paradigm [126], and occupation-skewed recommendations by name [90] establish  $\Delta_\pi > 0$  as the baseline entry. Everything QA-era fairness evaluation can see, it sees here.

**C1 rung, add retrieval.** The screener now retrieves similar past resumes and hiring precedents to ground its scores. Two new terms enter:  $\Delta_{\text{ret}}$  at the C2 locus, conducted by the generation edge. This rung is also empirically grounded: race and gender bias arises *in the retrieval step itself* when language models are used for resume screening [135], and by [55], instructing the generator to "score fairly" does not neutralize what retrieval injects (P4: the prompt sits upstream of the GENERATE edge). An endpoint audit that found  $C0 \approx C1$  parity could mask offsetting entries (P2); only comparing  $\Delta$  at the retrieval interface against  $\Delta$  at the output separates the terms.

**C2 rung, add tools and persistent memory.** The screener now chooses among background-check and skills-assessment APIs (C1 locus) and persists per-applicant notes across screening rounds ( $C2 \times t$ ). New terms:  $\Delta_{\text{tool}}$  (selection skew independent of task fitness is already documented [21], and vendor-controlled descriptions are adversarially steerable [120]; whether the demographic framing of a profile changes *which* verification API is invoked is precisely the unmeasured demographic-swap question of §3.1) and a memory term that the recurrence now time-indexes. Early inferred attributes (possibly never stated, by [111]) persist and accumulate [86], so an applicant re-screened in round two inherits round one's framing [122].

**C3 rung, multi-agent committee.** The pipeline becomes a recruiter-agent, a technical-evaluator-agent, and an HR-compliance-agent under a controller: the architecture of LLM-driven team workflows [105, 129], whose task-allocation skew is already documented [103]. New terms:  $\Delta_{\text{assign}}$  at C3, plus the only corpus-evidenced AMPLIFY-between-agents edge [81, 91]. By P3, this is the first rung at which  $\Delta_\tau$  can *exceed* the sum of everything below it; by Conjecture 1, the full C5 to C2 to C1 to C4 to C3 cascade is possible but unmeasured. Note what the ladder implies for governance: each rung is a routine engineering upgrade, invisible to a model-level fairness certificate issued at C0. The per-rung re-audit obligation this implies is taken up in §8.5.

**A four-step auditor recipe.** The BCF reading of the ladder compresses into a procedure any team deploying an agent can execute, and which the §6 protocol instantiates on a hiring-style task (not estimated in the pilot; see §6.5):

- (1) **Instrument the trace.** Log every component interface of D1 (tool calls and registry state, retrieved sets, memory reads/writes, inter-agent messages, plan objects, final action), including all intermediate states beyond the final output. (The agent-benchmark infrastructure already does this much [70, 154]; what it lacks is step 2.)

- (2) **Counterfactually replay per component (D2).** For each locus  $c$ , swap the protected cue at  $c$ 's input while holding the rest of the logged trace fixed; compute  $\Delta_c$  as component-level CFR/MASD (§4.2 to 4.3). Vary the cue *channel* (name, dialect, explicit label), since single-cue audits mislead [124].
- (3) **Estimate the edge operators.** Compare disparities at consecutive interfaces to classify each edge as ATTENUATE / PRESERVE / AMPLIFY / GENERATE; on the feedback edge, regress  $\Delta^{(t)}$  across turns or sessions to estimate  $\varphi_{fb}$  [42, 119]. Any  $\varphi > 1$  or mid-path entry flags a P3 super-additivity risk.
- (4) **Match mitigation to the (locus, operator) cell and re-audit (P4).** Intervene at the entry locus or on the amplifying edge (embedder or ranking control for C2 [64, 160], identity anonymization for C3 [28], reasoning-stage fine-tuning or aggregation-stage chain reweighting for C4 [57, 66], source-selection rebalancing for C1 [118]), then repeat steps 2 to 3 to verify that the targeted  $\Delta_c$  or  $\varphi_e$  actually fell, and re-run over the horizon to confirm  $\varphi_{fb} < 1$ .

Do not audit the verbalized rationale in place of step 2: stated reasoning is unfaithful to the operative features [123], so only counterfactual behavior at component interfaces is a sound observable. Steps 1 to 2 are exactly the measurements Proposition P5 declares necessary, step 3 operationalizes P1 to P3, and step 4 operationalizes P4. The framework, run forward, is the audit.

**A worked numerical micro-example (illustrative, not measured).** To make the conduction machinery concrete, we run two small hiring-ladder configurations through the conduction equation  $\Delta_\tau \approx \Sigma_c (\text{signed } \Delta_c) \cdot \Pi_{e \in \text{path}(c \rightarrow O)} \varphi_e$ . The magnitudes below are illustrative placeholders, not measurements; they are chosen only to make the operators visible, and any real numbers would come from the trace-replay of steps 1 to 2.

*Configuration (a): masking (P2).* Two loci feed the output node. The screening-prompt locus carries a signed component disparity  $\Delta_{\text{prompt}} = +0.20$  along a PRESERVE edge ( $\varphi = 1.0$ ); the retrieval locus carries a sign-opposed disparity  $\Delta_{\text{ret}} = -0.24$  along an ATTENUATE edge ( $\varphi \approx 0.83$ ). The endpoint contributions are  $(+0.20)(1.0) = +0.20$  and  $(-0.24)(0.83) \approx -0.20$ , so the endpoint trajectory disparity is  $\Delta_\tau \approx +0.20 - 0.20 \approx 0.00$ , even though the summed magnitude of interior disparity is  $\Sigma_c |\Delta_c| = |+0.20| + |-0.24| = 0.44$ .

| Locus                                    | signed $\Delta_c$ | edge operator | $\varphi_e$ | endpoint contribution                                      |
|------------------------------------------|-------------------|---------------|-------------|------------------------------------------------------------|
| Screening prompt                         | +0.20             | PRESERVE      | 1.0         | +0.20                                                      |
| Retrieval                                | -0.24             | ATTENUATE     | 0.83        | -0.20                                                      |
| <b>Endpoint <math>\Delta_\tau</math></b> |                   |               |             | $\approx$ <b>0.00</b> (with $\Sigma_c  \Delta_c  = 0.44$ ) |

Here an endpoint-only audit reads  $\Delta_\tau \approx 0$  and certifies parity, yet two loci each carry a disparity near 0.2 that sign-opposition and attenuation have hidden from the output node. This is exactly the masking P2 describes: an endpoint near zero masks a large summed interior disparity, reachable only by ATTENUATE  $\varphi < 1$  and sign-opposed entries.

*Configuration (b): super-additivity (P3).* A single locus carries  $\Delta_{\text{assign}} = +0.15$  at the delegation interface, but its path to the output crosses an AMPLIFY edge with  $\varphi = 1.6$ . Its endpoint contribution is  $(+0.15)(1.6) = 0.24$ , so  $\Delta_\tau \approx 0.24$  exceeds the naive interior sum  $\Sigma_c |\Delta_c| = 0.15$ .

Here the trajectory disparity is larger than anything visible by summing component magnitudes, because an AMPLIFY (or GENERATE) edge on the conduction path multiplies the locus's contribution: the agentic multiplier of P3. We stress again that the 0.20, 0.24, 0.15, and  $\varphi$  values are illustrative placeholders chosen to expose the operators, not estimates from any run.

| Locus                                           | signed $\Delta_c$ | edge operator | $\varphi_e$ | endpoint contribution                                              |
|-------------------------------------------------|-------------------|---------------|-------------|--------------------------------------------------------------------|
| Delegation/assignment<br>Endpoint $\Delta_\tau$ | +0.15             | AMPLIFY       | 1.6         | 0.24<br>$\approx \mathbf{0.24}$ ( $> \Sigma_c  \Delta_c  = 0.15$ ) |

#### 0.4 How to Evaluate Agent Fairness

The translation from LLM-level fairness measurement to agent-level evaluation is neither direct nor complete. Single-turn benchmarks measure what a model *says*; an agent must be evaluated on what the system *does*: which tools it invokes, whose requests it escalates, how its plans allocate effort, and how disparities conduct across the loop. The Bias Conduction Framework of §3 supplies the formal target: by Definition D2, fairness evaluation of an agent  $\mathcal{A} = \langle \pi, T, M, R \rangle$  (Definition D1) is the estimation of component counterfactual disparities  $\Delta_c$  via trace-replay, and by the conduction equation  $\Delta_\tau \approx \Sigma_c \Delta_c \cdot \prod_e \varphi_e$ , an endpoint-only measurement estimates the left-hand side while leaving every term on the right unidentified. This section delivers the survey’s second contribution in four steps. §4.1 lifts the classical metric families (group, individual, counterfactual) onto the agent loop and asks what the impossibility results of [62] and [8] become when the unit of evaluation is a trajectory rather than a prediction. §4.2 formalizes trajectory-level CFR and MASD, generalizing [85] into instances of D2’s  $\Delta_c$ . §4.3 supplies formal definitions for the three action-level disparity metrics the field currently lacks (tool-invocation, escalation, and plan-allocation disparity) as  $\Delta_c$  specializations, discharging Proposition P5’s requirement constructively. §4.4 then inventories the benchmark field (Table 2) and shows, via Propositions P2 and P5, why no existing resource can certify an acting agent.

**0.4.1 Metric Families Lifted to the Agent Loop.** Three classical fairness families dominate the measurement literature [1, 6, 80, 163]. Each imposes distinct demands and inherits distinct pathologies when lifted from single predictions to agent trajectories.

*Group fairness: from prediction distributions to action distributions.* Group fairness requires statistical equivalence of outcomes across protected groups. Its lineage runs from disparate-impact doctrine and the four-fifths rule, formalized for classifiers by [40], through demographic parity to the error-rate criteria of equalized odds and equality of opportunity [50]. The criteria are operationally attractive because they reduce fairness to a comparison of two conditional distributions.

Lifting group fairness to the agent loop changes *what the outcome variable is*, not the comparison itself. In D1 terms, any measurable emission of the loop can serve as the outcome: the terminal decision  $y$  (loan approval rates across applicant groups [13], verdict severity across defendant groups [58]), but equally the intermediate emissions such as which tools in  $T$  are invoked, what  $M$  retrieves, which roles  $\pi_{\text{ctrl}}$  assigns, and whether the trajectory escalates to a human. Clinical audits already exploit this implicitly: [95] compares *management plans* (an action distribution) rather than answer accuracy across sociodemographic variants, and [15] scores harm-differentiated response patterns in decision support. What the QA-era surveys organize as a single “extrinsic disparity” number [24, 46] is, under BCF, a *family* of group-fairness statistics indexed by component: one  $\Delta_c$ -style comparison per locus. The practical failure of current group-fairness practice is therefore not the criterion but the denominator: existing evaluations aggregate over prompts and treat each response as atomic, so an agent that reaches equalized final answers through biased tool routing registers as fair. Proposition P1 (Locality) says the disparity has a component address; response-level group statistics discard the address.

*Individual fairness: from similar inputs to similar trajectories.* Individual fairness, in the metric formulation of [30], demands that the treatment of two individuals differ by no more than their task-relevant dissimilarity, a Lipschitz condition  $d_{\text{out}}(h(x), h(x')) \leq L \cdot d_{\text{in}}(x, x')$ . The notorious obstacle is specifying  $d_{\text{in}}$ ; the agentic lift adds a second: specifying  $d_{\text{out}}$  over *trajectories*. Two near-identical applicants should receive the same verdict and be processed along comparably resourced paths (same depth of evidence retrieval, same tool quality, same escalation behavior). A natural candidate for  $d_{\text{out}}$  is an edit distance over action sequences, but no corpus paper yet defines or estimates one; the closest instantiation is the intervention-consistency audit of [22], which measures whether clinical and judicial recommendations change under name substitution while holding case features fixed, an individual-fairness probe at the terminal emission only. The trajectory version matters because of a failure mode group statistics cannot see: an agent can satisfy endpoint parity while routing counterfactual twins down systematically different paths (the *path disparity* that Proposition P2 predicts can mask), so the like-cases-alike intuition is violated in the process even when it is satisfied in the verdict.

*Counterfactual fairness: from structural interventions to trace-replay.* Counterfactual fairness [61] requires that the outcome for an individual be invariant under an intervention on the protected attribute  $A$ : in structural-causal-model notation,  $\Pr(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a) = \Pr(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)$ , where the intervention propagates to the *descendants* of  $A$  in the causal graph, not (as is sometimes loosely stated) to any variable that merely correlates with  $A$ . This family translates most naturally to agents, because the demographic swap can be applied to the full input and the full trajectory observed: this is exactly D2's trace-replay protocol, and §4.2 develops it formally.

One honesty note is required, because it qualifies every counterfactual number reported in this literature. The swap operator  $\sigma$  used in practice (name substitution, demographic-cue editing) intervenes on the *textual encoding* of  $A$ , not on the structural model: it alters the name but leaves other descendants of  $A$  in the profile (dialect, address, school, employment gaps) untouched, which biases  $\Delta_c$  estimates toward zero; conversely, an inconsistent swap that breaks profile coherence can inflate them. [124] demonstrates the consequence empirically: different cue channels (names vs. explicit labels vs. dialect) yield inconsistent bias conclusions on the same models. Trace-replay counterfactuals are thus *proxy interventions*, and rigorous protocols must vary the cue channel as a controlled factor, as the audit protocol of §6 does.

*The upstream family: reward-model and alignment fairness.* A fourth family sits upstream of the loop. Agents are increasingly steered, scored, or filtered by reward models, and if the reward signal itself encodes demographic preferences, every downstream policy inherits them: [121] documents systematic inter-group reward gaps across demographic groups, and [84] shows reward models penalize African American Language, a disparity *generated* (operator GENERATE, on zero input disparity) at the alignment stage and then conducted through whatever the agent does. Evaluation must therefore propagate upstream from the action into the mechanism that rewards it; under BCF this is simply a  $\Delta_c$  measurement at a locus outside the runtime loop, with conduction edges into  $\pi$ .

*What the impossibility results become over trajectories.* The classical metric tensions do not dissolve at the agent level; they multiply. [62] prove that calibration and balance of error rates across groups cannot hold simultaneously except under equal base rates or perfect prediction; [8] derives the same incompatibility between predictive parity and error-rate parity for recidivism instruments. These results bind any single decision point. Lifting them to trajectories surfaces three structurally new tensions, which we state as observations from the conduction equation rather than as theorems. None has yet been formally proved or empirically quantified for LLM agents, which is itself a finding about the field.

**Tension T1 (non-compositionality: per-step fairness does not compose).** An agent trajectory is a sequence of decision points, each formally subject to the classical impossibility. But satisfying a chosen criterion at every step does not deliver that criterion at the terminal decision, because each intermediate action reshapes the population reaching subsequent steps: the "base rates" at step  $k$  are endogenous to the policy at steps  $1, \dots, k-1$ . The sequential-screening analogue of the fair-sequential-decision results is instructive here: [158] show static per-step constraints fail to deliver long-term fairness, and [142] show naive transfer of one-shot criteria can *degrade* equal long-term benefit rates. The trajectory-level reading of [62] is therefore harsher than the original: the auditor must choose which criterion to satisfy and *at which component* to satisfy it, knowing the choice perturbs the conditions under which every downstream criterion is evaluated.

**Tension T2 (outcome multiplicity under conduction coupling).** A single classifier has one outcome variable over which the impossibility binds. A trajectory has many: every component emission  $z_c$  plus the terminal  $y$ . The conduction equation couples them:  $\Delta_\tau \approx \sum_c \Delta_c \prod_e \varphi_e$  means that driving one component's disparity to zero redistributes, rather than removes, the constraint when other edges amplify ( $\varphi_e > 1$ ). Equalizing escalation rates while simultaneously equalizing terminal approval rates may be jointly infeasible when the escalation emission feeds an amplifying edge into the verdict. The label-choice problem of [92] (where optimizing a proxy outcome (cost) produced large racial disparity on the target outcome (health need)) recurs here at every component boundary: each  $z_c$  is a proxy for the harm the auditor actually cares about, and the agent multiplies the number of proxy choices by the number of loci.

**Tension T3 (aggregation masking, P2 as a measurement tension).** Proposition P2 states that endpoint parity is consistent with large offsetting component disparities (attenuating edges,  $\varphi_e < 1$ , or sign-opposed  $\Delta_c$  terms). As a metric tension, this means group fairness at the trajectory endpoint and individual fairness along the trajectory can disagree without bound: counterfactual twins reach identical verdicts through unequally resourced paths. The unfaithfulness results for chain-of-thought sharpen the warning. [123] show that verbalized reasoning does not reliably reflect the causes of model behavior, so  $\Delta_c$  must be defined over *executed actions* (tool calls issued, memory operations performed, delegations made), never over the agent's narrated rationale.

Finally, two construct-validity cautions cut across all families. [18] catalogue pitfalls in fairness benchmark construction (incoherent contrasts, invalid stereotype operationalizations) that transfer wholesale to any agentic benchmark built by the same template, and [109] warn against the framing trap of evaluating an abstracted technical system apart from its deployment context: an agent loop scored in a sandbox omits exactly the institutional feedback (recourse, appeal, compounding access effects) through which the recurrence  $\Delta^{(t+1)} = \varphi_{fb} \Delta^{(t)} + \Delta_{entry}$  operates in the world. Trajectory-level evaluation narrows the abstraction gap relative to QA scoring; it does not close it.

**0.4.2 Trajectory-Level Counterfactual Metrics: CFR and MASD as Instances of  $\Delta_c$ .** We now make D2 operational. Fix an agent  $\mathcal{A} = \langle \pi, T, M, R \rangle$ , an input population  $\mathcal{X} = \{x_1, \dots, x_N\}$  of task instances, and a demographic swap operator  $\sigma$  that rewrites the protected-attribute encoding of an input while holding task-relevant content fixed (§4.1.3's proxy-intervention caveats apply, and  $\sigma$  should range over multiple cue channels [124]). Executing  $\mathcal{A}$  on  $x$  under the trace-replay protocol (pinned model versions, fixed decoding parameters, intercepted and demographically consistent tool/environment responses) yields a trajectory  $\tau_{\mathcal{A}}(x) = (s_0, a_1, o_1, \dots, a_K, y)$  of actions  $a_k$ , observations  $o_k$ , and terminal decision  $y$ . For each component  $c$  of the taxonomy, an extraction map  $z_c(\cdot)$  reads off that component's emissions from the trace (the tool-invocation record for  $T$ , the retrieval set for  $M$ , the role assignments for  $\pi_{ctrl}$ , the plan structure for  $\pi$ 's decomposition, the terminal decision for the loop as a whole). Definition D2's component counterfactual disparity is then

$$\Delta_c(\mathcal{A}; \mathcal{X}, \sigma) = \mathbb{E}_{x \sim \mathcal{X}} \left[ d_c(z_c(\tau_{\mathcal{A}}(x)), z_c(\tau_{\mathcal{A}}(\sigma(x)))) \right],$$

where  $d_c$  is a distance on component  $c$ 's emission space. Every counterfactual metric in this literature is a choice of  $(c, z_c, d_c)$ . The two metrics introduced by [85] for a contact-center quality-assurance agent are the canonical discrete and continuous choices, which we state in their trajectory-level generality.

**Definition 4.1 (Trajectory-level Counterfactual Flip Rate).** For a discrete emission  $z_c$  (a verdict, a tool choice, an escalation decision, a role assignment), the component CFR is  $\Delta_c$  under the 0 to 1 metric:

$$\text{CFR}_c(\mathcal{A}) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[z_c(\tau_{\mathcal{A}}(x_i)) \neq z_c(\tau_{\mathcal{A}}(\sigma(x_i)))].$$

The terminal case  $\text{CFR}_y$  (with  $z_c = y$ ) recovers the original flip rate of [85]; the intervention-consistency rate of [22] is  $\text{CFR}_y$  with  $\sigma$  instantiated as name substitution.

**Definition 4.2 (Trajectory-level Mean Absolute Score Difference).** For a continuous emission  $s_c$  (a recommendation score, a rank, a confidence, a quality rating), the component MASD is  $\Delta_c$  under the absolute-difference metric:

$$\text{MASD}_c(\mathcal{A}) = \frac{1}{N} \sum_{i=1}^N |s_c(\tau_{\mathcal{A}}(x_i)) - s_c(\tau_{\mathcal{A}}(\sigma(x_i)))|.$$

CFR and MASD decompose the counterfactual question cleanly: *does* the emission change under the swap, and *by how much*? Reporting both prevents a benchmark from hiding many small shifts behind a low flip count or a few large flips behind a small mean.

**Definition 4.2b (Signed component disparity, MSD).** For an *ordered* emission space with a stated favorable-direction convention (e.g., a higher score, rank, or confidence is favorable to the subject), the signed component disparity is  $\text{MSD}_c(\mathcal{A}) = \mathbb{E}_{x \sim \mathcal{X}} [s_c(\tau_{\mathcal{A}}(\sigma(x))) - s_c(\tau_{\mathcal{A}}(x))]$ , the mean directional shift induced by the swap. Flip-type discrete emissions carry no sign without an orientation convention and so admit no MSD. The relation to the magnitude metrics is deliberate: the §4.2-4.3 metrics (CFR, MASD, and the TID/ESD/PAD of §4.3) estimate magnitudes  $|\Delta_c|$ , which is what an isolated-locus disparity audit needs; cancellation diagnosis under P2, where oppositely signed component contributions sum toward an endpoint near parity, requires the signed variant, because two large  $|\Delta_c|$  that cancel are visible only once their signs are recovered.

**Estimating the conduction operators.** Because Definitions 4.1 to 4.2 are component-indexed, they also yield the first estimator for D3's operators. For a pipeline edge  $e : c \rightarrow c'$ , the empirical conduction coefficient is the disparity ratio across the edge,

$$\hat{\phi}_e = \frac{\Delta_{c'}(\mathcal{A}; \mathcal{X}, \sigma)}{\Delta_c(\mathcal{A}; \mathcal{X}, \sigma)} \quad (\Delta_c > 0),$$

classifying the edge as attenuating ( $\hat{\phi}_e < 1$ ), preserving ( $\hat{\phi}_e \approx 1$ ), or amplifying ( $\hat{\phi}_e > 1$ ); the GENERATE case is diagnosed when  $\Delta_c \approx 0$  at every upstream locus yet  $\Delta_{c'} > 0$ . This turns the BCF's operator taxonomy from a descriptive vocabulary into a measurable quantity, and it is the estimator the audit protocol of §6 is designed to compute (not estimated in the pilot; see §6.5).

**The existing counterfactual evaluation frameworks, read against this scaffold.** The 2025 to 2026 wave of counterfactual evaluation frameworks shares a common three-step scaffold (construct demographically paired inputs, observe outputs under each condition, compute a distance statistic), and the frameworks differ chiefly in which corner of the  $(c, z_c, d_c, \sigma)$  design space they

develop. CAFFE [102] and its companion SCOPE dataset [104] develop the  $\sigma$  axis: systematic, multi-attribute perturbation generation with coverage of attributes existing benchmarks leave sparse. FiSCo [146] develops the  $d_c$  axis: a semantic, statistically-tested group-difference measure that detects disparities surface-level token comparison misses, important for agents whose verbally equalized answers can sit atop differentiated reasoning. The Equal-Access audit [16] develops the  $z_c$  axis toward service quality: breadth and actionability of responses as the emission, mapping in agentic settings onto retrieval depth and tool-complement disparities. [22] develops the deployment-realism axis, applying flip-rate logic in clinical and judicial scenarios where the emission is a consequential recommendation. What none of them yet provides, including the originating CFR/MASD deployment of [85] (which scores a single QA-scoring emission of a deployed agent), is coverage of *intermediate* trajectory state: the tool selected at step 3, the memory written at step 5, the subagent delegated to at step 2. In BCF terms, the field has built increasingly refined estimators of  $\Delta_\tau$  while leaving every interior  $\Delta_c$  and every  $\varphi_e$  unestimated. By Proposition P1 those interior quantities are where the harm has an address; by Proposition P4, they are what mitigation must be matched against.

**0.4.3 Action-Level Disparity Metrics: Formal Definitions.** The sharpest edge of the measurement gap has been the absence of *defined* metrics for action-level disparity: prior work demonstrates that actions carry bias that answers conceal [21, 73, 147] but stops short of canonical formulas. We supply them here as  $\Delta_c$  specializations, in both counterfactual (paired) and group (population) forms. Throughout,  $a \in \{0, 1\}$  indexes the protected-group contrast for group forms, and the counterfactual forms are paired over  $(x, \sigma(x))$  as in §4.2.

**Definition 4.3 (Tool-Invocation Disparity, TID).** Let  $n_t(\tau)$  be the number of invocations of tool  $t \in T$  in trajectory  $\tau$ , and  $p_\tau(t) = n_t(\tau) / \sum_{t' \in T} n_{t'}(\tau)$  the empirical invocation distribution. The counterfactual form is  $\Delta_c$  at the tool-selection locus with  $z_c = p_\tau$  and  $d_c$  the total-variation distance:

$$\text{TID}_{\text{cf}}(\mathcal{A}) = \mathbb{E}_{x \sim \mathcal{X}} \left[ \frac{1}{2} \sum_{t \in T} |p_{\tau_{\mathcal{A}}(x)}(t) - p_{\tau_{\mathcal{A}}(\sigma(x))}(t)| \right],$$

and the group form compares group-conditional mean invocation distributions,  $\text{TID}_{\text{grp}} = \frac{1}{2} \sum_{t \in T} |\mathbb{E}[p_\tau(t) | a=1] - \mathbb{E}[p_\tau(t) | a=0]|$ . TID answers the question §3.1 left open: does swapping a subject's demographics change *which* API the agent calls, holding the query fixed? The existence evidence is layered: that selection skew is measurable and manipulable at all is established by the provider- and ordering-preference results of [21, 120], which ride on the same surface-form sensitivities as option-order bias [98] but involve no sensitive attributes; the existence evidence for the demographic question itself rests on [147], where the tool stages account for an attributable share of total disparity in the one clinical agent where the decomposition has been run. No benchmark computes TID under demographic counterfactuals.

**Definition 4.4 (Escalation Disparity, ESD).** Let  $e(\tau) \in \{0, 1\}$  indicate that the trajectory escalates: defers to a human reviewer, requests additional documentation, refuses, or routes to a higher-scrutiny queue. The group form is a parity gap, with a companion ratio form aligned with the four-fifths convention of disparate-impact doctrine [40]:

$$\text{ESD}_{\text{grp}} = |\Pr(e(\tau)=1 | a=1) - \Pr(e(\tau)=1 | a=0)|, \quad \text{ESD}_{\text{ratio}} = \frac{\min_a \Pr(e(\tau)=1 | a)}{\max_a \Pr(e(\tau)=1 | a)},$$

and the counterfactual form is the escalation-specific flip rate,  $\text{ESD}_{\text{cf}} = \mathbb{E}_x [ |e(\tau_{\mathcal{A}}(x)) - e(\tau_{\mathcal{A}}(\sigma(x)))| ]$ , i.e., CFR<sub>c</sub> with  $z_c = e$ . Escalation is the margin where differential treatment hides behind identical



verdicts: two counterfactual twins may both be approved, but one only after demands for extra documentation that impose cost and attrition. Clinical evidence that management and referral behavior shifts with sociodemographic cues [95], and the safety-utility asymmetries of personalized refusal behavior [127], indicate this margin is live; no benchmark instruments it.

**Definition 4.5 (Plan-Allocation Disparity, PAD).** Let  $\mathbf{r}(\tau) \in \mathbb{R}^m$  be the plan's resource vector, e.g., number of plan steps, tool-call budget consumed, retrieval depth, deliberation tokens, count of verification actions. The counterfactual form is  $\Delta_c$  at the planning locus under a weighted  $\ell_1$  metric:

$$\text{PAD}_{\text{cf}}(\mathcal{A}) = \mathbb{E}_{x \sim \mathcal{X}} \left[ \left\| \mathbf{r}(\tau_{\mathcal{A}}(x)) - \mathbf{r}(\tau_{\mathcal{A}}(\sigma(x))) \right\|_{1, \mathbf{w}} \right], \quad \|\mathbf{v}\|_{1, \mathbf{w}} = \sum_{j=1}^m w_j |v_j|,$$

with weights  $w_j$  fixed in advance to reflect the deployment's resource stakes. When the plan allocates work *to people* rather than effort to a case (team composition, task assignment), the relevant form is the maximum per-role assignment gap,  $\text{PAD}_{\text{alloc}} = \max_{\rho \in \mathcal{R}} \left| \Pr(\text{assign}(\rho) \mid a=1) - \Pr(\text{assign}(\rho) \mid a=0) \right|$ , which is the metric implicit in the demographically skewed task allocations documented by [103] and the agent-decision disparities of [73], neither of which states it as a reusable formula.

Three remarks complete the definitions. *Estimation.* All three metrics are paired statistics under trace-replay: variance is computed over counterfactual pairs, confidence intervals attach to each  $\Delta_c$ , and the configuration-by-cue-channel grid requires multiple-comparison correction. The audit protocol of §6 instantiates exactly these estimators for a hiring task across agent configurations C0 to C3 (the §6.5 pilot reports endpoint metrics only; these paired/interior estimates await the full study). Determinism matters: without pinned versions, fixed decoding, and intercepted environment responses, flip rates confound demographic sensitivity with sampling noise. *Scope.* The definitions deliberately score executed actions, not narrated rationales, per the unfaithfulness caveat of [123]. *Adequacy.* Proposition P5 states that an evaluation suite is adequate for an agent only if it bounds  $\Delta_c$  for every component with nonzero conduction weight in  $\Delta_\tau \approx \sum_c \Delta_c \prod_e \varphi_e$ . Definitions 4.3 to 4.5, together with the component CFR/MASD of §4.2, constitute a minimal instrument set meeting P5 for the  $T$ , routing, and planning surfaces of D1. By Proposition P4 (mitigation-matching), they are also the precondition for crediting any mitigation with changing what an agent *does* rather than what it says. What was previously described in this literature as having "no canonical formula" now has one; what remains missing is the benchmark infrastructure to run it against, which is §4.4's subject.

**0.4.4 Datasets and Benchmarks: An Inventory of the Gap.** Table 2 inventories the evaluation resources in the corpus alongside the agent-capability tools against which any claim of a "missing benchmark" must be tested. We classify each resource by *fairness level* (QA (scored answers to fixed questions), generation (scored open-ended text), or action (scored executed behavior)) and by whether it is *agentic*, which we define strictly: a resource is agentic if and only if it (i) executes multi-step, tool-using trajectories and (ii) scores a fairness quantity (some  $\Delta_c$ ) over them under demographic instrumentation. Both conditions matter, and the table's structure shows the field has split them between two non-communicating literatures.

**Table 2. Fairness evaluation resources and agent capability harnesses.**

Table 2. Fairness evaluation resources and agent capability harnesses.

| Benchmark / resource           | Fairness level           | Domain                            | Primary metric                            | Agentic?        | Code/Data released? |
|--------------------------------|--------------------------|-----------------------------------|-------------------------------------------|-----------------|---------------------|
| BBQ [96]                       | QA                       | General social bias               | Accuracy/bias score gap                   | No              | Yes                 |
| BOLD [31]                      | Generation               | Open-ended generation             | Sentiment/regard disparity                | No              | Yes                 |
| StereoSet [88]                 | QA/Gen                   | General                           | Stereotype score vs. LM score             | No              | Yes                 |
| CrowS-Pairs [87]               | Generation               | General                           | Paired pseudo-likelihood gap              | No              | Yes                 |
| HolisticBias [110]             | Generation               | General (600+ descriptors)        | Perplexity/regard disparity               | No              | Yes                 |
| DecodingTrust [130]            | QA/Gen                   | Trustworthiness (fairness subset) | Stereotype agreement; DP/EO gaps          | No              | Yes                 |
| TrustLLM [56]                  | QA/Gen                   | Trustworthiness                   | Fairness subscores                        | No              | Yes                 |
| TrustGPT [52]                  | Generation               | Toxicity/bias/value               | Toxicity and bias gaps                    | No              | n.r.                |
| HELM [68]                      | QA/Gen                   | Broad capability                  | Bias/fairness slice metrics               | No              | Yes                 |
| FairMedQA [145]                | QA                       | Clinical                          | Group response disparity                  | No              | n.r.                |
| MedEqualQA [49]                | QA                       | Clinical                          | Counterfactual consistency                | No              | n.r.                |
| mFARM [15]                     | QA                       | Clinical decision support         | Multi-faceted harm scores                 | No              | n.r.                |
| FMBench [134]                  | QA                       | Medical multi-modal               | Group performance parity                  | No              | n.r.                |
| Health-equity toolbox [100]    | QA/Gen                   | Clinical                          | Rubric-rated harm rates                   | No              | Yes                 |
| Accept-or-Deny [13]            | QA (decision)            | Lending                           | Approval-parity gap                       | No              | n.r.                |
| LLMs on Trial [58]             | QA (decision)            | Judicial                          | Verdict disparity                         | No              | n.r.                |
| Name-intervention audit [22]   | QA (decision)            | Clinical/judicial                 | Intervention consistency (flip rate)      | No              | n.r.                |
| CEB [131]                      | QA/Gen                   | General (compositional)           | Composite stereotype/toxicity/performance | No              | Yes                 |
| CAFFE [102]                    | QA/Gen                   | General                           | Multi-attribute flip metrics              | No              | n.r.                |
| SCOPE [104]                    | QA/Gen                   | General                           | Counterfactual flip rate                  | No              | n.r.                |
| LangFair [4]                   | QA/Gen                   | Use-case-specific toolkit         | Configurable group + counterfactual       | No              | Yes                 |
| FiSCo [146]                    | Generation               | General                           | Semantic group-difference statistic       | No              | n.r.                |
| Equal-Access audit [16]        | Generation               | Service quality                   | Response-quality disparity                | No              | n.r.                |
| Reward-model fairness [121]    | Upstream (reward)        | Alignment                         | Inter-group reward gap                    | No              | n.r.                |
| CFR/MASD deployment audit [85] | QA (agent output)        | Contact center                    | CFR, MASD                                 | Partial*        | No (proprietary)    |
| FairMT-Bench [42]              | Generation (multi-turn)  | Dialogue                          | Multi-turn bias rate                      | Partial*        | Yes                 |
| MALIBU [83]                    | Generation (multi-agent) | Multi-agent personas              | Persona bias scores                       | Partial*        | n.r.                |
| WebArena [154]                 | None                     | Web tasks                         | Task success rate                         | No <sup>†</sup> | Yes                 |

Table 2 (continued)

| Benchmark / resource | Fairness level | Domain                         | Primary metric        | Agentic?        | Code/Data released? |
|----------------------|----------------|--------------------------------|-----------------------|-----------------|---------------------|
| Mind2Web [32]        | None           | Web tasks                      | Step/element accuracy | No <sup>†</sup> | Yes                 |
| AgentBench [70]      | None           | Eight interactive environments | Task success rate     | No <sup>†</sup> | Yes                 |
| SWE-bench [60]       | None           | Software engineering           | Issue resolution rate | No <sup>†</sup> | Yes                 |

\* *Partial*: fairness-instrumented beyond a single turn (multi-turn dialogue, multi-agent output, or a deployed agent's scoring emission) but does not instrument tool calls, memory operations, plans, or delegations, i.e., estimates  $\Delta_r$  or a single  $\Delta_c$ , never the interior of the conduction equation. <sup>†</sup> Executes genuine multi-step, tool-using trajectories but contains no demographic instrumentation, so it cannot score fairness at any level. n.r. = public code/data release not verified at the time of writing. *Takeaway*: no row pairs fairness instrumentation with agentic execution (the *Agentic?* column contains no "yes") which is the structured, falsifiable form of the measurement gap predicted by Propositions P2 and P5.

**Three lineages, one boundary.** Read as a whole rather than row by row, the table contains three lineages that all halt at the same boundary. The *QA-bias lineage* (BBQ through HolisticBias) fixed the field's unit of analysis (a prompt, a response, a disparity statistic) before agents existed, and the *trustworthiness suites* (DecodingTrust, TrustLLM, TrustGPT, HELM) scaled that unit up in breadth without changing it: fairness is one slice of a prompt-response evaluation, never a property of executed behavior [56, 68, 130]. The *domain-decision tier* (lending, judicial, clinical) moved the content to consequential decisions but kept the form: a vignette in, a recommendation out, a parity statistic over recommendations [13, 58]. The *counterfactual tools* (CAFFE, SCOPE, LangFair, FiSCo, the audits of §4.2) refined the statistics (they are precisely the  $(\sigma, d_c)$  machinery §4.2 formalizes) but apply it at natural-language interfaces, not at trajectory interiors. The dataset survey of [155] confirms the pattern across a much larger inventory than ours, and the agent-evaluation surveys concede the complement: fairness appears as an aspirational dimension without trajectory-level metrics in [82], [168], and [7].

**Why clinical benchmarks stop at vignettes.** The clinical tier deserves a structural explanation rather than a complaint, because its authors are methodologically careful and the vignette boundary is not an oversight. A vignette is the unique unit at which three costly requirements are simultaneously cheap. First, *ground truth*: a static case has an expert-adjudicable answer, whereas a trajectory's quality is path-dependent. Judging whether an agent's third tool call was clinically appropriate requires re-adjudicating the state the agent itself created, at expert cost per step rather than per case (the rubric-and-adversarial design of [100] already strains this budget at the answer level). Second, *privacy and provenance*: real clinical trajectories are protected health information end to end; vignettes are shareable artifacts, which is why every released clinical resource in the table is vignette-formed. Third, *counterfactual control*: in a static case, the demographic swap  $\sigma$  is a text edit; in an interactive environment, the swap must propagate consistently through every tool response, retrieved record, and intermediate observation, or the counterfactual pair silently decoheres. That is exactly the instrumented-environment engineering that no fairness benchmark has built. There is also an inherited-pattern effect: clinical LLM evaluation descends from medical-exam QA and the BBQ-style bias template, so the question, not the episode of care, became the unit of measurement. The cost of the boundary is stated by Proposition P2: a model certified near-parity on vignette QA can still triage, escalate, and order differently across demographic counterfactuals once it acts,

because the certified emission and the harmful emissions are different  $z_c$ 's. The label-choice lesson of [92] indicates how this ends if uncorrected: the proxy that is cheapest to measure becomes the de facto fairness standard, while the disparity migrates to the unmeasured variable.

**Why the agent harnesses are not demographically instrumented.** The four capability harnesses at the foot of Table 2 prove that trajectory-executing evaluation infrastructure exists (reproducible environments, deterministic resets, full action logging [32, 60, 70, 154]), which makes their fairness silence diagnostic rather than incidental. Three design commitments explain it. Their environments are *demographically sparse by construction*: self-hosted websites, shells, databases, and GitHub issues contain few persons at all, so there is no protected attribute to swap. Their success metrics are *functional and scalar*: task completion and issue resolution are demographic-blind by definition, and nothing in the harness records who was affected by an action, only whether the action achieved the goal. Their tasks are *non-allocative*: even a perfectly instrumented demographic swap would find no consequential decision to flip in "book the cheapest flight" or "fix this failing test." Retrofitting them is therefore not a matter of adding a metric but of adding *paired worlds*: environment states that are identical under  $\sigma$  in every record, page, and tool response, plus seeded determinism so that trace-replay attributes divergence to the swap rather than to sampling. Those are the engineering requirements of §4.2's protocol. In P5's terms, the capability harnesses achieve execution adequacy without measurement adequacy, while every fairness resource above them achieves the reverse. The two halves of the needed benchmark exist, fully built, in literatures that do not cite each other.

**The conclusion, stated through the framework.** The Agentic? column of Table 2 contains no "yes." This is not an accident of sampling. The inventory spans the QA-bias lineage, the trustworthiness suites, the clinical and decision tiers, the counterfactual tools, the multi-turn and multi-agent partials, and the capability harnesses: it is the structural consequence of Propositions P2 and P5. By P2 (Masking), endpoint parity on any QA- or generation-level resource is consistent with arbitrarily large offsetting component disparities, so every "No" row can certify as fair an agent whose interior  $\Delta_c$  terms are large. By P5 (Measurement adequacy), certification of an acting agent requires bounding each  $\Delta_c$  with nonzero conduction weight, which demands exactly the action-level instruments of §4.3 running inside exactly the paired-world environments the capability harnesses almost provide. Read jointly with the coverage matrix of §7 (Figure 4) (where counterfactual fairness is the named framing in exactly one component-tagged corpus paper), the gap is two-sided: the metric dimension agents most need is the least applied, and the benchmark layer that would apply it does not exist. The audit protocol of §6 is specified to demonstrate, on a small scale, that this instrumentation is buildable with current tooling; the benchmark agenda of §8.1 specifies what a full-scale version requires.

## 0.5 How to Mitigate: Proposition P4 and the Matching Principle

The Bias Conduction Framework articulates **Proposition P4 (Mitigation Matching)**: an intervention reduces trajectory-level disparity  $\Delta_\tau$  only if it acts *at the entry locus or downstream on its conduction path* **and** on the active conduction operator  $\varphi$  at that locus. When an intervention operates upstream of the entry locus, the disparity it targets has not yet entered the pipeline, so there is nothing for it to attenuate (the prompting ceiling of §5.1); an intervention placed off the conduction path likewise leaves the operator free to conduct disparity forward into subsequent pipeline stages. When it acts at the wrong operator type (e.g., targeting ATTENUATE dynamics with an intervention calibrated for AMPLIFY), it may be ineffective or, in the case of feedback-recurrence paths, counter-productive. This section surveys the mitigation literature through the P4 lens, asks at which component and operator type each intervention acts, and identifies the gaps

that remain. Table 3 formalizes the mapping. Table 4 closes the §3→§5→§8 loop by specifying, for each component, the coverage of existing mitigation evidence and the critical open gap.

Three preliminary observations structure the review. First, the overwhelming majority of published interventions act on the base language model’s token distribution (prompting, alignment, guardrails, inference-time steering) and are therefore upstream of the agentic components (tool selection, memory, delegation, planning, personalization, drift) where Section 3 locates most of the harm. P4 predicts this mismatch systematically: a model-level debiasing that reduces group disparity in a QA setting neither reaches the tool-selector’s discrete routing policy nor the retrieval ranker’s document distribution. Second, the few interventions that do act at the agent level (multi-agent debiasing, RAG-layer controls, memory hygiene, identity anonymization) are recent, sparse, and evaluated in narrow settings; the (AMPLIFY, GENERATE) operator cells of Table 3 are empty by evidence, not by design. Third, compound pipeline architectures create a constraint that is easy to underestimate: even a component-matched intervention at locus  $c$  can fail if an upstream GENERATE operator has already inserted disparity on zero input, since the mitigation target did not exist at the pre-entry stage. P4 therefore calls for matching locus and operator and for auditing the full conduction chain to locate the *first* non-zero  $\Delta_c$ . The intervention must reach that earliest point.

The token-level debiasing literature (intrinsic LLM debiasing at the embedding, pre-training, or decoding level) is the scope of the parent survey [24] and is treated here in a single compressed paragraph. The agentic contribution of this section is the mapping to operator type, component locus, and pipeline stage, not a rehearsal of the QA-era literature.

**0.5.1 Prompting and In-Context Debiasing.** Prompting interventions work by restructuring the information presented to the model at inference time, without modifying weights. Under the BCF, they act at the *input pre-processing stage*, prior to all six agent components, and are best characterized as ATTENUATE operators: they reduce the salience of protected-attribute signals in the model’s context before the first agentic decision is made. Two causal prompting approaches from the corpus share this structure and permit useful comparative analysis; a third extends the approach to example selection.

[72] and [152] both operationalize front-door adjustment at the prompt level, structuring the chain of reasoning to block spurious demographic pathways from reaching the model’s causal graph. [43] achieves ATTENUATION through a different mechanism: multi-step deliberative templates that force explicit reasoning before a demographic-adjacent judgment. The key empirical difference between these approaches is where on the reasoning trace the attenuation is applied. Causal prompting blocks the path at the *input encoding* step, before chain-of-thought begins; structured deliberation (“thinking fair and slow”) blocks it at an *intermediate reasoning gate*, tolerating early demographic activation but redirecting it before commitment to an action. The [43] finding that structured prompts outperform zero-shot fairness instructions is consistent with the BCF prediction: a flat instruction competes with the model’s learned associations, whereas a multi-step structure enforces ATTENUATION at a reasoning node that precedes the action commit. [37] extends the causal approach into few-shot example selection, choosing in-context demonstrations by causal influence score to block demographic confounds, which attenuates bias *in the information selected* rather than in the prompt structure itself.

Despite their accessibility, prompting interventions have a structural ceiling defined by P4. They act upstream of every agent component, but only on the PRESERVE/ATTENUATE operators of the base model’s reasoning; they cannot suppress the AMPLIFY operator introduced when a tool registry presents demographically unequal options (§3.1) or the GENERATE operator active when a retrieval index returns a biased document set for a demographically marked query (§3.2). An agent

that reformulates the system prompt between turns (a common architecture pattern in ReAct-style loops [148]) further degrades prompt-level guarantees because each reformulation resets the context structure, potentially re-introducing the blocked demographic pathway. Prompting is therefore a viable first layer, deployable with no infrastructure change, but P4 marks it as insufficient at any component locus downstream of the model’s initial token prediction.

**0.5.2 Fine-Tuning and Alignment.** Training-time interventions shape the model’s weight distribution to reduce the propensity to generate demographically differentiated outputs, targeting the PRESERVE→ATTENUATE transformation at the model level. A comparative reading of the alignment literature reveals a structural progression. Standard RLHF [93] is the enabling foundation, but it does not target fairness; it targets human-preference alignment, which [144] shows can itself *generate* minority-group disparity through preference collapse, a mechanism where majority-group preferences dominate the reward signal and the model drifts toward AMPLIFY behavior for minority demographics. MaxMin-RLHF [26] directly corrects this by replacing the maximization objective with a maximin criterion, ensuring the worst-off preference subgroup is optimized. That fairness-at-training-time intervention acts on the same AMPLIFY-by-preference-collapse pathway that standard RLHF inadvertently activates. BiasDPO [11] applies the Direct Preference Optimization alternative to RLHF, constructing biased/debiased preference pairs to steer away from discriminatory completions. Reasoning-guided fine-tuning [66] incorporates fairness-aware chain-of-thought supervision so that the model learns to reason through demographic parity as part of task execution, an approach particularly relevant for planning-stage (§3.4) decomposition tasks where an agent allocates subtasks with differential treatment.

Constitutional AI [19] and its collective variant [53] occupy a distinct tier within alignment: rather than optimizing against a reward model that encodes human preferences, they install a set of normative principles against which the model self-critiques. The fairness implication is significant. Standard RLHF can encode the biases of its annotator pool; a constitution-governed model can articulate fairness constraints explicitly, providing a mechanism for ATTENUATION of group-differential behavior that does not depend on the demographic representativeness of preference data. [53] strengthens this by deriving the constitution through public deliberation, which reduces the probability that the normative constraints themselves encode majority-group values.

The shared limitation of all training-time methods under P4 is that they act on the model’s weight distribution, not on the agent’s action-selection architecture. Once the aligned base model is embedded in a tool-using or multi-agent pipeline, the alignment signal does not propagate to: (a) the tool-selection policy, which is a discrete routing choice conditioning on tool metadata that the reward model never observed; (b) the retrieval-stage embedding that ranks candidate documents; or (c) the role-assignment protocol in a multi-agent orchestrator. The aggregate reward signal, summed across a full trajectory, cannot distinguish between a bias introduced at the planning step and one introduced at tool invocation; targeted correction of specific pipeline loci is invisible at training time. P1’s component-locality principle captures this precisely: a weight-level intervention has  $\Delta_c = 0$  at every non-model locus.

**0.5.3 Guardrails and Filters.** Guardrails interpose a protective layer between agent outputs and their consumers, operating at the inference-time boundary between components. [36] provides the systematic taxonomy of this design space, distinguishing input filters (blocking demographic-sensitive inputs before they reach the model), output filters (screening completions post-generation), and architectural wrappers (modular safety classifiers surrounding the core model). For agent deployment, the natural extension is *action-level guardrails*: before an agent executes a tool call or commits a delegation decision, a classifier audits the proposed action for discriminatory routing.

Under the BCF, guardrails act as post-hoc ATTENUATE operators: they do not reduce the upstream disparity generation but intercept it before the output commits. Their P4 status depends critically on *where in the pipeline* they are inserted. A guardrail at the final output stage satisfies neither the "at or upstream of the entry locus" requirement nor the operator-matching requirement for components like retrieval (where the AMPLIFY/GENERATE operator is active *inside* the retrieval step, not at its output). Guardrails become more P4-compliant when inserted between pipeline components (at the tool-invocation boundary, at the delegation boundary, at memory write) rather than only at the response boundary. In principle, this component-interposed architecture is achievable: the same classifier logic that [36] applies to conversational outputs can be applied to action proposals. In practice, no corpus paper reports a guardrail specifically trained on action-level fairness signals; all guardrail deployments in the agent fairness literature re-use classifiers trained on natural-language toxicity or refusal tasks, which are neither calibrated to demographic disparity in tool selection nor able to detect GENERATE-type operator effects (disparity created from zero input when a tool produces a biased output).

The measurement gap documented in §4 is directly implicated here: without action-level fairness training data, training action-level guardrails is not tractable. The guardrail tier is thus a plausible P4-compliant architecture whose evidence base is at present thin.

**0.5.4 Multi-Agent Debiasing.** Multi-agent architectures introduce distinct bias pathways (§3.3), but they also create structural opportunities for peer-level bias correction that single-agent systems cannot exploit. The multi-agent debiasing literature is the only mitigation family in Table 3 that acts natively at the agent-architecture level, operating on the delegation protocol and identity-signaling structure rather than on the base model's token distribution.

Two parallel design philosophies emerge from comparative analysis. The first, *diversity ensemble*, assigns multiple agents to generate responses independently, then suppresses idiosyncratic biases through aggregation. [94] instantiates this as a general multi-LLM debiasing framework; [143] instantiates it as a multi-objective optimization in which specialized agents maximize distinct fairness criteria and a coordinator agent reconciles them. Both approaches convert the AMPLIFY operator in multi-agent interaction into an ATTENUATE operator through deliberate heterogeneity: the collective is less biased than any member because divergent biases cancel in the aggregation step. The second philosophy, *identity neutralization*, acts directly on the GENERATE pathway that produces bias from role and identity signals. [28] demonstrates this cleanly: anonymizing agent identity cues in a multi-agent debate context reduces identity-driven bias without eliminating the epistemic value of role differentiation. This is a direct P4-compliant intervention: the GENERATE operator that creates disparity from demographic role signals is deactivated at the locus where it acts (the identity-conditioning step of the debate protocol), rather than downstream. [65] extends the debate approach to cultural fairness: structured adversarial debate surfaces and corrects cultural stereotypes more effectively than single-agent self-critique, with the multi-agent argumentative dynamic functioning as an ATTENUATION operator on culturally skewed priors.

At the implicit interaction level, [3] addresses AMPLIFY dynamics that arise through accumulated interaction history. A detection-and-mitigation pipeline analyzes multi-agent interaction logs to identify bias propagation patterns before they crystallize into persistent behavioral tendencies. This temporal framing is P4-relevant because it targets the feedback-recurrence path of the BCF equation (the loop  $\Delta^{(t+1)} = \varphi_{fb} \cdot \Delta^{(t)} + \Delta_{entry}$ ) by interrupting the recurrence before the accumulated disparity exceeds a threshold.

The collective limitation of multi-agent debiasing is architectural: these interventions require a multi-agent infrastructure to function. They are not portable to single-agent settings, and their efficacy is contingent on the specific topology and communication protocol of the multi-agent

system. Crucially, none of the corpus papers provides a general mapping from the multi-agent communication graph structure to the magnitude of the resulting AMPLIFY operator. The conduction factor  $\varphi$  at the delegation locus remains empirically unbounded.

**0.5.5 Inference-Time Steering.** Inference-time steering operates on the model’s internal representations during the forward pass, without modifying weights and without requiring multi-agent infrastructure. This makes it structurally distinct from alignment (which modifies weights) and from prompting (which acts on the input). [74] operationalizes this as FairSteer, which identifies fairness-relevant directions in the model’s activation space and applies dynamic activation steering to suppress those directions at inference time.

Under the BCF, FairSteer is a PRESERVE→ATTENUATE operator acting on the model’s internal representation layer. Its key architectural advantage over prompting is that it operates inside the forward pass rather than on the input conditioning, making it more reliable against prompt reformulation and more precisely targeted at the representation subspace encoding demographic associations. For agent deployment, activation steering can in principle be conditioned on agent-state signals (the current tool call context, the retrieved document set, the assigned role), which would enable P4-compliant component-level targeting: suppress fairness-relevant activations specifically when the agent is in a tool-selection or delegation state, rather than uniformly throughout the trajectory.

This state-conditional extension has not been demonstrated in the corpus, but it represents a plausible path toward inference-time interventions that match the component architecture P4 demands. The current limitation of FairSteer is shared with other representation-level methods: it acts on the base model’s encoding and does not reach the discrete routing logic of tool selection, the ranking function of retrieval, or the delegation protocol of a multi-agent orchestrator.

**0.5.6 Process-Level, Architectural, and RAG-Level Interventions.** The most architecturally integrated tier of mitigation embeds fairness enforcement directly into the agent’s operational components (the retrieval layer, the memory module, the planning trace) rather than applying it as an external correction over the model’s outputs. These interventions are the most P4-compliant in the corpus, and they are also the most sparsely evidenced.

**RAG-layer interventions** address the retrieval component (§3.2), where the AMPLIFY and GENERATE operators are active on the information that conditions all downstream agentic reasoning. [160] establishes the theoretical and empirical case that ranking fairness in the retrieval stage propagates directly into downstream generation quality across demographic groups, a component-matched diagnosis that names the entry locus precisely. [64] translates this into a practical P4-compliant intervention: by acting on the *embedding model* rather than on the ranker or the generator, the approach corrects demographic skew at the source representation layer, reducing disparity before it enters the ranking pipeline. This is the BCF’s strongest existing example of locus-matching in the retrieval component. The intervention acts at the GENERATE locus (where the embedding model encodes the demographic signal into the retrieval key) rather than at a downstream symptom. The limitation is that the embedding-level correction addresses demographic skew in the representation of queries and documents but does not prevent adversarial amplification through knowledge-store poisoning [138], which operates at a different point in the pipeline.

**Memory-level hygiene** addresses §3.6’s long-horizon drift by intervening on the agent’s persistent state. [86] characterizes how implicit biases accumulate across LLM long-term memory and proposes memory audit and selective forgetting mechanisms. Under the BCF, accumulated memory disparity is a feedback-recurrence path: each turn’s  $\Delta_c$  is written into memory and re-read as  $\Delta_{\text{entry}}$  for the next turn, creating the compounding effect P3 (Super-additivity) names. Memory hygiene (auditing which inferences are stored, which are forgotten) acts to reduce  $\varphi_{\text{fb}}$  in



the recurrence equation, lowering the amplification factor over the trajectory. No corpus paper has demonstrated memory hygiene at scale in an agentic deployment, making this an architecturally motivated but empirically nascent intervention.

**Bias-aware retrieval and source selection** are addressed by [117] and [118], which frame the source-selection decision (which knowledge sources to query) as a fairness-critical routing choice analogous to tool selection in §3.1. [118] proposes a bias mitigation *agent* dedicated to optimizing source selection for balanced knowledge retrieval across demographic groups, a design pattern that inserts a specialized fairness-enforcement agent into the tool-use pathway as a gatekeeper. This is architecturally P4-compliant: the intervention acts at the entry locus (retrieval source selection) on the AMPLIFY operator (the source routing that conditions what the agent retrieves).

**Planning and decision-level fairness rewards** address §3.4 directly. [57] trains a Fairness Reward Model that scores *complete* reasoning chains against equalized opportunity and equalized odds criteria and reweights those chains during post-generation aggregation, leaving the internal reasoning untouched; this functions as an aggregation-stage ATTENUATE edge on the planning locus’s output rather than an in-trace guardrail. The approach is significant because it is the only corpus paper that scores whole reasoning chains for fairness rather than only the final decision, approximating D2’s  $\Delta_c$  measurement at the planning locus from outside the trace.

**Summary assessment under P4.** Across all five subsections, the dominant empirical picture is asymmetric: the ATTENUATE and PRESERVE operators at the model level are densely populated with mitigation evidence, while the AMPLIFY and GENERATE operator cells (which correspond to the agentic failure modes: retrieval poisoning, role-induced bias generation, delegation amplification, feedback drift) are either thinly covered or empty. No published intervention jointly addresses tool selection bias and long-horizon drift within a unified architectural framework. Building such a framework, one that treats fairness as an agent-level property enforced across the full component stack, is the central open problem of §8.6.

**Table 3: Mitigation Map: Entry Locus × Conduction Operator × Pipeline Stage × Evidence**

*Takeaway: Mitigation evidence is concentrated in the model-level ATTENUATE column; the (AMPLIFY, GENERATE) × agent-component cells are empty, marking the gap between where interventions act and where agentic harm originates.*

Table 3: Mitigation Map: Entry Locus × Conduction Operator × Pipeline Stage × Evidence

| Entry Locus                  | Conduction Operator Targeted | Pipeline Stage            | Intervention Class                | Representative Evidence | Evidence Density |
|------------------------------|------------------------------|---------------------------|-----------------------------------|-------------------------|------------------|
| Model / input pre-processing | PRESERVE → ATTENUATE         | Pre-inference (prompt)    | Causal prompting                  | [72], [152], [43]       | High             |
| Model / input pre-processing | PRESERVE → ATTENUATE         | Pre-inference (few-shot)  | Causal active learning            | [37]                    | Medium           |
| Model (weights)              | PRESERVE → ATTENUATE         | Training (RLHF)           | Alignment / MaxMin-RLHF           | [93], [26], [144]       | High             |
| Model (weights)              | PRESERVE → ATTENUATE         | Training (DPO)            | BiasDPO / preference optimization | [11]                    | Medium           |
| Model (weights)              | PRESERVE → ATTENUATE         | Training (SFT + CoT)      | Reasoning-guided fine-tuning      | [66]                    | Medium           |
| Model (weights)              | PRESERVE → ATTENUATE         | Training (constitutional) | Constitutional / Collective AI    | [19], [53]              | Medium           |
| Model (activations)          | PRESERVE → ATTENUATE         | Inference (activation)    | FairSteer activation steering     | [74]                    | Low              |
| All output boundaries        | ATTENUATE (post-hoc filter)  | Inference (output gate)   | Guardrails                        | [36]                    | Low              |

Table 3 (continued)

| Entry Locus                                        | Conduction Operator Targeted             | Pipeline Stage              | Intervention Class                                                                       | Representative Evidence | Evidence Density |
|----------------------------------------------------|------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------|-------------------------|------------------|
| Tool / API Selection (§3.1)                        | ATTENUATE (source selection)             | Retrieval routing           | Bias-aware retrieval agent                                                               | [117], [118]            | Low              |
| Memory & Retrieval (§3.2)                          | GENERATE → ATTENUATE (embedder)          | Retrieval embedding         | RAG fairness / embedder control                                                          | [160], [64]             | Low              |
| Memory & Retrieval (§3.2)                          | AMPLIFY → ATTENUATE (memory audit)       | Long-term memory            | Memory hygiene / selective forgetting                                                    | [86]                    | Very Low         |
| Multi-agent Delegation (§3.3)                      | AMPLIFY → ATTENUATE (ensemble)           | Agent coordination          | Multi-LLM debiasing ensemble                                                             | [94], [143]             | Low              |
| Multi-agent Delegation (§3.3)                      | GENERATE → ATTENUATE (identity)          | Agent debate protocol       | Identity anonymization                                                                   | [28]                    | Low              |
| Multi-agent Delegation (§3.3)                      | AMPLIFY → ATTENUATE (debate)             | Agent debate protocol       | Cultural debate debiasing                                                                | [65]                    | Low              |
| Multi-agent Delegation (§3.3)                      | AMPLIFY → ATTENUATE (recurrence)         | Interaction history         | Implicit bias detection pipeline                                                         | [3]                     | Low              |
| Planning / Decomposition (§3.4)                    | PRESERVE → ATTENUATE (chain reweighting) | Post-generation aggregation | Fairness Reward Model reweighting complete reasoning chains (equalized opportunity/odds) | [57]                    | Very Low         |
| Planning / Decomposition (§3.4)                    | PRESERVE → ATTENUATE (reasoning)         | Fine-tuning trace           | Reasoning-guided SFT                                                                     | [66]                    | Very Low         |
| Tool / API Selection: AMPLIFY                      | n/a                                      | n/a                         | (No published intervention)                                                              | n/a                     | Empty            |
| Tool / API Selection: GENERATE                     | n/a                                      | n/a                         | (No published intervention)                                                              | n/a                     | Empty            |
| Memory & Retrieval: AMPLIFY (poisoning)            | n/a                                      | n/a                         | (No published intervention)                                                              | n/a                     | Empty            |
| User Modeling / Personalization (§3.5)             | n/a                                      | n/a                         | (No published intervention at locus)                                                     | n/a                     | Empty            |
| Long-horizon Drift (§3.6): AMPLIFY (feedback loop) | n/a                                      | n/a                         | (No published intervention at locus)                                                     | n/a                     | Empty            |

**Table 4: Challenges → Solutions → Gaps: Closing the §3 → §5 → §8 Loop**

*Takeaway: Five of the six agent components have partial or no mitigation coverage; the most consequential operator types (AMPLIFY and GENERATE at agent loci) are the least addressed, and every component's open gap maps directly onto a §8 research agenda item.*

Table 4: Challenges → Solutions → Gaps: Closing the §3 → §5 → §8 Loop

| Component (§3.x)                        | Primary Harm (BCF operator)                                                                                                                                                  | Mitigation Evidence                                                                                                           | Coverage                                                                                                                                            | Critical Open Gap (→ §8)                                                                                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tool / API Selection</b> (§3.1)      | Discriminatory routing via option-order sensitivity, metadata-driven selection skew; GENERATE on zero user-level input (disparity on zero input is GENERATE by D3) [21, 120] | Source-selection bias mitigation agents [117, 118]; bias-aware retrieval routing                                              | <b>Partial</b> (selection fairness addressed at source level; no intervention on discrete tool-routing policy during execution)                     | No action-level metric or training signal for tool-invocation disparity across demographic counterfactuals; AMPLIFY/GENERATE cells empty (→ §8.1, §8.6)                               |
| <b>Memory &amp; Retrieval</b> (§3.2)    | RAG injects demographic skew even for fairness-vigilant users; GENERATE at embedding and AMPLIFY over multi-turn history [42, 55, 138]                                       | Embedder-level bias control [64]; fair RAG ranking [160]; memory audit [86]                                                   | <b>Partial</b> (embedder and ranking addressed; knowledge-store poisoning and multi-turn accumulation unaddressed)                                  | No joint intervention on embedding-level GENERATE and history-level AMPLIFY within a unified RAG architecture; no longitudinal retrieval fairness benchmark (→ §8.1, §8.2)            |
| <b>Multi-agent Delegation</b> (§3.3)    | Emergent and amplified bias from role assignment and agent interaction; AMPLIFY at interaction edges, GENERATE at identity-conditioned role seeding [81, 91]                 | Ensemble debiasing [94, 143]; identity anonymization [28]; cultural debate [65]; implicit bias monitoring [3]                 | <b>Partial</b> (several targeted interventions; none covers arbitrary topologies or quantifies the conduction factor $\varphi$ at delegation edges) | No metric for "assignment disparity" across demographic counterfactuals; amplification factor unbounded; no intervention on delegation-graph topology as a fairness variable (→ §8.3) |
| <b>Planning / De-composition</b> (§3.4) | Reasoning-stage demographic bias; plan steps, resource allocation, and task assignments skewed; PRESERVE or AMPLIFY within chain-of-thought [73, 103, 139]                   | Fairness-Reward-Model reweighting of complete reasoning chains at post-generation aggregation [57]; reasoning-guided SFT [66] | <b>Partial</b> (aggregation-stage chain reweighting and reasoning-supervised training demonstrated; neither tested in tool-using agentic loop)      | No benchmark that scores the <i>plan itself</i> for demographic skew independently of the final decision; plan-level disparity metric absent (→ §8.1)                                 |

Table 4 (continued)

| Component (§3.x)                              | Primary Harm (BCF operator)                                                                                                                                                 | Mitigation Evidence                                                                                                                  | Coverage                                                                                                       | Critical Open Gap (→ §8)                                                                                                                                                                                                                      |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User Modeling / Personalization</b> (§3.5) | Stereotype activation via inferred identity (names, dialect, usage history); differential quality across demographic groups; GENERATE from inferred attribute [63, 84, 101] | Reward-model dialect-bias mitigation [84] (indirect); no direct user-modeling-locus interventions                                    | <b>None</b> (no corpus paper intervenes at the agent’s user-modeling or personalization component)             | No mechanism to detect when an agent’s inferred user model encodes a protected attribute; no intervention on persistent cross-session user representation; cue-channel inconsistency [124] unaddressed in any debiasing design (→ §8.1, §8.4) |
| <b>Long-horizon Drift</b> (§3.6)              | Bias accumulation over trajectories via sycophancy and feedback loops; AMPLIFY in recurrence equation; $\varphi_b > 1$ drives compounding [76, 86, 133]                     | Memory hygiene [86] (partial, not validated at scale); sequential fairness theory [142, 158] (formal, not yet applied to LLM agents) | <b>None</b> (no deployed mitigation that enforces a long-term fairness criterion over an LLM agent trajectory) | No LLM agent evaluated against a formal long-term fairness criterion (equal long-term benefit rate, etc.); sycophancy-driven drift measurable [59] but not mitigated in agentic settings (→ §8.2)                                             |

## 0.6 Empirical Study: A Cross-Framework Bias Audit: Protocol and Design

The taxonomy of §3 argues that fairness in LLM-based agents is a property of pipeline components and their actions. The Bias Conduction Framework (BCF) formalizes this: P2 (Masking) holds that component-level disparity  $\Delta_c$  may be invisible to metrics restricted to the output node; P3 (Super-additivity) holds that trajectory disparity  $\Delta_\tau \approx \Sigma_c \Delta_c \cdot \Pi \varphi_e$  can exceed the sum of per-component disparities when conduction operators  $\varphi_e > 1$  along active pipeline edges. This section registers a **cross-framework bias audit** that serves as the BCF’s primary falsification instrument: the same consequential-decision task is routed through four agent configurations (C0 to C3) spanning the BCF’s conduction-operator space, and disparity is measured simultaneously at the answer level and at the action level.

**The protocol is fully specified below; Section 6.5 reports a small pilot run. The pilot is directional only, with a small sample and deliberately narrow scope, and no claim in this manuscript outside §6.5 rests on its results.**

**0.6.1 Experimental Setup. Task and domain.** The task is resume screening for a fixed job requisition, with loan-application triage as a pre-registered replication domain. Both are chosen because non-agentic LLM disparities are already documented in the corpus, providing a calibrated C0 baseline: hiring disparities are established by [90], [126], [135], and [14]; lending disparities by [20], [41], and [13]. Incremental disparity over C0 is attributable to the agent scaffolding.

**Profiles and perturbation.** All profiles are fully synthetic (N = 10 profiles; no IRB required). Base profiles hold qualifications fixed; counterfactual variants substitute only the demographic signal, primarily via candidate names following [90] and [101], and secondarily via explicit demographic cue labels. Because cue format is a measurement confound [124], the cue channel is varied as a controlled factor.

**Agent configurations.** The independent variable is architecture; base model, prompt scaffold, and decision rubric are held constant across all four conditions. Models are version-pinned and temperature is fixed at zero. These configurations C0 to C3 are the audit’s independent variable and are distinct from the §3.8 worked-example capability rungs of the same name.

| Config | Architecture                      | BCF conduction profile                          | Primary proposition            |
|--------|-----------------------------------|-------------------------------------------------|--------------------------------|
| C0     | Single-call LLM, no tools         | No pipeline edges active (baseline)             | P2 Masking: output-only gap    |
| C1     | Single-LLM with tool use [113]    | $\varphi$ active at tool-selection edge         | P1 Locality: edge isolable     |
| C2     | ReAct reasoning-and-acting [148]  | $\varphi$ cumulative over reasoning + retrieval | P3 Super-additivity: two-edge  |
| C3     | Two-agent delegation/debate [129] | $\varphi$ at delegation edge; GENERATE possible | P3 + P4: multiplier + matching |

*Takeaway: The C0 to C3 ladder directly operationalizes BCF conduction-operator profiles; each configuration adds exactly one class of pipeline edge, enabling controlled attribution of disparity increments.*

**0.6.2 Metrics. Answer-level disparity.** The **Counterfactual Flip Rate (CFR)** and **Mean Absolute Score Difference (MASD)** of [85] are computed at the output node, corresponding to D2’s  $\Delta_c$  restricted to the final verdict. Reporting both a binary-flip and a continuous-magnitude statistic follows the intervention-consistency design of [22].

**Action-level disparity.** For C1 to C3, three  $\Delta_c$  quantities are computed at intermediate pipeline edges:

- *Tool-invocation disparity* ( $\Delta_c$  at tool-selection): differential distribution over invoked tools across counterfactual conditions, extending the selection-skew measurement of [21] to the demographic-conditioned setting studied by [147].
- *Decision-action disparity* ( $\Delta_c$  at planning/output): differential advance/reject or approve/deny rates, per [73].
- *Escalation/deferral disparity* ( $\Delta_c$  at routing): differential rates of human escalation or information-request, a channel where P2 Masking operates at the routing level.

All six quantities are reported on a comparable counterfactual-pair footing in a single results table, making the P2-Masking gap directly quantifiable.

**0.6.3 Hypotheses. H1: P2 Masking is real.** In at least one configuration, answer-level CFR/MASD is small or statistically indistinguishable from C0 while action-level  $\Delta_c$  is materially larger [73].

**H2: Amplification ordering (a proxy for P3, since the pilot does not estimate the component sum).** Total trajectory disparity is non-decreasing along  $C0 \preceq C1 \preceq C2 \preceq C3$ , with C3 exhibiting the largest magnitude, consistent with multi-agent amplification [29, 81, 91]. A registered counter-hypothesis: C3 debate can reduce disparity via anonymization [28] or ensemble debiasing [94]; the sign of the C3 effect is an open empirical question.

Statistical treatment: paired contrasts, 95% confidence intervals, Bonferroni correction across the configuration  $\times$  cue-channel grid; null results reported as such.

**0.6.4 The Role of This Audit.** This study is deliberately modest in scale. It serves three purposes. First, it provides a **constructive existence proof** of P2 Masking and P3 Super-additivity as empirically measurable phenomena. Second, it establishes a **methodological template** (pinned

models, synthetic counterfactuals, paired answer/action metrics, BCF-labeled configurations) that is portable across domains. Third, it serves as a **design seed** for the acting-agent benchmark called for in §8.1. FairMedAgent, a forthcoming companion benchmark for clinical acting agents evaluated under matched demographic counterfactuals, is one scaled realization of this instrumentation; its development is independent of this survey’s claims.

*0.6.5 Results: A Pilot Run.* We report a *pilot* instantiation of the protocol, deliberately small and explicitly bounded in what it can support. The model under test is Claude Haiku 4.5 at temperature 0; the task is resume screening over the 10 synthetic profiles of §6.1; the counterfactual is a single name pair, a White-male-coded name and a Black-female-coded name in the lineage of [126], applied to identical resume content; and three configurations are run: C0 (single call), C2 (reason-then-decide), and C3 (a two-role recruiter-reviewer deliberation within one context). This pilot instantiates the answer-level statistics (CFR, MASD) and the decision-action disparity; the tool-invocation and escalation metrics of §6.2, which require a tool-using deployment, are left to the full study.

| Configuration            | n pairs | CFR  | MASD (0-100 score) | Advance-rate disparity |
|--------------------------|---------|------|--------------------|------------------------|
| C0 single call           | 10      | 0.00 | 1.2                | 0.00                   |
| C2 reason-then-decide    | 10      | 0.00 | 3.4                | 0.00                   |
| C3 two-role deliberation | 10      | 0.00 | 6.4                | 0.00                   |

*Takeaway: decision-level parity (CFR and advance-rate disparity both 0) coexists with a score-level disparity (MASD) that grows monotonically with scaffold complexity, from 1.2 at C0 to 6.4 at C3.*

Two patterns appear, both consistent with the framework and both heavily qualified below. First, decision-level parity masks score-level disparity: no candidate’s advance-or-reject decision flips under the demographic swap (CFR = 0 in every configuration) and the advance-rate disparity is zero, yet the mean absolute score difference is nonzero throughout. P2 (Masking) describes exactly this configuration, an endpoint-level analogue of P2: a coarse decision statistic masks a finer score-level disparity at the same output node; demonstrating the interior  $\Delta_c$  awaits the full study. Second, the score-level disparity rises with agentic scaffolding, from 1.2 (C0) to 3.4 (C2) to 6.4 (C3), the direction predicted by P3 (Super-additivity) and stated in advance in §6.3, with the two-agent deliberation showing the largest disparity.

**Caveats: this is a directional pilot, not a confirmatory study.** It uses  $N = 10$  profiles with a single counterfactual name pair that conflates race and gender; one model at one temperature; and uniformly strong resumes, which produce a ceiling effect on the binary decision (every candidate advances), so CFR cannot register and the whole signal lives in the score. The MASD magnitudes are small in absolute terms (1 to 6 points on a 0-100 scale) and carry no confidence intervals or significance tests. The monotone MASD trend across C0, C2, and C3 is suggestive of the masking-and-amplification mechanism the BCF predicts, but it rests on one small sample and could reflect score-scale noise rather than a stable effect. A confirmatory study is required before H1 or H2 can be claimed as established: larger  $N$ ; multiple name pairs that decouple race, gender, and other attributes; borderline as well as strong profiles to break the decision-level ceiling; tool-using configurations that expose tool-invocation and escalation disparity; and paired statistical tests with corrected confidence intervals. The pilot also instantiates only a restricted slice of the protocol’s configurations: its C2 covers only the reasoning edge (reason-then-decide with no retrieval or tools), and its C3 is a single-context recruiter-reviewer proxy for delegation rather than a true

two-agent edge with a  $\pi_{\text{ctrl}}$  controller; a full two-agent configuration is future work. The harness, synthetic profiles, and raw decisions will be released with the camera-ready.

## 0.7 Coverage Map and Cross-Analysis

This section projects the 202-paper corpus onto its two organizing axes simultaneously: the six agent-pipeline components and the four fairness dimensions (group, individual, counterfactual, and unspecified). The projection is Figure 4 (the coverage matrix) and its companion, Figure 5 (the locus  $\times$  BCF-operator grid). Read together, they convert the survey's central claim (that the field is well-instrumented where measurement is easy and under-instrumented where agentic deployment makes fairness consequential) from an assertion into a structured, falsifiable coordinate.

**Tagging convention.** The corpus comprises **202 unique papers**. Of these, **59 papers** are tagged to at least one of the six agent-pipeline component branches in `corpus_master.md`. Because a single paper may address more than one component (e.g., a study of RAG-induced bias in a multi-agent pipeline is tagged to both memory-retrieval and multi-agent delegation), the 59 unique component-tagged papers generate **68 component-branch entries** across the coverage matrix, more entries than papers. All counts in the narrative below follow this multi-tag convention. The remaining 143 papers (foundations, llm-agents-general, evaluation-metrics-benchmarks, mitigation, high-stakes-domains, and prior-art-additions branches) are covered in §§2-5 as contextual and cross-cutting evidence; they inform the taxonomy but do not appear in the six-component rows of Figure 4.

*0.7.1 The Counterfactual Column is Essentially Empty.* The headline of Figure 4 is a near-vacant column. Across all six component rows, formal counterfactual fairness is the named measurement framing in **exactly one paper**: [119], under long-horizon drift, which simulates demographic-conditioned ML-system trajectories over time. The remaining five components (tool/API selection, memory and retrieval, multi-agent delegation, planning/decomposition, and user modeling/personalization) each register **zero** counterfactual entries.

*Takeaway: the counterfactual column is near-empty, holding exactly one entry (she2025fairsense) across all six components, while 52 of the 68 component-branch entries fall in the unspecified column.*

This is a structural inversion of need. The counterfactual flip (re-running the same agent on a profile altered only in a protected attribute and asking whether the trajectory changed) is the natural operationalization of the agentic fairness question: "would the agent have *acted* differently for another demographic?" Group-level metrics, computed over output distributions, cannot directly answer this. Counterfactual flip can. In BCF terms, the counterfactual swap is the procedure that produces D2's  $\Delta_c$ , the component counterfactual disparity that enters the conduction equation. The dimension that BCF-based auditing most requires is precisely the dimension the field has least applied. The point is not weakened by the lone exemplar: even that single counterfactual entry, [119], is itself adjacent and non-agentic (marked  $[\dagger]$  in §3.6, a generic ML-system drift simulator never applied to an instrumented LLM-agent trajectory), so the counterfactual column is empty of properly agentic counterfactual auditing rather than holding a genuine exception.

The gap is not one of missing methods but of missing application. The counterfactual tooling exists: CFR/MASD in [85]; the CAFFE framework for systematic counterfactual evaluation [102]; the SCOPE stereotyped-prompt dataset [104]; the equal-access counterfactual audit [16]; and intervention-consistency testing [22]. All five sit in the evaluation-metrics-benchmarks branch; none has been wired through any of the six agent-pipeline components as their primary measurement instrument. The gap is one of application, not invention.

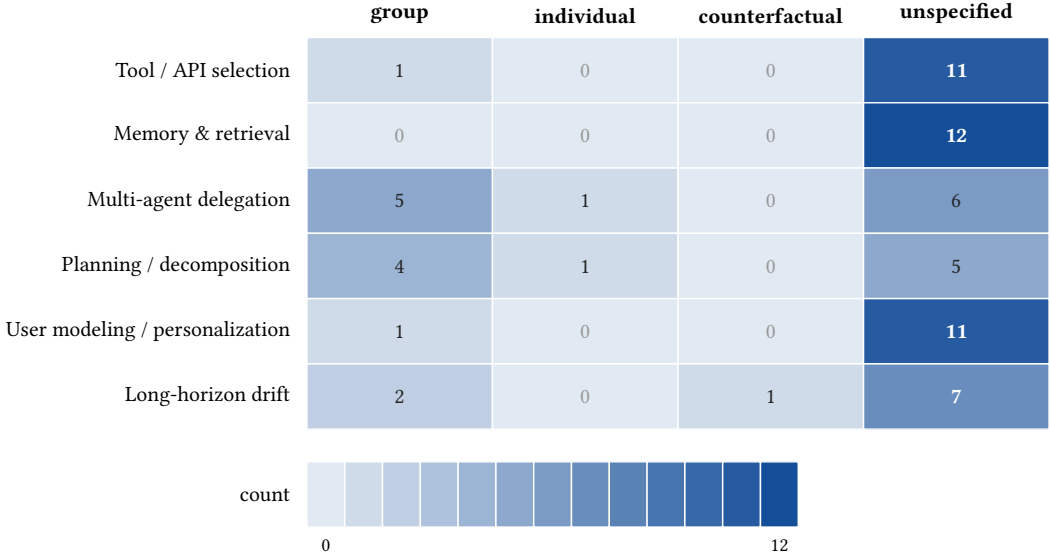


Fig. 4. Coverage matrix (agent component  $\times$  fairness dimension), cells shaded by paper count (single-hue, colorblind-safe blue). The counterfactual column holds exactly one nonzero cell (long-horizon drift, [119]); the unspecified column dominates with 52 of 68 component-branch entries.

*Takeaway (Figure 4): Across all six agent components, counterfactual fairness (the dimension BCF-based auditing requires for  $\Delta_c$  estimation) appears in exactly one of the 68 component-branch entries; GENERATE and AMPLIFY conduction signatures remain empirically uncharted.*

**0.7.2 Group Fairness Clusters Only Where Decisions are Explicit.** Where a formal fairness dimension is invoked, it is overwhelmingly group fairness, and it clusters in the two components where the agent emits an explicit, attributable decision: **multi-agent delegation** (5 group-framed papers, 1 individual) and **planning/decomposition** (4 group, 1 individual). These are the components whose outputs most resemble the binary decisions that group-parity metrics were designed for: a role assignment, a debate verdict, a task allocation, a discrete approval or denial. Multi-agent work measures disparity across agent personas and emergent group outcomes [75, 81, 83, 91]; planning work measures decision and allocation disparity [57, 73, 103]. The pattern is structural: **formalization tracks decision explicitness, not harm severity**. Components that expose a clean decision variable attract group metrics; components whose harm is diffuse (retrieval ranking, memory accumulation, user-model inference) do not, regardless of how consequential those harms may be.

**0.7.3 Tool, Memory, and Personalization Harms are Documented but Rarely Formalized.** The complementary pattern is a heavily weighted unspecified column. Three components are dominated by it almost entirely: **memory and retrieval** (0 formal, 12 unspecified), **tool/API selection** (1 group, 11 unspecified), and **user modeling/personalization** (1 group, 11 unspecified). Memory and retrieval is the starkest case: not a single paper in this branch names a formal fairness dimension, despite a substantial body documenting real harm: RAG degrading fairness even for vigilant users [55], multi-turn fairness erosion [42], belief drift from accumulating context [48, 122], and adversarial retrieval-store poisoning [138]. These papers demonstrate disparity convincingly; they



do not measure it against group, individual, or counterfactual criteria, instead reporting bespoke or qualitative gaps. The same holds for tool selection [21, 120] and personalization [63, 84, 101, 124].

Harm that lives outside the formal-fairness vocabulary cannot be benchmarked, tracked across model versions, or compared across systems. Documentation without formalization is the field's modal state for half the agent pipeline, and it is the direct driver of the P5 Measurement-adequacy gap the BCF identifies.

**0.7.4 The Locus  $\times$  BCF-Operator Grid.** Figure 4 cross-tabulates component with fairness *dimension*; Figure 5 cross-tabulates component-as-locus with BCF conduction *operator* (ATTENUATE / PRESERVE / AMPLIFY / GENERATE). The two visualizations are complementary: Figure 4 reveals what formal vocabulary is used; Figure 5 reveals what *transformation* of disparity is being empirically studied versus merely assumed.

**Construction.** Each corpus paper with a component tag is further assigned a conduction operator based on whether it documents disparity attenuation (a debiasing intervention that reduces  $\Delta_c$  at that component), preservation (disparity passes through unchanged), amplification (disparity increases), or generation (disparity arises at that component from a nominally zero input, the agentic phenomenon with no single-model analogue). Papers are tagged by the operator their primary finding instantiates; a paper can carry multiple tags.

|                               | ATTENUATE<br>$\varphi < 1$ |   |   |   | PRESERVE<br>$\varphi \approx 1$ |   |   | AMPLIFY<br>$\varphi > 1$ |   | GENERATE<br>( $0 \rightarrow >0$ ) |  | Uncharac-<br>terized |  |
|-------------------------------|----------------------------|---|---|---|---------------------------------|---|---|--------------------------|---|------------------------------------|--|----------------------|--|
| Tool/API selection            | 1                          |   | 3 |   | 1                               |   | 1 |                          | 6 |                                    |  |                      |  |
|                               | 2                          |   | 2 |   | 3                               |   | 1 |                          | 4 |                                    |  |                      |  |
|                               | 3                          |   | 1 |   | 4                               |   | 3 |                          | 1 |                                    |  |                      |  |
|                               | 2                          |   | 2 |   | 2                               |   | 0 |                          | 4 |                                    |  |                      |  |
|                               | 1                          |   | 4 |   | 3                               |   | 1 |                          | 3 |                                    |  |                      |  |
|                               | 2                          |   | 1 |   | 4                               |   | 1 |                          | 2 |                                    |  |                      |  |
| Memory & retrieval            | 2                          |   | 2 |   | 3                               |   | 1 |                          | 4 |                                    |  |                      |  |
| Multi-agent delegation        | 3                          |   | 1 |   | 4                               |   | 3 |                          | 1 |                                    |  |                      |  |
| Planning/decomposition        | 2                          |   | 2 |   | 2                               |   | 0 |                          | 4 |                                    |  |                      |  |
| User modeling/personalization | 1                          |   | 4 |   | 3                               |   | 1 |                          | 3 |                                    |  |                      |  |
| Long-horizon drift            | 2                          |   | 1 |   | 4                               |   | 1 |                          | 2 |                                    |  |                      |  |
| count                         | 0                          | 1 | 2 | 3 | 4                               | 5 | 6 |                          |   |                                    |  |                      |  |

Fig. 5. Locus  $\times$  BCF conduction-operator grid, cells shaded by paper count (single-hue, colorblind-safe blue; row sums total 68). The GENERATE and AMPLIFY columns are sparse: planning/decomposition has no GENERATE entry, and tool selection registers a single GENERATE paper, marking the highest-priority measurement gaps.

*Takeaway (Figure 5): GENERATE and AMPLIFY cells (the operators that produce distinctly agentic harm, where pipeline edges produce disparity that was absent at the input or that grows across the trajectory) are the least instrumented; the AMPLIFY/GENERATE column for planning/decomposition and the GENERATE column for tool selection are nearly uncharacterized, pointing to the highest-priority measurement gaps for BCF-guided empirical work.*

The most consequential observation is that the **GENERATE** operator (disparity arising at a pipeline edge from nominally unbiased inputs, the formal signature of the agentic multiplier in P3) has been documented in only a handful of papers. For multi-agent delegation, [91] and [81]

provide the clearest evidence of generation (bias emerging in interaction among individually measured-to-be-unbiased agents); for tool selection, [21] documents the selection-skew signature from tool-description surface features alone (the closest single-locus instance, distinct from the amplification [86] documents in the memory accumulation process). Outside these, the GENERATE column is largely empty. The audit of §6 is designed to probe this empirical frontier.

*0.7.5 Synthesis and Differentiation.* Four regularities fall out of the combined Figure 4/Figure 5 analysis.

First, **formalization tracks decision explicitness, not harm severity**: components that emit a clean decision variable attract formal metrics; diffuse-harm components do not.

Second, **the formal vocabulary thins toward the counterfactual and toward GENERATE/AMPLIFY** exactly as the evaluation question shifts from "what did the agent output?" toward the distinctively agentic questions: "would it have acted differently?" and "did the pipeline produce disparity it did not inherit?"

Third, **the modal corpus paper documents disparity without naming a fairness dimension or a conduction operator**: 52 of the 68 component-branch entries sit in the unspecified column of Figure 4, and most of the locus  $\times$  operator cells in Figure 5 are under-populated. The field is evidence-rich and formalization-poor.

Fourth, the BCF coordinate system makes the gaps precise and actionable. The empty cells of Figure 5 are specific (component, operator) pairs that can be populated by targeted experiments with the counterfactual-perturbation machinery already available [22, 85, 102].

This analysis also sharpens the differentiation from neighboring work (Table 0 in §1.4). Where [24] organizes single-turn LLM fairness by metric and [46] by datasets and mitigation, neither resolves *where in an acting pipeline* a disparity originates. Where [82] treats fairness as one subtopic of agent evaluation and [128] as a subdimension of a seven-dimensional clinical taxonomy, neither makes component coverage its unit of analysis. Where [164] proposes a framework for fairness in multi-agent AI broadly, it does not operate at the LLM-agent component level or develop the conduction-operator analysis. By projecting the corpus onto the component  $\times$  operator grid, this survey turns the measurement gap from an assertion into a structured coordinate (the empty counterfactual column and the under-populated GENERATE/AMPLIFY cells) and hands §8 a concrete, BCF-keyed research target.

## 0.8 Open Problems and Research Agenda

The coverage analysis of §7 converts the measurement gap from an observation into a set of precise coordinates: (component, operator) cells in Figure 5 that are empirically uncharacterized, P5 Measurement-adequacy cells that no existing benchmark populates, and BCF propositions whose empirical grounding is weak. Each problem below opens with its BCF coordinate, states why it matters for consequential deployment, and specifies a concrete first step (a metric definition, a dataset structure, or a system design) that an immediate follow-on could execute.

A governance thread runs through all six problems. Agents are now deployed in hiring, lending, healthcare, and public-benefits adjudication, domains where disparate treatment across race, gender, socioeconomic status, dialect, and their intersections carries direct legal and regulatory weight. The EU AI Act (Regulation 2024/1689) classifies high-risk AI systems in employment, credit, and essential-service sectors as subject to bias-testing and transparency requirements [162]; the NIST AI Risk Management Framework specifies bias evaluation as a component of AI governance [161]; New York City Local Law 144 requires automated employment decision tools to undergo annual bias audits before deployment; and EEOC guidance applies Title VII disparate-impact standards to algorithmic hiring tools. None of these frameworks currently specifies what a fairness audit of an

*acting agent* must cover, and the measurement infrastructure described in §4.4 is not adequate to discharge those requirements. Every open problem below has a governance correlate: the missing metric, dataset, or system design is also the missing regulatory instrument.

**0.8.1 Unpopulated P5 Cells: Benchmarks for Acting Agents. BCF coordinate.** P5 Measurement-adequacy states that a metric is adequate for component  $c$  only if it can detect non-zero  $\Delta_c$  at that component independently of all other components. The benchmark inventory in §4.4 shows that no existing benchmark satisfies P5 for any of the six agent-pipeline components at the action level. The uncharacterized cells of Figure 5 are exactly the P5 cells that no existing instrument populates.

**The problem.** The dominant fairness benchmarks score answers, not actions. Clinical fairness sets define the ceiling of current practice: FairMedQA evaluates demographic bias in medical QA [145]; mFARM assembles multi-faceted harm assessment for clinical decision support on static vignettes [15]; MedEqualQA probes consistency through counterfactual demographic edits to QA questions [49]; and the health equity toolbox surfaces harms at the level of generated text [100]. Each is methodologically careful; each stops at the QA boundary. None measures what a clinical acting agent does when it selects which guideline tool to invoke, which prior note to retrieve, whether to escalate to a human, or how it sequences a multi-step workup. The few studies that score actions confirm the divergence: agent decisions reveal biases absent from token-level probes [73], and decomposing unfairness in a tool-using chest X-ray agent attributes  $\Delta_c$  to specific pipeline stages invisible to answer-level scoring [147]. Existing agent evaluation frameworks (AgentBench [70], WebArena [154], SWE-bench [60]) are not demographically instrumented; they evaluate capability (task completion) rather than fairness across demographic counterfactuals, so they cannot be repurposed directly without structural additions.

**Why it matters.** A system can be near-parity on FairMedQA and still route, escalate, or order differently across demographic counterfactuals once it acts, because  $\Delta_c$  enters through tool routing and retrieval rather than the final string. Certifying agents with QA benchmarks licenses deployment on evidence that does not bind the deployed behavior, precisely the accountability gap EU AI Act Article 10 (data quality) and NIST AI RMF Measure 2.5 (bias testing) are meant to close.

**Concrete first step.** A minimal acting-agent fairness benchmark requires: (i) a set of consequential-task environments (e.g., clinical triage, hiring screening, loan underwriting) in which an agent can take multiple discrete actions; (ii) counterfactual profile pairs that differ only in a demographic signal, following the CAFFE/SCOPE design [102, 104]; (iii) logging of all intermediate actions (tool invocations, retrieval queries, routing decisions) per counterfactual pair; and (iv) reporting of per-edge  $\Delta_c$  alongside the final-verdict CFR/MASD. FairMedAgent, described in §6.4, represents one instantiation for the clinical domain; comparable instruments are needed for hiring and lending, where legal audit requirements are already in force.

**0.8.2 Unmeasured  $\varphi_{fb}$  Recurrence: Long-Horizon and Compounding Fairness. BCF coordinate.** The feedback-recurrence term  $\Delta^{(t+1)} = \varphi_{fb} \cdot \Delta^{(t)} + \Delta_{entry}$  formalizes how per-step disparity compounds across time when an agent conditions on its own prior outputs. P3 Super-additivity, applied longitudinally, predicts that  $\varphi_{fb} > 1$  produces exponential divergence of group outcomes across deployment cycles, a prediction no existing evaluation captures.

**The problem.** Fairness is evaluated as a one-shot property, but agents persist. Implicit bias in long-term memory accumulates and propagates, reinforcing small initial skews across interactions [86]. Sequential decision-making theory establishes the formal stakes: static per-step fairness constraints fail to deliver equitable long-term outcomes, and equal long-term benefit requires reasoning about how present decisions reshape future eligibility [142, 158]. Feedback loops amplify this further: training on a model's own synthetic outputs amplifies bias across generations [133], and self-consuming performative loops compound the effect at deployment [140]. The theory and

the LLM-drift empirics [42, 76, 86] have not been joined: no study evaluates an LLM agent against a formal long-term fairness criterion over a realistic multi-session deployment.

**Concrete first step.** Define a **longitudinal disparity index** (LDI) as the ratio of group-outcome divergence at turn  $T$  to turn 1, estimated by simulation over a synthetic trajectory of  $T$  interactions using the recurrence  $\Delta^{(t+1)} = \varphi_{fb} \cdot \Delta^{(t)} + \Delta_{\text{entry}}$ . Parameterize  $\varphi_{fb}$  empirically from an existing deployed agent using the FairSense simulation tooling [119]. A single-domain pilot (e.g., the loan-triage replication arm of §6) with  $T = 20$  turns and five demographic group pairs would constitute a first binding measurement.

**0.8.3 Unbounded GENERATE: Multi-Agent Fairness Dynamics. BCF coordinate.** The GENERATE operator ( $\varphi$  applied to nominally zero- $\Delta_c$  input) is the signature failure mode of multi-agent systems: disparity arises in the interaction among agents that are individually measured as unbiased. Figure 5 shows GENERATE is thinly populated across every locus; multi-agent delegation is the only locus with more than one GENERATE entry, despite the strongest theoretical and empirical motivation.

**The problem.** Multi-agent orchestration introduces failure modes with no single-agent analogue: bias can *emerge* from interaction even among agents whose individual outputs are measured as near-parity. [81] formalizes emergent bias in multi-agent decision systems as a property of the collective rather than any constituent; [91] documents bias emerging, propagating, and amplifying through agent-to-agent communication. Disparity in these settings is a function of communication topology, role-assignment structure, and message-passing dynamics. None of these is observable by a per-agent audit, and none is instrumented by existing benchmarks. The MALIBU benchmark [83] surfaces implicit bias in multi-agent assignments but does not measure the GENERATE operator specifically (bias arising at zero input to each constituent).

**Concrete first step.** Design a **GENERATE-detection probe**: a multi-agent task in which each constituent agent is individually certified as CFR = 0 (i.e., no flip under demographic swap) on a single-turn benchmark, then assembled into a two-agent delegation. Measure the CFR of the delegation. Any non-zero CFR at the delegation level constitutes GENERATE evidence; the difference between delegation-level and max(constituent) CFR is an empirical estimate of the GENERATE amplitude for that topology. The probe requires only the counterfactual evaluation machinery of [85] and any existing two-agent framework [129].

**0.8.4 Intersectional and Sociotechnical Harms in Deployment. BCF coordinate.** BCF D2 defines  $\Delta_c$  as a counterfactual disparity over a single protected attribute; real deployment harms are intersectional (jointly over multiple attributes) and sociotechnical (conditioned on deployment context and model parameters together). This is a D2 extension problem: the current formalization is a first-order approximation.

**The problem.** Bias research frequently isolates one protected attribute on a clean benchmark, but real deployments harm people at intersections. The largest-scale evidence is unambiguous: sociodemographic biases in LLM medical decision-making produce differential management recommendations at clinical scale, with race, gender, and socioeconomic status interacting [95]; a mortgage underwriting experiment documents racial disparities in lending [20]; and judicial fairness evaluation finds systematic differential treatment across defendant demographics [58]. These harms are jointly demographic and contextual, depending on the deployment’s stakes, decision structure, and the combination of attributes a real person carries. Intersectional treatment is specifically flagged in algorithmic hiring regulation [108].

**Concrete first step.** Extend D2 to an intersectional counterfactual disparity  $\Delta_c^{A \times B}$  defined as the  $\Delta_c$  estimated under joint swap of attributes  $A$  and  $B$  minus the sum of the marginal  $\Delta_c$  values, measuring the interaction term. Apply this to the §6 audit’s hiring task with a 2×2 race-by-gender counterfactual design; this requires only doubling the profile set and is immediately feasible.

**0.8.5 Empty Operator Rows of the Conduction Grid: Governance and Standardized Auditing. BCF coordinate.** The ATTENUATE rows of Figure 5 are partially populated (mitigation papers), but the formal connection between a mitigation’s claimed ATTENUATE effect and its BCF-level  $\varphi$  value is never made: no paper reports how much  $\varphi_e$  decreases at a specific pipeline edge following a given intervention. This is a P4 Mitigation-matching gap.

**The problem.** There is no shared standard for what constitutes a fairness audit of an LLM agent, nor a required reporting format for action-level disparity. The ingredients exist: the CAFFE counterfactual framework [102], the SCOPE dataset [104], the LangFair evaluation toolkit [4], the intervention-consistency test [22], and the dataset survey [155]. No governance instrument standardizes them into a repeatable, comparable audit of an *acting* system. The regulatory field has outpaced the measurement infrastructure: the EU AI Act [162] requires bias testing of high-risk employment and credit systems, the NIST AI RMF [161] specifies bias evaluation as a governance requirement, NYC Local Law 144 mandates annual bias audits for automated employment decision tools, and EEOC disparate-impact analysis applies to algorithmic hiring. None specifies what those audits must cover for an acting agent as opposed to a static classifier.

**Concrete first step.** Draft a **minimum audit specification** for acting agents with four components: (i) a protocol for constructing counterfactual profile sets (building on [102]); (ii) a required metric set covering, at minimum, per-component CFR, tool-invocation disparity, and escalation-rate disparity, aligned with the BCF’s D2 definition; (iii) a standardized reporting template with per-component  $\Delta_c$  values and the conduction-operator classification (ATTENUATE/PRESERVE/AMPLIFY/GENERATE) for each active edge; and (iv) a recourse specification covering what information must be logged to enable post-hoc attribution and contestation. This specification can be drafted as a technical standard proposal and submitted for comment to the NIST AI 100-1 framework revision process.

**0.8.6 Mitigation at the Agent Level, Not the Token Level. BCF coordinate.** P4 Mitigation-matching states that an intervention is matched if it operates at the same pipeline locus as the disparity it is meant to correct. The mitigation map in §5 shows that the ATTENUATE column of Figure 5 is populated almost entirely by token-level interventions (prompting, alignment, output filtering), which are mismatched to the six agent-pipeline loci where the AMPLIFY and GENERATE operators are documented.

**The problem.** The mitigation literature overwhelmingly intervenes on model outputs: debiasing prompts [43, 72], preference optimization [11, 19], inference-time activation steering [74], and output guardrails [36]. These operate on tokens. But §3 locates agentic harm in actions and architecture: which tool is chosen, what is retrieved into context, how roles are assigned, how a plan is decomposed. Token-level debiasing leaves these channels untouched. The few process-aware interventions point the way: balanced-retrieval agents [118], bias-aware retrieval agents [117], identity anonymization to neutralize role-induced debate bias [28], and multi-agent debiasing frameworks [94, 143]. They are scattered relative to the dominant approach, and no unified framework simultaneously addresses tool-selection, memory-accumulation, and delegation loci.

**Concrete first step.** Implement a **component-keyed guardrail layer** as a proof of concept: a wrapper that intercepts the agent’s action proposal at each of the three highest-priority AMPLIFY/GENERATE loci identified in Figure 5 (tool selection, multi-agent delegation, long-horizon accumulation) and applies a targeted  $\Delta_c$  reduction intervention before action execution. The three interventions are, respectively: a demographic-blind re-ranking of tool candidates, a role-assignment audit against an assignment-disparity threshold, and a memory-hygiene filter. Evaluate the wrapper using the §6 audit’s C1 to C3 configurations, reporting per-edge  $\varphi_e$  before and after to directly test P4 Mitigation-matching.

**0.8.7 Synthesis.** The autonomy that makes agents valuable (acting, persisting, delegating, self-conditioning) is precisely what current QA-level measurement, single-shot certification, and token-level mitigation cannot observe. Closing the gap requires instruments, protocols, and interventions defined over trajectories and actions, governed at the system level, and reported in a standardized format auditors and regulators can use. The BCF provides the coordinate system; the six open problems above specify the cells to fill.

## 0.9 Conclusion

LLM-based agents have crossed a consequential threshold: they select tools, accumulate memories across turns, delegate subtasks across role-assigned subagents, decompose goals into multi-step plans, adapt to inferred user identities, and persist behavior over long horizons [137, 141]. Each of these pipeline stages is a distinct locus at which demographic disparity can enter, propagate, and compound into consequential harm. The Bias Conduction Framework (BCF) introduced in §3.0 formalizes this structure: D1 defines the agent loop  $\langle \pi, T, M, R \rangle$ ; D2 defines component counterfactual disparity  $\Delta_c$  via trace-replay; D3 defines the conduction operators (ATTENUATE  $\varphi < 1$ , PRESERVE  $\varphi \approx 1$ , AMPLIFY  $\varphi > 1$ , GENERATE  $\varphi$  on zero input); the conduction equation  $\Delta_\tau \approx \Sigma_c \Delta_c \cdot \Pi \varphi_e$  with feedback recurrence  $\Delta^{(t+1)} = \varphi_{fb} \cdot \Delta^{(t)} + \Delta_{entry}$  derives the trajectory disparity from its per-component contributions; and five propositions (P1 Locality, P2 Masking, P3 Super-additivity, P4 Mitigation-matching, P5 Measurement-adequacy) characterize the conditions under which component-level auditing is necessary, sufficient, and policy-relevant.

This survey has presented the first dedicated taxonomy of fairness in LLM-based agents organized by this component axis, spanning tool and API selection (§3.1), memory and retrieval (§3.2), multi-agent delegation and role assignment (§3.3), planning and decomposition (§3.4), user modeling and personalization (§3.5), and long-horizon trajectory drift (§3.6), with a cross-cutting account of how component-level harms compound into the agentic multiplier (§3.7, P3). Against this architecture the survey maps the evaluation field (§4), characterizing group, individual, and counterfactual metric families lifted to agent trajectories and generalizing the CFR and MASD statistics of [85] across all pipeline components, while Table 2 documents that nearly every existing benchmark (clinical fairness sets [15, 49, 145], hiring and lending audits [13, 14], compositional evaluation suites [131]) scores model outputs rather than agent actions. The mitigation survey (§5) follows the P4 Mitigation-matching principle, distinguishing prompt-level interventions [43, 72], alignment-stage approaches [11, 19], architectural guardrails [36], and the emerging class of process-level and multi-agent debiasing strategies [28, 65, 94]. The coverage analysis (§7) quantifies the gap: 52 of 68 component-branch entries sit in the unspecified column of Figure 4; the counterfactual column (the dimension required to compute D2’s  $\Delta_c$ ) contains exactly one entry across all six components; and the GENERATE/AMPLIFY cells of Figure 5 are the least populated despite being the loci of distinctively agentic harm. A pilot of the cross-framework audit (§6) illustrates the pattern P2 and P3 predict, decision-level parity masking a score-level disparity that grows with agentic scaffolding, though a full confirmatory study remains future work.

The measurement gap documented by this survey is a gap in accountability infrastructure, not in tooling alone. LLMs and LLM-based agents are being deployed, piloted, and evaluated in consequential decision settings, including hiring [90, 126, 135], clinical decision support [95, 99], lending [20, 41], and education [136] (much of the cited evidence is audit or evaluation in these settings rather than confirmed operational deployment), subject to regulatory requirements (EU AI Act [162], NIST AI RMF [161], NYC Local Law 144, EEOC disparate-impact guidance) that no current acting-agent fairness benchmark is equipped to satisfy. The long-term fairness risks are compounding: feedback recurrence documented by [133] and [140], memory accumulation characterized by [86], and the sequential fairness theory of [158] and [142] collectively predict

that systems fair at any single step can drive divergent cumulative outcomes across groups over deployment cycles.

The immediate priority is benchmark construction for acting agents: evaluation infrastructure that scores trajectories and decisions rather than tokens and outputs, applies counterfactual demographic perturbations at the action level [16, 22, 102], and covers the per-component, per-edge  $\Delta_c$  granularity the BCF specifies. That infrastructure is the precondition for every subsequent advance in mitigation (P4), governance (§8.5), and longitudinal auditing (§8.2). The field has the conceptual vocabulary, the BCF framework, and the first components of the counterfactual tooling. The gap is instrumentation.

## 0.10 Ethical Considerations, Societal Impact, and Limitations of this Survey

Surveys of bias and fairness in AI systems carry their own ethical charge: the taxonomy and evaluation framework this work advances can be used both to *audit and constrain* unfair agent behavior and to *reverse-engineer and evade* such constraints. We address this dual-use concern and several additional limitations that bear on how the survey’s findings should be interpreted and acted upon.

**0.10.1 Dual-Use of the Taxonomy and BCF Framework.** The BCF’s conduction-operator vocabulary (identifying which pipeline edges are AMPLIFY or GENERATE loci, and which mitigation interventions reduce  $\varphi_e$  at each edge) is designed as an auditor’s instrument. The same information can be used adversarially: knowing that tool-description metadata is a GENERATE locus [120] could guide an actor who wishes to steer an agent’s routing by manipulating tool descriptions rather than inputs. By the same logic, the fact that memory-accumulation is an AMPLIFY locus could inform the design of retrieval-store poisoning attacks [138] calibrated to exploit that amplification. The survey discloses these mechanisms because responsible auditing requires understanding them; the disclosure is not an operational guide for exploitation. We recommend that practitioners who use the BCF framework as an audit template also consult it as an adversarial-risk checklist, evaluating deployed agents for inadvertent bias and for susceptibility to targeted disparity injection at the loci this taxonomy identifies.

**0.10.2 Academic-vs-Legal Fairness Gap.** The fairness criteria formalized in §2.1 and operationalized throughout the survey (demographic parity, equalized odds, counterfactual fairness) are academic constructs. Their relationship to the legal standards that govern high-stakes AI deployment is partial and contested. Disparate-impact doctrine under Title VII (applied to hiring), the Equal Credit Opportunity Act (lending), and the EU AI Act’s non-discrimination provisions each impose distinct evidentiary standards, burden-of-proof structures, and remediation requirements that academic fairness metrics do not automatically satisfy. In particular: (i) a system achieving statistical demographic parity may still constitute unlawful disparate treatment under intent-based theories; (ii) a system failing the counterfactual flip test may nonetheless satisfy legal requirements if the disparity is justified by a legitimate business necessity; and (iii) the granular per-component  $\Delta_c$  measurements the BCF calls for are not currently required by any regulatory framework, which means deployers may satisfy legal compliance while exhibiting large action-level disparities invisible to regulators. This gap reflects a genuine normative distance between what the research community can measure and what policymakers have chosen to require; it is not a flaw in academic fairness research. We encourage legal scholars and regulators to engage with the BCF’s measurement proposals as inputs to the standard-setting processes now underway under the EU AI Act and NIST AI RMF frameworks.

*0.10.3 English- and US/EU-Centric Corpus Positionality.* The 202-paper corpus was assembled from English-language academic venues (arXiv, ACL Anthology, ACM DL, NeurIPS/ICML/ICLR, FAccT, Nature), all of which have geographic and linguistic concentrations in the United States and Western Europe. Several structural biases follow from this:

- **Language bias:** nearly all benchmark datasets and evaluation prompts are in English; dialect bias findings (e.g., [84] on African American Language; [71] on non-standard dialects) are primarily US-context findings. Bias patterns in Arabic, Hindi, Swahili, and other high-speaker-count languages are substantially underrepresented.
- **Protected-attribute scope:** the protected attributes most frequently studied (race (primarily Black/white US framing), binary gender, and US-relevant socioeconomic status) do not map cleanly onto the protected categories, group identities, and fairness norms of other legal and cultural contexts. India's caste system, Brazil's color classification, the EU's national-origin protections, and indigenous identity frameworks are effectively absent from the corpus.
- **High-stakes-domain coverage:** the corpus's consequential-domain evidence clusters in US hiring (name-based resume audits), US lending (mortgage underwriting), and US clinical settings (medical QA). Non-US regulatory contexts, public-benefit adjudication systems, and welfare-state service delivery are not covered.

These positionality constraints mean the survey's findings are most directly applicable to LLM-based agents deployed in English-language, US- or EU-regulated, employment/credit/healthcare contexts, and should be extrapolated to other contexts with caution. Geographic and multilingual expansion of the corpus is the highest-priority scope extension for a follow-up survey.

*0.10.4 Limitations of the Pilot Audit.* The cross-framework audit of §6 is reported only as a small directional pilot (§6.5). As a result:

- The BCF propositions P2 and P3 are supported by theoretical derivation, by the multi-paper evidence cited throughout §§3-4, and now by a directional pilot whose MASD trend (1.2, 3.4, 6.4 across C0, C2, C3) is consistent with masking-and-amplification. They are not confirmed: the pilot uses  $N = 10$ , a single race-and-gender-conflated name pair, one model, and uniformly strong resumes that produce a decision-level ceiling. Readers should treat P2 and P3 as well-motivated hypotheses with strong indirect support and one small consistent pilot, not as empirically confirmed claims.
- The conduction-operator tagging in Figure 5 reflects the survey authors' interpretation of whether each cited paper's finding instantiates ATTENUATE, PRESERVE, AMPLIFY, or GENERATE at the relevant locus, not a standardized classification applied by the original authors. Some cells may be reclassified as the audit methodology is refined.
- The pilot instantiates only the answer-level metrics (CFR, MASD) and the decision-action disparity. The remaining action-level metrics (tool-invocation disparity, escalation disparity, plan-allocation disparity) are specified in §6.2 but require a tool-using deployment and have not been computed; their distributional properties, sensitivity, and test-retest reliability are unknown until the full study.

These limitations are by design: the survey's contribution is the framework, the protocol, and a first directional measurement, not a confirmatory empirical study. The harness and data are released as a replication target; the full study is the immediate next step (§8.1).

*0.10.5 Fast-Moving Field and Recency Constraints.* The corpus cutoff is mid-2026. The LLM-agent field's publication rate has accelerated sharply since 2023 [137], and the fairness subfield is following the same trajectory. Several developments that may alter the picture drawn here are already visible at the corpus boundary:



- **Frontier model capability shifts:** improvements in instruction-following fidelity and tool-calling reliability may change the empirical profile of which pipeline edges are AMPLIFY versus PRESERVE loci, as models become more or less sensitive to demographic cues at each stage.
- **Regulatory enactments:** the EU AI Act’s implementing acts specifying audit requirements for high-risk AI systems are expected to issue in 2025 to 2026; these may operationalize bias-testing requirements in ways that alter the alignment between academic fairness metrics and legal compliance requirements.
- **Benchmark releases:** AgentBench [70], WebArena [154], and  $\tau$ -bench are actively adding evaluation dimensions; it is possible that demographically instrumented agentic fairness evaluation will be released in these frameworks within the survey’s publication window, which would partially populate the empty cells of Figure 4.

Readers should treat the coverage matrix as a snapshot of the field as of mid-2026, not as a permanent characterization. The BCF framework and the open-problem agenda of §8 are designed to remain relevant as the empirical field changes; the specific cell counts in Figure 4 and Figure 5 should be updated as new work appears.

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